

MOHSIN IBRAHIM HOSSAIN
AIUB, MATH 3 NOTE.

MATH-3

Complex Variable
Laplace and Z-Transformation.

Table of Contents

SL.NO	Topic	Pages
1	Laplace Transformation	03-18
2	Pice-Wise Function	19-29
3	Inverse Laplace Transformation	30-47
4	Application of Laplace Transformation	48-62
5	System of ODE	63-77
6	Dirac's Delta Function	78-80
7	Unit Step Function	81-99
8	Inverse Unit Step Function	100-108
9	Complex Variables	109-121
10	Complex Plane	122-129
11	Polar Form of the Complex Numbers	130-141
12	Power & Roots	142-154
13	Differentiability & Analyticity	155-170

Laplace Transformation

Choo

Countdown

Laplace Transformation:-

⇒ Let, $f(t)$ be a function of t defined for $0 \leq t < \infty$ then Laplace transformation of $f(t)$ is denoted by $\mathcal{L}\{f(t)\}$ or $F(s)$.

$$\therefore \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$$

Direct Formula:-

$$\textcircled{1} \mathcal{L}\{c\} = \frac{c}{s} \quad ; \text{ c is any constant, } s > 0$$

$$\textcircled{2} \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad ; \text{ where } n = 0, 1, 2, 3, \dots$$

$$\textcircled{3} \mathcal{L}\{e^{at}\} = \frac{1}{s-a} \quad ; \text{ where } s > a$$

$$\textcircled{4} \mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$$

$$\textcircled{5} \mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$$

$$\textcircled{6} \mathcal{L}\{\sinh at\} = \frac{a}{s^2 - a^2}$$

$$\textcircled{7} \mathcal{L}\{\cosh at\} = \frac{s}{s^2 - a^2}$$

P.T.O

Derivation of Direct Formula:-

(*) $f(t) = c$

we know,
Laplace transform

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$$

$$= \int_0^{\infty} e^{-st} \cdot c \cdot dt$$

$$= c \int_0^{\infty} e^{-st} \cdot dt$$

$$= c \left[-\frac{1}{s} e^{-st} \right]_0^{\infty}$$

$$= -\frac{c}{s} (0 - 1)$$

$$= \frac{c}{s}$$

$$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{c\} = \frac{c}{s}$$

(*) $f(t) = t$

$$\therefore \mathcal{L}\{f(t)\} = \int_0^{\infty} t \cdot e^{-st} \cdot dt$$

$$= t \int_0^{\infty} e^{-st} \cdot dt - \int_0^{\infty} \frac{d}{dt} t \times \int_0^{\infty} e^{-st} \cdot dt \cdot dt$$

$$= t \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} - \int_0^{\infty} -\frac{1}{s} e^{-st} \cdot dt$$

$$= \left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^{\infty}$$

$$= \left[-\frac{\infty}{s} e^{-\infty} - \frac{0}{s} e^{-s \cdot 0} \right] - \frac{1}{s^2} [e^{-\infty} - e^0]$$

$$= -[0 - 0] - \frac{1}{s^2} [0 - 1]$$

$$= \frac{1}{s^2}$$

$$\therefore \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$* f(t) = e^{at}$$

$$\begin{aligned} \therefore \mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-st} \cdot e^{at} \cdot dt \\ &= \int_0^{\infty} e^{-st+at} \cdot dt \\ &= \int_0^{\infty} e^{t(-s+a)} \cdot dt \\ &= \frac{1}{-s+a} \left[e^{t(-s+a)} \right]_0^{\infty} \\ &= \frac{1}{-s+a} (0 - 1) \\ &= \frac{1}{s-a} \end{aligned}$$

$$\therefore \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

Ans:-

Find the transform, Assume that a, b, ω, θ are constant.

Page: 210

$$\textcircled{1} f(t) = 3t + 12$$

$$\begin{aligned} \Rightarrow \mathcal{L}\{f(t)\} &= \mathcal{L}\{3t + 12\} \\ &= \mathcal{L}\{3t\} + \mathcal{L}\{12\} \\ &= 3 \times \mathcal{L}\{t\} + \mathcal{L}\{12\} \\ &= 3 \times \frac{1}{s^2} + \frac{12}{s} \\ &= \frac{3}{s^2} + \frac{12}{s} \\ &= \frac{3 + 12s}{s^2} \end{aligned}$$

Ans:-

$$\textcircled{2} f(t) = (a - bt)^2$$

$$\Rightarrow f(t) = a^2 - 2abt + b^2t^2$$

$$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{a^2 - 2abt + b^2t^2\}$$

$$= \mathcal{L}\{a^2\} - \mathcal{L}\{2abt\} + \mathcal{L}\{b^2t^2\}$$

$$= \frac{a^2}{s} - \frac{2ab}{s^2} + \frac{2b^2}{s^3}$$

Ans:—

$$\textcircled{3} f(t) = \cos \pi t$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos \pi t\}$$

$$= \frac{s}{s^2 + \pi^2}$$

Ans:—

$$\textcircled{4} f(t) = \cos^2 \omega t$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos^2 \omega t\}$$

$$= \mathcal{L}\left\{\frac{1}{2}(1 + \cos 2\omega t)\right\}$$

$$= \frac{1}{2} \mathcal{L}\{1\} + \frac{1}{2} \mathcal{L}\{\cos 2\omega t\}$$

$$= \frac{1}{2s} + \frac{s}{2(s^2 + 4\omega^2)}$$

Ans:—

$$(7) f(t) = \sin(\omega t + \theta)$$

$$\Rightarrow f(t) = \sin \omega t \cdot \cos \theta + \cos \omega t \cdot \sin \theta \quad [\sin(a+b) = \sin a \cdot \cos b + \cos a \cdot \sin b]$$

$$\begin{aligned} \therefore \mathcal{L}\{f(t)\} &= \mathcal{L}\{\sin \omega t \cdot \cos \theta + \cos \omega t \cdot \sin \theta\} \\ &= \mathcal{L}\{\sin \omega t \cdot \cos \theta\} + \mathcal{L}\{\cos \omega t \cdot \sin \theta\} \\ &= \cos \theta \cdot \mathcal{L}\{\sin \omega t\} + \sin \theta \cdot \mathcal{L}\{\cos \omega t\} \\ &= \cos \theta \times \frac{\omega}{s^2 + \omega^2} + \sin \theta \times \frac{s}{s^2 + \omega^2} \end{aligned}$$

$$= \frac{\omega \cdot \cos \theta + s \cdot \sin \theta}{s^2 + \omega^2}$$

Ans:-

$$(8) f(t) = 1.5 \sin(3t - \pi/2)$$

$$\Rightarrow \mathcal{L}\{f(t)\} = \mathcal{L}\{1.5 \sin(3t - \pi/2)\} \quad [\sin(a-b) = \sin a \cdot \cos b - \cos a \cdot \sin b]$$

$$= \mathcal{L}\{1.5 [\sin 3t \cdot \cos \frac{\pi}{2} - \cos 3t \cdot \sin \frac{\pi}{2}]\}$$

$$= \frac{3}{2} \mathcal{L}\{\sin 3t \cdot \cos \frac{\pi}{2}\} - \frac{3}{2} \mathcal{L}\{\cos 3t \cdot \sin \frac{\pi}{2}\}$$

$$= \frac{3}{2} \times \cos \frac{\pi}{2} \times \mathcal{L}\{\sin 3t\} - \frac{3}{2} \times \sin \frac{\pi}{2} \times \mathcal{L}\{\cos 3t\}$$

$$= \frac{3}{2} \times 0 \times \frac{3}{s^2 + 9} - \frac{3}{2} \times 1 \times \frac{s}{s^2 + 9}$$

$$= - \frac{3s}{2(s^2 + 9)}$$

(11) $f(t) = 5 \cosh 2t - 3 \sinh t$

$$\begin{aligned} \Rightarrow \mathcal{L}\{f(t)\} &= 5 \mathcal{L}\{\cosh 2t\} - 3 \mathcal{L}\{\sinh t\} \\ &= 5 \times \frac{s}{s^2 - 2^2} - 3 \times \frac{1}{s^2 - 1^2} \\ &= \frac{5s}{s^2 - 4} - \frac{3}{s^2 - 1} \end{aligned}$$

Ans:—

(12) $f(t) = \sin^2\left(\frac{1}{2}\pi t\right)$

$$\begin{aligned} \Rightarrow \mathcal{L}\{f(t)\} &= \mathcal{L}\left\{\sin^2\left(\frac{1}{2}\pi t\right)\right\} \left[\because \sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)\right] \\ &= \mathcal{L}\left\{\frac{1}{2}(1 - \cos 2 \times \frac{1}{2}\pi t)\right\} \\ &= \mathcal{L}\left\{\frac{1}{2}(1 - \cos \pi t)\right\} \\ &= \frac{1}{2} \mathcal{L}\{1\} - \frac{1}{2} \mathcal{L}\{\cos \pi t\} \\ &= \frac{1}{2s} - \frac{s}{2(s^2 + \pi^2)} \end{aligned}$$

Ans:—

(13) $f(t) = \sin \omega t \cdot \cos \omega t$

$$\begin{aligned} \Rightarrow \mathcal{L}\{f(t)\} &= \mathcal{L}\{\sin \omega t \cdot \cos \omega t\} \left[\because 2 \sin \theta \cdot \cos \theta = \sin 2\theta\right] \\ &= \mathcal{L}\left\{\frac{1}{2} \sin 2\omega t\right\} \\ &= \frac{2\omega}{2(s^2 + 4\omega^2)} \\ &= \frac{\omega}{s^2 + 4\omega^2} \end{aligned}$$

Ans:—

First Shifting or Translation property:-

Formula:-

① if $\mathcal{L}\{f(t)\} = F(s)$ then,

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$

Example:-

$$\text{Q} \quad f(t) = \frac{e^{-2t} \cdot t^{10}}{e^{2t} \cdot f(t)}$$

$$= F(s+2)$$

$$= \mathcal{L}\{f(t+2)\}$$

$$= \mathcal{L}\{(t+2)^{10}\}$$

$$= \frac{10!}{(s+2)^{11}}$$

Ans:-

Page:- 210:-

Application of s-shifting:-

$$\text{Q} \quad f(t) = \sinh t \cdot e^{2t}$$

we know:-

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$

$$\therefore \mathcal{L}\{e^{2t} \cdot \sinh t\} = F(s-2) \quad \text{--- (1)}$$

Now, $f(t) = \sinh t$

$$F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{\sinh t\}$$
$$= \frac{1}{s^2 - 1^2}$$

From eq (1) =

$$\mathcal{L}\{e^{2t} \cdot \sinh 4t\} = \frac{1}{(s-2)^2 - 1}$$

Ans:—

⑥ $f(t) = e^{-t} \cdot \sinh 4t$

⇒ We know:—

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$

$$\therefore \mathcal{L}\{e^{-t} \cdot \sinh 4t\} = F(s+1) \quad \text{--- (1)}$$

Now, $f(t) = \sinh 4t$

$$F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{\sinh 4t\}$$

$$= \frac{4}{s^2 - 4^2}$$

From eq (1) =

$$\mathcal{L}\{e^{-t} \cdot \sinh 4t\} = \frac{4}{(s+1)^2 - 16}$$

Ans:—

Page no.:- 211:-

Q33. $f(t) = t^2 e^{-3t}$

We know:-

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$

$$\therefore \mathcal{L}\{e^{-3t} \cdot t^2\} = F(s+3) \quad \text{--- (1)}$$

Now,

$$\begin{aligned} F(s) &= \mathcal{L}\{f(t)\} \\ &= \mathcal{L}\{t^2\} \\ &= \frac{2}{s^3} \end{aligned}$$

from eq (1):

$$\mathcal{L}\{e^{-3t} \cdot t^2\} = \frac{2}{(s+3)^3}$$

Ans:-

Q34. $f(t) = k e^{-at} \cdot \cos \omega t$

We know:-

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$

$$\therefore \mathcal{L}\{k e^{-at} \cdot \cos \omega t\} = F(s+a) \quad \text{--- (1)}$$

Now,

$$\begin{aligned} F(s) &= \mathcal{L}\{k \cdot \cos \omega t\} \\ &= k \mathcal{L}\{\cos \omega t\} \end{aligned}$$

from eq (1):

$$\mathcal{L}\{k e^{-at} \cdot \cos \omega t\} = k \cdot \frac{s}{s^2 + \omega^2}$$

$$\mathcal{L}\{k e^{-at} \cdot \cos \omega t\} = \frac{k(s+a)}{(s+a)^2 + \omega^2}$$

Ans:-

Q5) $0.5e^{-4.5t} \cdot \sin 2\pi t = f(t)$

we know:-

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$

$$\therefore \mathcal{L}\{0.5e^{-4.5t} \cdot \sin 2\pi t\} = F(s+4.5) \quad \text{--- (1)}$$

Now, $F(s) = \mathcal{L}\{0.5 \sin 2\pi t\}$
 $= 0.5 \times \mathcal{L}\{\sin 2\pi t\}$
 $= 0.5 \times \frac{2\pi}{s^2 + 4\pi^2}$

from eq (1) :-

$$\mathcal{L}\{0.5e^{-4.5t} \cdot \sin 2\pi t\} = 0.5 \times \frac{2\pi}{(s+4.5)^2 + 4\pi^2}$$

Ans:-

Q6) $\sinh t \cdot \cosh t = f(t)$

$$\Rightarrow f(t) = \left(\frac{e^t - e^{-t}}{2}\right) \cosh t$$

$$\left[\begin{aligned} \because \sinh at &= \frac{e^{at} - e^{-at}}{2} \\ \because \cosh at &= \frac{e^{at} + e^{-at}}{2} \end{aligned} \right]$$

$$\Rightarrow f(t) = \frac{1}{2} e^t \cdot \cosh t - \frac{1}{2} e^{-t} \cdot \cosh t$$

we know:-

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$

$$\therefore \mathcal{L}\left\{\frac{1}{2}(e^t - e^{-t}) \cosh t\right\} = F_1(s-1) - F_2(s+1) \quad \text{--- (1)}$$

Now $F_1(s) = \mathcal{L}\left\{\frac{1}{2} \cosh t\right\}$
 $= \frac{1}{2} \mathcal{L}\{\cosh t\}$
 $= \frac{1}{2} \times \frac{s}{s^2 + 1}$

$F_2(s) = \mathcal{L}\left\{\frac{1}{2} \cosh t\right\}$
 $= \frac{1}{2} \times \frac{s}{s^2 + 1}$

from eq (v) =

$$\mathcal{L}\left\{\frac{1}{2}(e^t - e^{-t})\cos t\right\} = \frac{1}{2} \times \frac{(s-1)}{(s-1)^2+1} - \frac{1}{2} \times \frac{(s+1)}{(s+1)^2+1}$$
$$= \frac{1}{2} \left\{ \frac{(s-1)}{(s-1)^2+1} - \frac{(s+1)}{(s+1)^2+1} \right\}$$

Ans:—

Page No: 2518—

(12) $f(t) = e^{-t}(\cos 4t - 2\sin 4t)$

$$\Rightarrow f(t) = e^{-t}\cos 4t - e^{-t} \cdot 2\sin 4t$$

We know:—

$$\mathcal{L}\{f(t) \cdot e^{at}\} = F(s-a)$$

$$\therefore \mathcal{L}\{e^{-t}(\cos 4t - 2\sin 4t)\} = F_1(s+1) - F_2(s+1) \quad \text{--- (1)}$$

Now:—

$$F_1(s) = \mathcal{L}\{\cos 4t\}$$
$$= \frac{s}{s^2+4^2}$$

$$F_2(s) = \mathcal{L}\{2\sin 4t\}$$
$$= 2 \times \frac{4}{s^2+4^2}$$
$$= \frac{8}{s^2+4^2}$$

Now, From eq (1) =

$$\mathcal{L}\{e^{-t}(\cos 4t - 2\sin 4t)\} = \frac{(s+1)}{(s+1)^2+16} - \frac{8}{(s+1)^2+16}$$

$$= \frac{(s+1-8)}{(s+1)^2+16} = \frac{s-7}{(s+1)^2+16}$$

Ans:—

Property of Multiplication by t^n :

Formula:

① if $\mathcal{L}\{f(t)\} = F(s)$,

then $\mathcal{L}\{t^n \cdot f(t)\} = (-1)^n \cdot \frac{d^n}{ds^n} [F(s)]$

Example:

$\mathcal{L}\{f(t)\} = \frac{1}{s^n} \cdot \sin 2t$

we know:

$\mathcal{L}\{t^n \cdot f(t)\} = (-1)^n \cdot \frac{d^n}{ds^n} [F(s)]$

$\therefore \mathcal{L}\{t \cdot \sin 2t\} = (-1)^1 \cdot \frac{d}{ds} \mathcal{L}\{f(t)\}$

$= - \frac{d}{ds} \mathcal{L}\{\sin 2t\}$

$= - \frac{d}{ds} \times \frac{2}{s^2 + 4}$

$= \frac{2 \cdot 2s}{(s^2 + 4)^2}$

$= \frac{4s}{(s^2 + 4)^2}$

Ans:

$$\begin{aligned} &= \frac{d}{ds} 2 \times (s^2 + 4)^{-1} \\ &= 2 \times -1 (s^2 + 4)^{-1-1} \times 2s \\ &= \frac{-4s}{(s^2 + 4)^2} \end{aligned}$$

Page No. 251:-

(17) $f(t) = t \cos t + \sin t$ ——— (1)

We know:-
mom

$$\mathcal{L}\{t^n \cdot f(t)\} = (-1)^n \cdot \frac{d^n}{ds^n} [F(s)]$$

$$\therefore \mathcal{L}\{t \cdot \cos t\} = (-1)^1 \frac{d}{ds} \mathcal{L}\{\cos t\}$$

$$= - \frac{d}{ds} \times \frac{s}{s^2+1}$$

$$= - \left[\frac{(s^2+1) \frac{d}{ds} s - s \frac{d}{ds} (s^2+1)}{(s^2+1)^2} \right]$$

$$= - \left[\frac{(s^2+1) - s \cdot 2s}{(s^2+1)^2} \right]$$

$$= \frac{2s^2 - s^2 - 1}{(s^2+1)^2}$$

$$= \frac{s^2 - 1}{(s^2+1)^2}$$

Again mom $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$

So, from eq (1) =

$$\mathcal{L}\{t \cdot \cos t + \sin t\} = \frac{s^2 - 1}{(s^2+1)^2} + \frac{1}{s^2+1}$$

Ans:-

Q) $f(t) = 12t \cdot e^{-3t}$

We know:-

$$\mathcal{L}\{t^n \cdot f(t)\} = (-1)^n \cdot \frac{d^n}{ds^n} [F(s)]$$

$$\therefore \mathcal{L}\{12t \cdot e^{-3t}\} = (-1)^1 \times 12 \times \frac{d}{ds} \mathcal{L}\{e^{-3t}\}$$

$$= -12 \frac{d}{ds} \times \frac{1}{s+3}$$

$$= -12 \times \frac{-1}{(s+3)^2}$$

$$= \frac{12}{(s+3)^2}$$

Ans:-

For More practice:-

* $f(t) = t \cdot \cosh 2t$

We know:-

$$\mathcal{L}\{t^n \cdot f(t)\} = (-1)^n \cdot \frac{d^n}{ds^n} [F(s)]$$

$$\therefore \mathcal{L}\{t \cdot \cosh 2t\} = (-1)^1 \frac{d}{ds} \mathcal{L}\{\cosh 2t\}$$

$$= - \frac{d}{ds} \times \left(\frac{s}{s^2 - 4} \right)$$

$$= - \left[\frac{(s^2 - 4) \frac{d}{ds} s - (s) \frac{d}{ds} (s^2 - 4)}{(s^2 - 4)^2} \right]$$

$$= - \left[\frac{s^2 - 4 - s \cdot 2s}{(s^2 - 4)^2} \right]$$

$$= \frac{s^2 + 4}{(s^2 - 4)^2} \quad \underline{\underline{\text{Ans:-}}}$$

⑩ $f(t) = t \cdot \cos bt$

We know, —

$$\mathcal{L}\{t^n \cos bt\} = (-1)^n \cdot \frac{d^n}{ds^n} \mathcal{L}\{\cos bt\}$$

$$\therefore \mathcal{L}\{t \cdot \cos bt\} = (-1)^1 \cdot \frac{d}{ds} \left(\frac{s}{s^2 + b^2} \right)$$

$$= - \frac{d}{ds} \times \frac{s}{s^2 + b^2}$$

$$= - \left[\frac{(s^2 + b^2) \cdot \frac{d}{ds} s - s \cdot \frac{d}{ds} (s^2 + b^2)}{(s^2 + b^2)^2} \right]$$

$$= - \left[\frac{s^2 + b^2 - s \cdot 2s}{(s^2 + b^2)^2} \right]$$

$$= \frac{2s^2 - (s^2 + b^2)}{(s^2 + b^2)^2}$$

$$= \frac{2s^2}{(s^2 + b^2)^2} - \frac{(s^2 + b^2)}{(s^2 + b^2)^2}$$

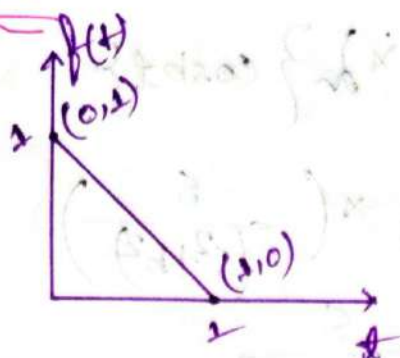
$$= \frac{2s^2}{(s^2 + b^2)^2} - \frac{1}{(s^2 + b^2)}$$

Ans: —

piece-wise function:-

Page No. 2101-

①



⇒ We know:-

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

$$\Rightarrow \frac{x-0}{0-1} = \frac{y-1}{1-0}$$

$$\Rightarrow -y+1 = x$$

$$\Rightarrow y = 1-x$$

Now:-

$$f(t) = \begin{cases} 1-t & ; 0 \leq t < 1 \\ 0 & ; t \geq 1 \end{cases}$$

We know:-

$$\therefore \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$$

$$= \int_0^1 e^{-st} \cdot (1-t) \cdot dt + \int_1^{\infty} e^{-st} \cdot 0 \cdot dt$$

$$= (1-t) \int_0^1 e^{-st} \cdot dt - \int_0^1 \left\{ \frac{d}{dt} (1-t) \cdot \int e^{-st} \cdot dt \right\} dt + 0$$

$$= \left[(1-t)x - \frac{1}{s} e^{-st} \right]_0^1 - \left[\int_0^1 1x - \frac{1}{s} e^{-st} \cdot dt \right]$$

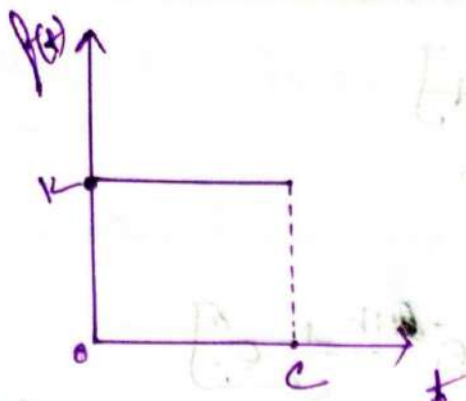
$$= \frac{1}{s} - \left[-\frac{1}{s^2} e^{-st} \right]_0^1$$

$$= \frac{1}{s} + \frac{1}{s^2} [e^{-st}]_0^1$$

$$= \frac{1}{s} + \frac{1}{s^2} e^{-s} - \frac{1}{s^2}$$

$$= \frac{1}{s} + \frac{e^{-s}}{s^2} - \frac{1}{s^2}$$

10



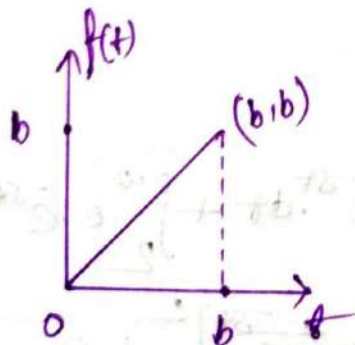
$$\therefore f(t) = \begin{cases} k & ; 0 < t < c \\ 0 & ; t > c \end{cases}$$

we know—

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt \\ &= \int_0^c e^{-st} \cdot k \cdot dt + \int_c^{\infty} e^{-st} \cdot 0 \cdot dt \\ &= k \cdot \left[-\frac{1}{s} e^{-st} \right]_0^c + 0 \\ &= -\frac{k}{s} [e^{-cs} - e^0] \\ &= \frac{k}{s} - \frac{ke^{-cs}}{s} \end{aligned}$$

Ans:-

11



$$\therefore f(t) = \begin{cases} t & ; 0 < t < b \\ 0 & ; t > b \end{cases}$$

Now:-

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^b e^{-st} \cdot t \cdot dt + \int_b^{\infty} e^{-st} \cdot 0 \cdot dt \\ &= t \cdot \int_0^b e^{-st} \cdot dt - \int_0^b \left(\frac{d}{dt} t \right) \cdot t \cdot \int e^{-st} \cdot dt \end{aligned}$$

$$\Rightarrow \left[-\frac{t}{s} e^{-st} \right]_0^b - \left[\int_0^{b\infty} -\frac{1}{s} e^{-st} \cdot dt \right]$$

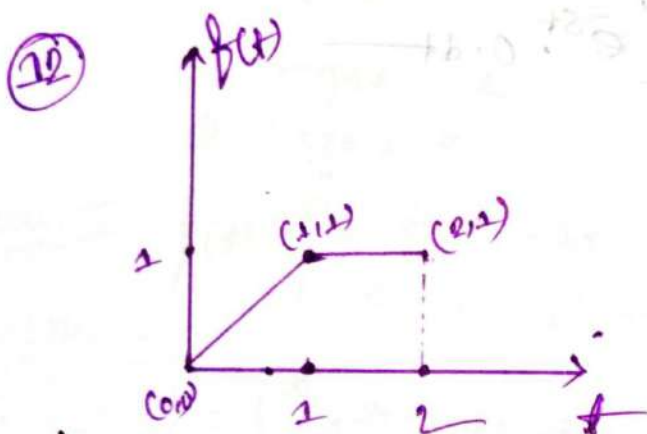
$$= \left[-\frac{t}{s} e^{-st} \right]_0^b - \left[\frac{1}{s^2} e^{-st} \right]_0^b$$

$$= -\left[\frac{b}{s} e^{-bs} - \frac{0}{s} e^{-0} \right] - \left[\frac{1}{s^2} e^{-bs} - \frac{1}{s^2} e^{-0} \right]$$

$$= -\frac{b}{s} e^{-bs} + \frac{1}{s^2} - \frac{1}{s^2}$$

$$= \frac{1}{s^2} - \frac{b}{s} e^{-bs} - \frac{1}{s^2} e^{-bs}$$

Ans:-



$$\therefore f(t) = \begin{cases} 0 & ; 0 < t < 1 \\ 1 & ; 1 <= t <= 2 \\ 0 & ; t > 2 \end{cases}$$

Now, $\mathcal{L}\{f(t)\} = \int_0^1 t \cdot e^{-st} \cdot dt + \int_1^2 1 \cdot e^{-st} \cdot dt + \int_2^\infty 0 \cdot e^{-st} \cdot dt$

$$= \left[t \cdot -\frac{1}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^1 + \left[-\frac{1}{s} e^{-st} \right]_1^2 + 0$$

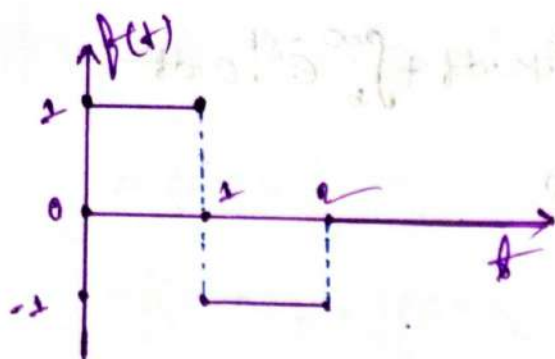
$$= -\left[\frac{1}{s} e^{-s} - \frac{0}{s} e^{-0} \right] + \left[-\frac{1}{s} e^{-2s} + \frac{1}{s} e^{-s} \right] - \left[\frac{1}{s^2} e^{-s} - \frac{1}{s^2} e^{-0} \right]$$

$$= -\frac{e^{-s}}{s} - \frac{1}{s} e^{-2s} + \frac{1}{s} e^{-s} - \frac{1}{s^2} e^{-s} + \frac{1}{s^2}$$

$$= \frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-2s}}{s^2}$$

Ans:-

13



$$\therefore f(t) = \begin{cases} 2 & ; 0 < t < 1 \\ -1 & ; 1 < t < 2 \\ 0 & ; t > 2 \end{cases}$$

$$\therefore \mathcal{L}\{f(t)\} = \int_0^1 2 \cdot e^{-st} \cdot dt + \int_1^2 -1 \cdot e^{-st} \cdot dt + \int_2^{\infty} 0 \cdot e^{-st} \cdot dt$$

$$= +\frac{2}{s} [e^{-st}]_0^1 + \frac{1}{s} [e^{-st}]_1^2 + 0$$

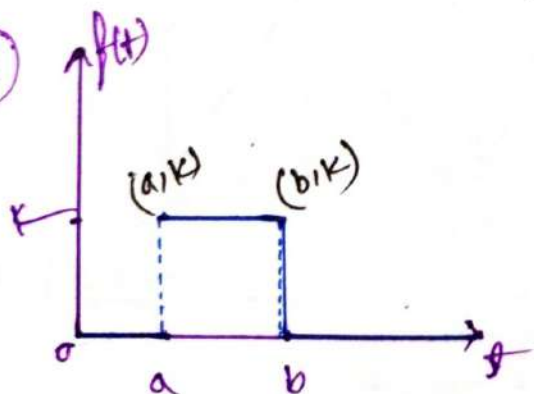
$$= -\frac{2}{s} [e^{-s} - e^0] + \frac{1}{s} [e^{-2s} - e^{-s}]$$

$$= -\frac{2e^{-s}}{s} + \frac{2}{s} + \frac{1}{s} e^{-2s} - \frac{e^{-s}}{s}$$

$$= \frac{2}{s} + \frac{e^{-2s}}{s} - \frac{3e^{-s}}{s}$$

Ans:-

14

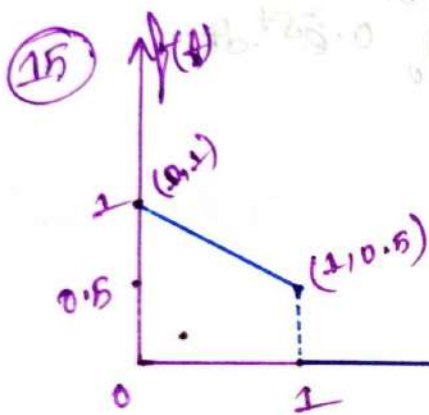


We know, -

$$f(t) = \begin{cases} 0 & ; 0 < t < a \\ k & ; a < t < b \\ 0 & ; t > b \end{cases}$$

$$\begin{aligned}
 \therefore \mathcal{L}\{f(t)\} &= \int_0^a e^{-st} \cdot 0 \cdot dt + \int_a^b e^{-st} \cdot k \cdot dt + \int_b^\infty e^{-st} \cdot 0 \cdot dt \\
 &= 0 + kx \cdot \frac{1}{s} e^{-st} \Big|_a^b + 0 \\
 &= -\frac{k}{s} [e^{-bs} - e^{-as}] \\
 &= \frac{ke^{-as}}{s} - \frac{ke^{-bs}}{s}
 \end{aligned}$$

Ans:—



$\Rightarrow$ we know—
norm

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

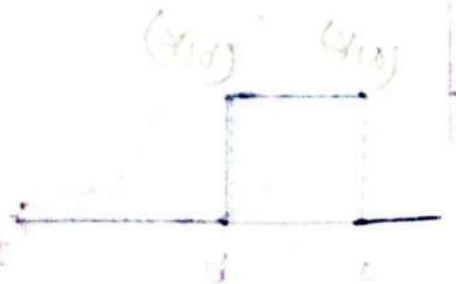
$$\Rightarrow \frac{x-0}{0-1} = \frac{y-1}{1-0.5}$$

$$\Rightarrow \frac{x}{-1} = \frac{y-1}{0.5}$$

$$\Rightarrow 0.5x = -y + 1$$

$$\Rightarrow y = 1 - 0.5x$$

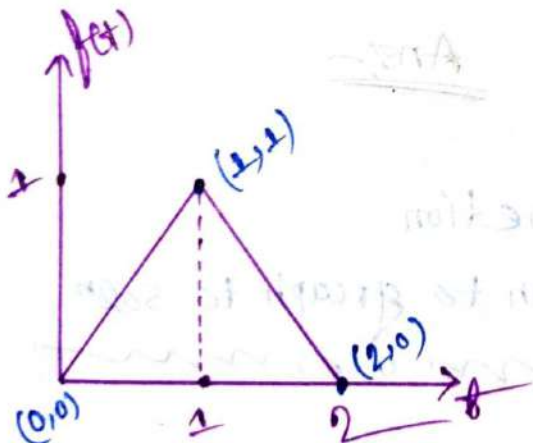
$$\therefore f(t) = \begin{cases} 1 - 0.5t & ; 0 \leq t \leq 1 \\ 0 & ; t > 1 \end{cases}$$



$$\begin{aligned}
 \therefore \mathcal{L}\{f(t)\} &= \int_0^1 (1-0.5t) \cdot e^{-st} \cdot dt + \int_1^\infty e^{-st} \cdot 0 \cdot dt \\
 &= (1-0.5t) \int_0^1 e^{-st} \cdot dt - \int_0^1 \frac{d}{dt} (1-0.5t) \cdot e^{-st} \cdot dt \\
 &= \left[(1-0.5t) \cdot \frac{1}{s} e^{-st} \right]_0^1 - \int_0^1 -0.5 \cdot \frac{1}{s} e^{-st} \cdot dt \\
 &= \left[(1-0.5) \cdot \frac{1}{s} e^{-s} - 1 \cdot \frac{1}{s} e^0 \right] + \frac{1}{2s^2} [e^{-st}]_0^1 \\
 &= \frac{1}{2s} e^{-s} + \frac{1}{s} + \frac{1}{2s^2} e^{-s} - \frac{1}{2s^2} \\
 &= \frac{1}{s} - \frac{1}{2s} e^{-s} + \frac{1}{2s^2} e^{-s} - \frac{1}{2s^2} \\
 &= \frac{1}{s} - \frac{e^{-s}}{2s} + \frac{e^{-s}}{2s^2} - \frac{1}{2s^2}
 \end{aligned}$$

Ans:-

(16)



We know:-

$$\frac{x-y_1}{x_1-y_1} = \frac{y-y_1}{y_1-y_2}$$

$$\Rightarrow \frac{x-1}{1-2} = \frac{y-1}{1-0}$$

$$\Rightarrow x-1 = -y+1$$

$$\Rightarrow y = 2-x$$

$$\therefore f(t) = \begin{cases} t & ; 0 < t < 1 \\ 2-t & ; 1 < t < 2 \\ 0 & ; t > 2 \end{cases}$$

$$\begin{aligned} \therefore \mathcal{L}\{f(t)\} &= \int_0^1 t \cdot e^{-st} dt + \int_1^2 (2-t) \cdot e^{-st} dt + \int_2^\infty 0 \cdot e^{-st} dt \\ &= \left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^1 + \left[(2-t) \cdot -\frac{1}{s} e^{-st} \right]_1^2 - \int_1^2 -1 \cdot -\frac{1}{s} e^{-st} dt \\ &= -\frac{1}{s} e^{-s} - \frac{1}{s^2} [e^{-s} - 1] + \frac{1}{s} e^{-s} - \left[-\frac{1}{s^2} e^{-st} \right]_1^2 \\ &= -\frac{1}{s} e^{-s} - \frac{1}{s^2} e^{-s} + \frac{1}{s^2} + \frac{1}{s} e^{-s} + \frac{1}{s^2} e^{-2s} - \frac{1}{s^2} e^{-s} \\ &= \frac{e^{-2s}}{s^2} - \frac{2e^{-s}}{s^2} + \frac{1}{s^2} \\ &= \frac{e^{-2s} - 2e^{-s} + 1}{s^2} \end{aligned}$$

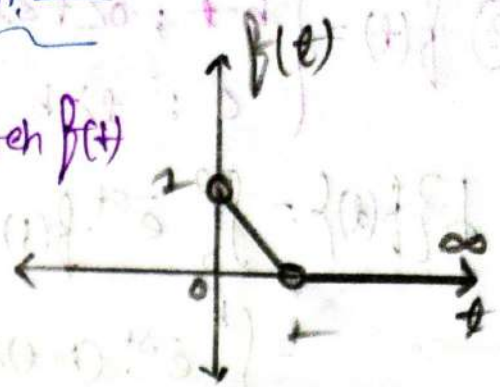
Ans:-

Piece-wise function
from function to graph to solve

For more practice!

Piece-wise function.

① $f(t) = \begin{cases} 1-t; & 0 < t < 1 \\ 0; & t > 1 \end{cases}$ Also sketch $f(t)$



$$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) dt$$

$$= \int_0^1 e^{-st} \cdot (1-t) dt + \int_1^{\infty} e^{-st} \cdot 0 \cdot dt$$

$$= \left[(1-t) \times -\frac{1}{s} e^{-st} + \frac{1}{s^2} \times e^{-st} \right]_0^1 + 0$$

$$= \left[(1-1) \times -\frac{1}{s} e^{-s} - (1-0) \times -\frac{1}{s} e^0 \right] + \left[\frac{1}{s^2} \times e^{-s} - \frac{1}{s^2} \times e^0 \right]$$

$$= 0 + \frac{1}{s} \times 1 + \frac{1}{s^2} \times e^{-s} - \frac{1}{s^2}$$

$$= \frac{1}{s} + \frac{e^{-s}}{s^2} - \frac{1}{s^2}$$

Ans:—

s	D	I
+	1-t	e^{-st}
-	-1	$-\frac{1}{s} e^{-st}$
+	0	$\frac{1}{s^2} e^{-st}$

② $f(t) = \begin{cases} k; & 0 < t < c \\ 0; & t > c \end{cases}$ Also sketch $f(t)$

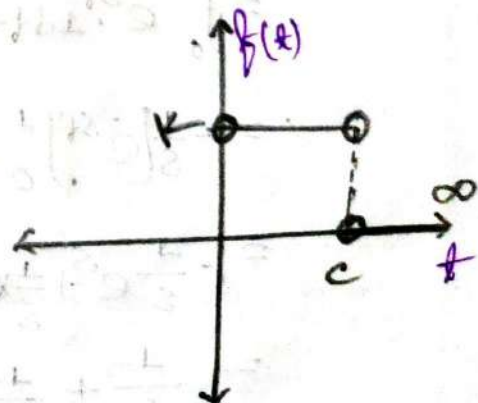
$$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$$

$$= \int_0^c e^{-st} \cdot k \cdot dt + \int_c^{\infty} e^{-st} \cdot 0 \cdot dt$$

$$= k \times -\frac{1}{s} [e^{-st}]_0^c + 0$$

$$= -\frac{k}{s} [e^{-sc} - e^0]$$

$$= \frac{k}{s} (1 - e^{-sc})$$



③ $f(t) = \begin{cases} b-t; & 0 \leq t \leq b \\ 0; & t > b \end{cases}$ Also sketch $f(t)$

$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$

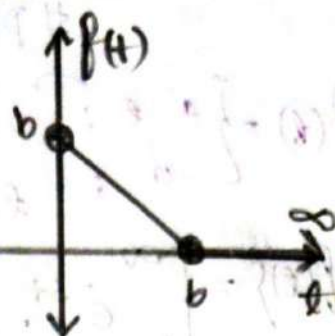
$= \int_0^b e^{-st} (b-t) dt + \int_b^{\infty} e^{-st} \cdot 0 \cdot dt$

$= \left[(b-t) \cdot \frac{-1}{s} e^{-st} + \frac{1}{s^2} e^{-st} \right]_0^b + 0$

$= \left[+\frac{b}{s} + \frac{1}{s^2} e^{-bs} - \frac{1}{s^2} \right]$

$= \frac{e^{-bs}}{s^2} + \frac{b}{s} - \frac{1}{s^2}$

Ans:-



S	D	I
+	$b-t$	e^{-st}
-	-1	$-\frac{1}{s} e^{-st}$
+	0	$\frac{1}{s^2} e^{-st}$

④ $f(t) = \begin{cases} 1; & 0 \leq t < 1 \\ -1; & 1 \leq t < 2 \\ 0; & t \geq 2 \end{cases}$ Also sketch $f(t)$

$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$

$= \int_0^1 e^{-st} \cdot 1 dt + \int_1^2 e^{-st} \cdot (-1) dt + \int_2^{\infty} e^{-st} \cdot 0 \cdot dt$

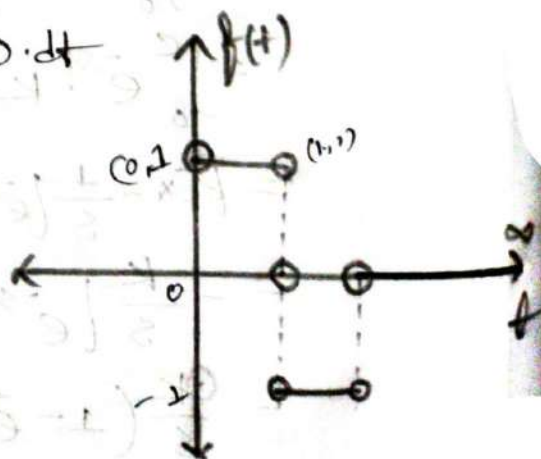
$= -\frac{1}{s} [e^{-st}]_0^1 + \frac{1}{s} [e^{-st}]_1^2 + 0$

$= -\frac{1}{s} e^{-s} + \frac{1}{s} \cdot 1 + \frac{1}{s} e^{-2s} - \frac{1}{s} e^{-s}$

$= \frac{1}{s} + \frac{1}{s} e^{-2s} - 2 \times \frac{1}{s} e^{-s}$

$= \frac{1}{s} + \frac{1}{s} e^{-2s} - \frac{2}{s} e^{-s}$

Ans:-



⑤ $f(t) = \begin{cases} t; & 0 \leq t \leq 4 \\ 1; & t > 4 \end{cases}$ Also sketch $f(t)$

$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$

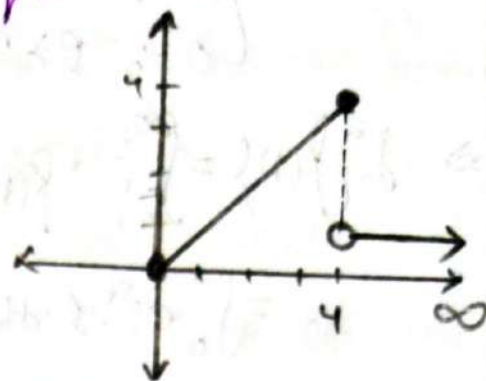
$= \int_0^4 e^{-st} \cdot t \cdot dt + \int_4^{\infty} e^{-st} \cdot 1 \cdot dt$

$= \left[t \cdot \left(-\frac{1}{s}\right) e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^4 + \left[-\frac{1}{s} e^{-st} \right]_4^{\infty}$

$= -\frac{1}{s} [4e^{-4s} - 0 \times e^0] - \frac{1}{s^2} [e^{-4s} - e^0] - \frac{1}{s} [e^{-\infty s} - e^{-4s}]$
 $= -\frac{4e^{-4s}}{s} - \frac{e^{-4s}}{s^2} + \frac{1}{s^2} + \frac{1}{s} e^{-4s}$

$= \frac{1}{s^2} - \frac{3e^{-4s}}{s} - \frac{e^{-4s}}{s^2}$

Ans:-



⑥ $f(t) = \begin{cases} t; & 0 \leq t < 1 \\ 0; & t > 1 \end{cases}$ Also sketch $f(t)$

$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$

$= \int_0^1 e^{-st} \cdot t \cdot dt + \int_1^{\infty} e^{-st} \cdot 0 \cdot dt$

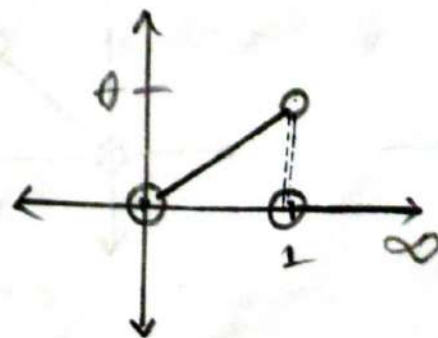
$= \left[t \cdot \left(-\frac{1}{s}\right) e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^1 + 0$

$= -\frac{1}{s} [te^{-st}]_0^1 - \frac{1}{s^2} [e^{-st}]_0^1$

$= -\frac{e^{-s}}{s} + \frac{1}{s} \times 0 - \frac{1}{s^2} e^{-s} + \frac{1}{s^2}$

$= \frac{1}{s^2} - \frac{e^{-s}}{s} - \frac{e^{-s}}{s^2}$

Ans:-



$$\textcircled{7} f(t) = \begin{cases} t & ; 0 < t < 1 \\ 1 & ; 1 < t < 2 \\ 0 & ; t > 2 \end{cases}$$

$$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) \cdot dt$$

$$= \int_0^1 e^{-st} \cdot t \cdot dt + \int_1^2 e^{-st} \cdot 1 \cdot dt + \int_2^{\infty} e^{-st} \cdot 0 \cdot dt$$

$$= \left[t \cdot \frac{1}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^1 + \left[-\frac{1}{s} e^{-st} \right]_1^2 + 0$$

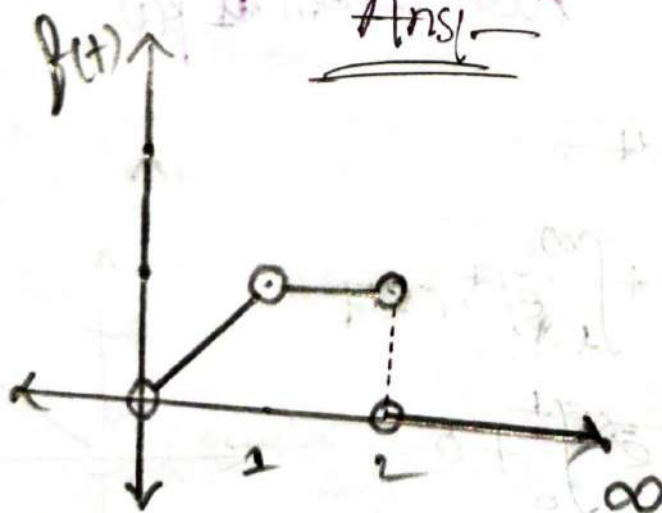
$$= -\frac{1}{s} \left[t e^{-st} \right]_0^1 - \frac{1}{s^2} \left[e^{-st} \right]_0^1 - \frac{1}{s} \left[e^{-st} \right]_1^2$$

$$= -\frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} + \frac{1}{s^2} - \frac{e^{-2s}}{s} + \frac{e^{-s}}{s}$$

$$= \frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-2s}}{s}$$

$$= \frac{1 - e^{-s} - s e^{-2s}}{s^2}$$

Ans.



Inverse Laplace
Transformation

Choo
Choo

#Formula of Inverse Laplace Transformation:

$$1. \mathcal{L}^{-1} \left\{ \frac{c}{s} \right\} = c.$$

$$2. \mathcal{L}^{-1} \left\{ \frac{1}{s^{n+1}} \right\} = \frac{t^n}{n!}$$

$$3. \mathcal{L}^{-1} \left\{ \frac{1}{s-a} \right\} = e^{at}$$

$$4. \mathcal{L}^{-1} \left\{ \frac{s}{s^2+a^2} \right\} = \cos at$$

$$5. \mathcal{L}^{-1} \left\{ \frac{a}{s^2+a^2} \right\} = \sin at$$

$$6. \mathcal{L}^{-1} \left\{ \frac{s}{s^2-a^2} \right\} = \cosh at$$

$$7. \mathcal{L}^{-1} \left\{ \frac{a}{s^2-a^2} \right\} = \sinh at$$

$$8. \mathcal{L}^{-1} \left\{ \frac{1}{s^2+a^2} \right\} = \frac{1}{a} \sin at$$

$$9. \mathcal{L}^{-1} \left\{ \frac{1}{s^2-a^2} \right\} = \frac{1}{a} \sinh at$$

Formula of Partial fraction:-

$$(1) \frac{s+k}{(s+a)(s+b)} = \frac{A}{(s+a)} + \frac{B}{(s+b)}$$

$$(2) \frac{s+k}{(s+a)(s^2+b)} = \frac{A}{(s+a)} + \frac{Bs+c}{(s^2+b)}$$

$$(3) \frac{s+k}{(s+a)^2(s+b)} = \frac{A}{(s+a)} + \frac{B}{(s+a)^2} + \frac{C}{(s+b)}$$

Page 211 # Find $f(t)$ using Direct formula:-

(25) $\frac{0.2s + 18}{s^2 + 3.24}$

$$\Rightarrow \alpha^{-1} \left\{ \frac{0.2s}{s^2 + 3.24} \right\} + \alpha^{-1} \left\{ \frac{18}{s^2 + 3.24} \right\}$$

$$\Rightarrow \alpha^{-1} \left\{ \frac{0.2s}{s^2 + (1.8)^2} \right\} + \alpha^{-1} \left\{ \frac{18}{s^2 + (1.8)^2} \right\}$$

$$\Rightarrow 0.2 \alpha^{-1} \left\{ \frac{s}{s^2 + (1.8)^2} \right\} + 18 \alpha^{-1} \left\{ \frac{1}{s^2 + (1.8)^2} \right\}$$

$$\Rightarrow 0.2 \cos 1.8t + 18 \times \frac{1}{1.8} \times \sin 1.8t$$

$$\Rightarrow 0.2 \cos 1.8t + 10 \sin 1.8t$$

Ans:-

(26) $\frac{5s + 1}{s^2 - 5^2}$

$$\Rightarrow \alpha^{-1} \left\{ \frac{5s}{s^2 - 5^2} \right\} + \alpha^{-1} \left\{ \frac{1}{s^2 - 5^2} \right\}$$

$$\Rightarrow 5 \alpha^{-1} \left\{ \frac{s}{s^2 - 5^2} \right\} + \alpha^{-1} \left\{ \frac{1}{s^2 - 5^2} \right\}$$

$$\Rightarrow 5 \cosh 5t + \frac{1}{5} \sinh 5t$$

Ans:-

(27)

$$\frac{s}{L^2 s^2 + n^2 \pi^2}$$

$$\Rightarrow \alpha^{-1} \left\{ \frac{s}{L^2 (s^2 + \frac{n^2 \pi^2}{L^2})} \right\}$$

$$\Rightarrow \alpha^{-1} \left\{ \frac{s}{L^2 (s^2 + (\frac{n\pi}{L})^2)} \right\}$$

$$\Rightarrow \frac{1}{L^2} \alpha^{-1} \left\{ \frac{s}{s^2 + (\frac{n\pi}{L})^2} \right\}$$

$$= \frac{1}{L^2} \cos \frac{n\pi}{L} \theta$$

Ans:-

(28)

$$\frac{12}{s^4} - \frac{228}{s^6}$$

$$\Rightarrow \alpha^{-1} \left\{ \frac{12}{s^{4+1}} \right\} - \alpha^{-1} \left\{ \frac{228}{s^{6+1}} \right\}$$

$$\Rightarrow 12 \alpha^{-1} \left\{ \frac{1}{s^{4+1}} \right\} - 228 \alpha^{-1} \left\{ \frac{1}{s^{6+1}} \right\}$$

$$\Rightarrow 12 \times \frac{\theta^3}{3!} - 228 \times \frac{1}{5!} \times \theta^5$$

$$\Rightarrow 2\theta^3 - 1.9\theta^5$$

Ans:-

(29)

$$\frac{4s+32}{s^2-16}$$

$$\Rightarrow \alpha^{-1} \left\{ \frac{4s}{s^2-4^2} \right\} + \alpha^{-1} \left\{ \frac{32}{s^2-4^2} \right\}$$

$$\Rightarrow 4 \cosh 4\theta + 8 \sinh 4\theta$$

Ans:-

Inverse Laplace transformation
by using partial fraction.

(28)

$$\frac{1}{(s+\sqrt{2})(s-\sqrt{3})}$$

$$\Rightarrow \frac{1}{(s+\sqrt{2})(s-\sqrt{3})} = \frac{A}{s+\sqrt{2}} + \frac{B}{s-\sqrt{3}} \quad (1)$$

using cover up rule:-

$$s = -\sqrt{2}; A = \frac{1}{s-\sqrt{3}} = \sqrt{2}-\sqrt{3}$$

$$s = \sqrt{3}; B = \frac{1}{s+\sqrt{2}} = \frac{1}{\sqrt{3}+\sqrt{2}}$$

Substitute the value of A and B in eq (1) $\Rightarrow$

$$\Rightarrow \frac{\sqrt{2}-\sqrt{3}}{s+\sqrt{2}} + \frac{1}{s-\sqrt{3}}$$

$$\Rightarrow \alpha^{-1} \left\{ \frac{\sqrt{2}-\sqrt{3}}{s+\sqrt{2}} \right\} + \alpha^{-1} \left\{ \frac{1}{s-\sqrt{3}} \right\}$$

$$\Rightarrow (\sqrt{2}-\sqrt{3}) \alpha^{-1} \left\{ \frac{1}{s+\sqrt{2}} \right\} + \alpha^{-1} \left\{ \frac{1}{s-\sqrt{3}} \right\}$$

$$\Rightarrow (\sqrt{2}-\sqrt{3}) \times e^{-\sqrt{2}t} + \frac{1}{\sqrt{3}+\sqrt{2}} \times e^{+\sqrt{3}t}$$

Ans:-

Q1 $\frac{s+10}{s^2-s-2}$

$$\Rightarrow \frac{s+10}{s^2-2s+1-2}$$

$$\Rightarrow \frac{s+10}{s(s-2)+1(s-2)}$$

$$\Rightarrow \frac{s+10}{(s-2)(s+1)} = \frac{A}{s-2} + \frac{B}{s+1} \quad \text{--- (1)}$$

Using Cover up rule:

$$s=2; A = \frac{s+10}{s+1} = 4$$

$$s=-1; B = \frac{s+10}{s-2} = -3$$

Put the value in eq (1):

$$= \frac{4}{s-2} + \frac{-3}{s+1}$$

$$= 2^{-1} \left\{ \frac{4}{s-2} \right\} - 2^{-1} \left\{ \frac{3}{s+1} \right\}$$

$$= 4e^{2t} - 3e^{-t}$$

Ans.

Q2 $\frac{1}{(s+a)(s+b)}$

$$\Rightarrow \frac{1}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b} \quad \text{--- (1)}$$

Using Cover up rule:

$$s=-a; A = \frac{1}{s+b} = \frac{1}{b-a}$$

$$s=-b; B = \frac{1}{s+a} = \frac{1}{a-b}$$

put the value of eq (1) ⇒

$$= \frac{1/b-a}{s+a} + \frac{1/a-b}{s+b}$$

$$= \alpha^{-1} \left\{ \frac{1}{b-a} \times \frac{1}{s+a} \right\} + \alpha^{-1} \left\{ \frac{1}{a-b} \times \frac{1}{s+b} \right\}$$

$$= \frac{1}{b-a} \alpha^{-1} \left\{ \frac{1}{s+a} \right\} + \frac{1}{a-b} \alpha^{-1} \left\{ \frac{1}{s+b} \right\}$$

$$= \frac{1}{b-a} e^{-at} + \frac{1}{a-b} e^{-bt}$$

Ans:-

Page 251:-

Q20 $\frac{7.5}{s^2-2s-8}$

$$= \frac{7.5}{s^2-4s+2s-8}$$

$$= \frac{7.5}{s(s-4)+2(s-4)}$$

$$\Rightarrow \frac{7.5}{(s+2)(s-4)} = \frac{A}{s+2} + \frac{B}{s-4} \quad \text{--- (1)}$$

Using Cover Up rule:-

$$s = -2 ; A = \frac{7.5}{s-4} = -\frac{5}{4}$$

$$\textcircled{1} s = 4 ; B = \frac{7.5}{s+2} = \frac{5}{4}$$

Put the value in eq (1) =

$$2 = \frac{5}{4} \times \frac{1}{s+2} + \frac{5}{4} \times \frac{1}{s-4}$$

$$2 \alpha^{-1} \left\{ \frac{5}{4} \times \frac{1}{s-4} \right\} - \alpha^{-1} \left\{ \frac{5}{4} \times \frac{1}{s+2} \right\}$$

$$= \frac{5}{4} \alpha^{-1} \left\{ \frac{1}{s-4} \right\} - \frac{5}{4} \alpha^{-1} \left\{ \frac{1}{s+2} \right\}$$

$$= \frac{5}{4} e^{4t} - \frac{5}{4} e^{-2t}$$

Ans:-

For more practice:-

$$\frac{3s+1}{(s+1)(s^2+1)}$$

$$\Rightarrow \frac{3s+1}{(s+1)(s^2+1)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+1} \quad \text{--- (1)}$$

Multiply (s+1)(s^2+1) from both side:-

$$3s+1 = A(s^2+1) + (Bs+C)(s+1) \quad \text{--- (1)}$$

if s+1=0

$$\therefore s = -1$$

put in eq (1) =

$$-3+1 = A(1+1) + 0$$

$$\therefore A = -1$$

Now:-

$$3s+1 = As^2 + A + Bs^2 + Bs + Cs + C$$

equating the co-efficients of s^2, s^1, s^0 :-

$$A + C = 1$$

$$Bs + Cs = 3s$$

$$s(B+C) = 3s$$

$$\left. \begin{aligned} B &= -C+3 \\ &= -(-1)+3 \\ &= 4 \end{aligned} \right\} C=1$$

put value of A, B, c in eq. (1) $\Rightarrow$

$$\Rightarrow \frac{-1}{s+1} + \frac{s+2}{s^2+1}$$

$$\Rightarrow \alpha^{-1} \left\{ -\frac{1}{s+1} \right\} + \alpha^{-1} \left\{ -\frac{s}{s^2+1} \right\} + \alpha^{-1} \left\{ \frac{2}{s^2+1} \right\}$$

$$\Rightarrow -e^{-t} + \cos t + 2\sin t$$

$$\Rightarrow \cos t + 2\sin t - e^{-t}$$

Ans:—

* $\frac{s}{(s+1)^2}$

$$\Rightarrow \frac{s}{(s+1)^2} = \frac{A}{s+1} + \frac{B}{(s+1)^2} \quad \text{--- (1)}$$

multiply $(s+1)^2$ from both sides

$$s = A(s+1) + B \quad \text{--- (ii)}$$

$$s = As + A + B \quad \text{--- (iii)}$$

if $s+1=0$

$$\therefore s = -1$$

put in eq. (ii) $\Rightarrow$

$$-1 = B$$

$$\therefore B = -1$$

equating like terms

$$As = s$$

$$\therefore A = 1$$

put the value of eq. (1) $\Rightarrow$

$$\Rightarrow \frac{A}{s+1} + \frac{B}{(s+1)^2} = \frac{1}{s+1} + \frac{-1}{(s+1)^2}$$

$$= d^{-1} \left\{ \frac{1}{s+1} \right\} - d^{-1} \left\{ \frac{1}{(s+1)^2} \right\}$$

$$= e^{-t} - e^{-t} d^{-1} \left\{ \frac{1}{s^2} \right\}$$

$$= e^{-t} - e^{-t} d^{-1} \left\{ \frac{1}{s+1} \right\}$$

$$= e^{-t} - e^{-t}$$

Ans:-

⊗

$$\frac{4s+5}{(s+1)^2(s+2)}$$

$$\Rightarrow \frac{4s+5}{(s+1)^2(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2} + \frac{C}{(s+2)} \quad \text{--- (i)}$$

multiply $(s+1)^2(s+2)$ from both sides

$$4s+5 = A(s+1)(s+2) + B(s+2) + C(s+1)^2 \quad \text{--- (ii)}$$

Let $s+1=0$

$$\therefore s = -1$$

Put the value of s in eq (ii)

$$-4+5 = B(-1+2)$$

$$\therefore B = 1$$

if $s+2=0$

$$\therefore s = -2$$

Put the value of s in eq (ii)

$$-8+5 = C(-2+1)^2$$

$$C = -3$$

Now put $s=1, B=1, C=-3$ in eq (ii)

$$0 = A(1+1)(1+2) + (1+2) \times 1 + (-3) \times (1+1)^2$$

$$0 = 6A + 3 - 12$$

$$\Rightarrow 6A = 0 + 12 - 3$$

$$\Rightarrow 6A = 9$$

$$\therefore A = \frac{9}{6}$$

$$= 3$$

Put the value of A, B, C in eq (1) =

$$= \frac{3}{(s+1)} + \frac{1}{(s+1)^2} + \frac{-3}{(s+2)}$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{3}{(s+1)} + \frac{1}{(s+1)^2} - \frac{3}{(s+2)} \right\}$$

$$\Rightarrow 3\mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2} \right\} - 3\mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\}$$

$$\Rightarrow 3e^{-t} + e^{-t} \cdot t - 3e^{-2t}$$

Ans: —

First-shifting or Translation property! —

Formula: —

1: if $\alpha^{-1}\{F(s)\} = f(t)$ then,

$$\alpha^{-1}\{F(s-a)\} = e^{at} \alpha^{-1}\{F(s)\}$$

For Example: —

$$\text{if } F(s) = \frac{1}{s}$$

$$\text{then, } F(s-5) = \frac{1}{s-5}$$

$$\therefore \alpha^{-1}\{F(s-5)\} = e^{5t} \alpha^{-1}\{F(s)\}$$

$$= e^{5t} \alpha^{-1}\left\{\frac{1}{s}\right\}$$

$$= e^{5t} \times 1$$

$$= e^{5t}$$

Ans: —

Page: 211: —

$$\textcircled{37} \quad \frac{\pi}{(s+\pi)^2}$$

$$= \alpha^{-1}\left\{\frac{\pi}{(s+\pi)^2}\right\}$$

$$= \pi e^{-\pi t} \alpha^{-1}\left\{\frac{1}{s^2}\right\}$$

$$= \pi e^{-\pi t} \cdot t$$

Ans: —

$$\textcircled{38} \quad \frac{6}{(s+1)^3}$$

$$= \alpha^{-1}\left\{\frac{6}{(s+1)^3}\right\}$$

$$= 6 \alpha^{-1}\left\{\frac{1}{(s+1)^3}\right\}$$

$$= 6 \times e^{-t} \alpha^{-1}\left\{\frac{1}{s^3}\right\}$$

$$= 6 e^{-t} \alpha^{-1}\left\{\frac{1}{s^2+1}\right\}$$

$$= 6 e^{-t} \cdot \frac{t^2}{2}$$

Ans: —

$$(39) \quad \frac{21}{(s+\sqrt{2})^4}$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{21}{(s+\sqrt{2})^4} \right\}$$

$$\Rightarrow 21 e^{-\sqrt{2}t} \mathcal{L}^{-1} \left\{ \frac{1}{s^3+1} \right\}$$

$$\Rightarrow 21 e^{-\sqrt{2}t} \times \frac{t^3}{3!}$$

$$\Rightarrow \frac{7}{2} e^{-\sqrt{2}t} \cdot t^3$$

Ans. -

$$(40) \quad \frac{4}{s^2-2s-3}$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{4}{s^2-2s-3} \right\}$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{2 \cdot 2 \cdot 1}{(s-1)^2-4} \right\}$$

$$\Rightarrow e^t \mathcal{L}^{-1} \left\{ \frac{2 \times 1 \times 2}{s^2-2^2} \right\}$$

$$\Rightarrow$$

$$\Rightarrow 2e^t \sinh 2t$$

$$\Rightarrow 2e^t \times \left(\frac{e^{2t} - e^{-2t}}{2} \right)$$

$$\Rightarrow \frac{2e^t}{2} (e^{2t} - e^{-2t})$$

$$\Rightarrow e^{3t} - e^{-t}$$

Ans. -

We can also solve this 40 no. math using partial fraction:-

$$(40) \quad \frac{4}{s^2-2s-3}$$

$$= \frac{4}{s^2-3s+s-3}$$

$$= \frac{4}{s(s-3)+1(s-3)}$$

$$\Rightarrow \frac{4}{(s-3)(s+1)} = \frac{A}{(s-3)} + \frac{B}{(s+1)} \quad \text{--- (1)}$$

Using Cover Up rule:-

$$s=3; A = \frac{4}{s+1} = 1$$

$$s=-1; B = \frac{4}{s-3} = -1$$

Put it in eq (1) =

$$= \frac{1}{s-3} - \frac{1}{s+1}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s-3} \right\} - \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\}$$

$$= e^{3t} - e^{-t}$$

Ans. -

41

$$\frac{\pi}{s^2 + 10\pi s + 25\pi^2}$$

$$= \frac{\pi}{s^2 + 2 \times 5\pi + (5\pi)^2 - \pi^2}$$

$$= \frac{\pi}{(s + 5\pi)^2 - \pi^2}$$

$$\Rightarrow \alpha^{-1} \left\{ \frac{\pi}{(s + 5\pi)^2 - \pi^2} \right\}$$

$$\Rightarrow e^{-5\pi t} \alpha^{-1} \left\{ \frac{\pi}{s^2 - \pi^2} \right\}$$

$$\Rightarrow e^{-5\pi t} \cdot \sinh \pi t$$

$$\Rightarrow e^{-5\pi t} \left(\frac{e^{\pi t} - e^{-\pi t}}{2} \right)$$

$$\Rightarrow \frac{1}{2} (e^{-4\pi t} - e^{-6\pi t})$$

Ans:-

42

$$\frac{2s-1}{s^2-6s+18}$$

$$\Rightarrow \frac{2s-1}{s^2-2 \cdot 3 \cdot 3 + 3^2 + 9}$$

$$\Rightarrow \frac{2s-1}{(s-3)^2 + 9}$$

$$\Rightarrow \frac{2(s-3)+5}{(s-3)^2 + 9}$$

Now,

$$\alpha^{-1} \left\{ \frac{2(s-3)+5}{(s-3)^2 + 9} \right\}$$

$$\Rightarrow 2\alpha^{-1} \left\{ \frac{s-3}{(s-3)^2 + 9} \right\} + \alpha^{-1} \left\{ \frac{5}{(s-3)^2 + 9} \right\}$$

$$\Rightarrow 2e^{3t} \alpha^{-1} \left\{ \frac{s}{s^2 + 3^2} \right\} + e^{3t} \alpha^{-1} \left\{ \frac{5}{s^2 + 3^2} \right\}$$

$$\Rightarrow 2e^{3t} \cos 3t + e^{3t} \times 5 \sin 3t / 3$$

$$\Rightarrow 2e^{3t} \cos 3t + \frac{5}{3} e^{3t} \sin 3t$$

Ans:-

Page: 254

11

(22)

$$\frac{\frac{1}{16}}{s^2 + s + \frac{1}{2}}$$

$$= \frac{\frac{1}{16}}{s^2 + 2s \cdot \frac{1}{2} + (\frac{1}{2})^2 + \frac{1}{2} - (\frac{1}{2})^2}$$

$$= \frac{4/16}{(s + \frac{1}{2})^2 + \frac{1}{4}}$$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{1/16}{(s + \frac{1}{2})^2 + \frac{1}{4}} \right\}$$

$$= \frac{1}{16} \times e^{-\frac{1}{2}t} \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + (\frac{1}{2})^2} \right\}$$

$$= \frac{1}{16} e^{-\frac{1}{2}t} \sin \frac{1}{2}t$$

$$= \frac{1}{8} e^{-\frac{1}{2}t} \sin \frac{1}{2}t$$

Ans:-

(25)

$$\frac{6(s+2)}{s^4}$$

$$= \mathcal{L}^{-1} \left\{ \frac{6s+6}{s^4} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{6s}{s^4} \right\} + \mathcal{L}^{-1} \left\{ \frac{6}{s^4} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{6}{s^3} \right\} + 6 \mathcal{L}^{-1} \left\{ \frac{1}{s^4} \right\}$$

$$= 6 \times \frac{t^2}{2!} + 6 \times \frac{t^3}{3!}$$

$$= 3t^2 + t^3$$

Ans:-

(27)

$$\frac{3s+4}{s^2+4s+5}$$

$$= \mathcal{L}^{-1} \left\{ \frac{3s+4}{s^2+2 \cdot s \cdot 2 + 2^2+1} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{3s+4}{(s+2)^2+1} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{3(s+2)+4-6}{(s+2)^2+1} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{3(s+2)-2}{(s+2)^2+1} \right\}$$

$$= 3e^{-2t} \mathcal{L}^{-1} \left\{ \frac{s}{s^2+1} \right\} - e^{-2t} \mathcal{L}^{-1} \left\{ \frac{2}{s^2+1} \right\}$$

$$= 3e^{-2t} \cdot \cos t - 2e^{-2t} \cdot \sin t$$

$$= 3e^{-2t} \cos t - 2e^{-2t} \sin t$$

Ans:-

(28)

$$\frac{3s}{s^2-2s+2}$$

$$= \mathcal{L}^{-1} \left\{ \frac{3s}{s^2-2 \cdot s \cdot 1 + 1^2+1} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{3s}{(s-1)^2+1} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{3(s-1)+3}{(s-1)^2+1} \right\}$$

P.T.O

$$\begin{aligned}
 &= \alpha^{-1} \left\{ \frac{3(s-1)}{(s-1)^2+1} \right\} + \alpha^{-1} \left\{ \frac{3}{(s-1)^2+1} \right\} \quad \left\{ \begin{array}{l} u+2.0 \\ -2u+2 \end{array} \right\} \\
 &= 3\alpha^{-1} \left\{ \frac{(s-1)}{(s-1)^2+1} \right\} + 3\alpha^{-1} \left\{ \frac{1}{(s-1)^2+1} \right\} \quad \left\{ \begin{array}{l} u+2.0 \\ -2u+2 \end{array} \right\} \\
 &= 3e^t \alpha^{-1} \left\{ \frac{s}{s^2+1} \right\} + 3e^t \left\{ \frac{1}{s^2+1} \right\} \quad \left\{ \begin{array}{l} u+2.0 \\ -2u+2 \end{array} \right\} \\
 &= 3e^t \cos t + 3e^t \sin t \quad \left\{ \begin{array}{l} u+2.0 \\ -2u+2 \end{array} \right\}
 \end{aligned}$$

Ans:-

For more practice:-

⊛ $\frac{2s-137}{s^2+2s+400}$

$$\begin{aligned}
 &= \alpha^{-1} \left\{ \frac{2s-137}{s^2+2s+400} \right\} \quad \left\{ \begin{array}{l} 2s-137 \\ s^2+2s+400 \end{array} \right\} \\
 &= \alpha^{-1} \left\{ \frac{2s-137}{(s+1)^2+400} \right\} \quad \left\{ \begin{array}{l} 2s-137 \\ s^2+2s+400 \end{array} \right\} \\
 &= \alpha^{-1} \left\{ \frac{2(s+1)-140}{(s+1)^2+400} \right\} \quad \left\{ \begin{array}{l} 2s-137 \\ s^2+2s+400 \end{array} \right\} \\
 &= 2\alpha^{-1} \left\{ \frac{s+1}{(s+1)^2+400} \right\} - 140\alpha^{-1} \left\{ \frac{1}{(s+1)^2+(20)^2} \right\} \quad \left\{ \begin{array}{l} 2s-137 \\ s^2+2s+400 \end{array} \right\} \\
 &= 2e^{-t} \cos 20t - 140e^{-t} \frac{\sin 20t}{20} \quad \left\{ \begin{array}{l} 2s-137 \\ s^2+2s+400 \end{array} \right\} \\
 &= 2e^{-t} \cos 20t - 7e^{-t} \sin 20t \quad \left\{ \begin{array}{l} 2s-137 \\ s^2+2s+400 \end{array} \right\}
 \end{aligned}$$

Ans:-

$$* \quad \frac{3s - 1000}{s^2 - 2s + 500}$$

$$= \alpha^{-1} \left\{ \frac{3s - 1000}{s^2 - 2 \cdot s \cdot 1 + 1^2 + 500} \right\}$$

$$= \alpha^{-1} \left\{ \frac{3s - 1000}{(s-1)^2 + 500} \right\}$$

$$= \alpha^{-1} \left\{ \frac{3(s-1) - 1000 + 3}{(s-1)^2 + 500} \right\}$$

$$= 3\alpha^{-1} \left\{ \frac{(s-1)}{(s-1)^2 + 500} \right\} - 1000 \alpha^{-1} \left\{ \frac{1}{(s-1)^2 + 500} \right\}$$

$$= 3e^t \alpha^{-1} \left\{ \frac{s}{s^2 + (10\sqrt{5})^2} \right\} - 1000 e^t \alpha^{-1} \left\{ \frac{1}{s^2 + (10\sqrt{5})^2} \right\}$$

$$= 3e^t \cos 10\sqrt{5}t - 1000 e^t \sin 10\sqrt{5}t / 10\sqrt{5}$$

$$= 3e^t \cos 10\sqrt{5}t - 20\sqrt{5} \sin 10\sqrt{5}t$$

Ans:

Application of Laplace Transformation

Ch: 06

Derivative formula of Laplace transformation

Formula:

① if $\mathcal{L}\{f(t)\} = F(s)$ then,

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0) - s^{-1}f'(0) - s^{-2}f''(0) - \dots$$

For Example:

Red Alert:

⇒ This is the hardest one so just avoid it.

* Solve $y''' - 3y'' + 3y' - y = t^2 e^t$

given that $y(0) = 1, y'(0) = 0, y''(0) = -2$

⇒ Taking Laplace transform on both sides.

$$\mathcal{L}\{y'''\} - 3\mathcal{L}\{y''\} + 3\mathcal{L}\{y'\} - \mathcal{L}\{y\} = \mathcal{L}\{t^2 e^t\} \quad \text{--- (1)}$$

So, P.H.S. —

$$\mathcal{L}\{t^2 e^t\} = F(s-1) \quad \text{--- (11)}$$

$$f(t) = t^2$$

$$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{t^2\}$$

$$= \frac{2}{s^3}$$

from eq (11) ⇒

$$\mathcal{L}\{t^2 e^t\} = \frac{2}{(s-1)^3}$$

from eq (1) ⇒

$$= [s^3 \mathcal{L}\{y\} - s^2 y(0) - s y'(0) - y''(0)] - 3[s^2 \mathcal{L}\{y\} - s y(0) - y'(0)] + 3[s \mathcal{L}\{y\} - y(0)] - \mathcal{L}\{y\} = \frac{2}{(s-1)^3}$$

put the value of $y(0)=1, y'(0)=0, y''(0)=-2$

$$\Rightarrow s^3 \mathcal{L}\{y\} - s^2 - 0 + 2 - 3s^2 \mathcal{L}\{y\} + 3s + 3s \mathcal{L}\{y\} - 3 - \mathcal{L}\{y\} = \frac{2}{(s-1)^3}$$

$$\Rightarrow \mathcal{L}\{y\} (s^3 - 3s^2 + 3s - 1) = (s^2 - 3s + 1) + \frac{2}{(s-1)^3}$$

$$\Rightarrow \mathcal{L}\{y\} (s-1)^3 = \frac{(s^2 - 3s + 1)(s-1)^3 + 2}{(s-1)^3}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{s^2 - 3s + 1 - s}{(s-1)^3} + \frac{2}{(s-1)^6}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{(s-1)^2 - s}{(s-1)^3} + \frac{2}{(s-1)^6}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{(s-1)^2}{(s-1)^3} - \frac{s}{(s-1)^3} + \frac{2}{(s-1)^6}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{1}{s-1} - \left(\frac{(s-1)+1}{(s-1)^3} \right) + \frac{2}{(s-1)^6}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3} + \frac{2}{(s-1)^6}$$

$$\Rightarrow y = \mathcal{L}^{-1} \left\{ \frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3} + \frac{2}{(s-1)^6} \right\}$$

$$\therefore y = e^t - e^t t - e^t \frac{t^2}{2} + 2e^t \frac{t^5}{120}$$

$$y = e^t \left(1 - t - \frac{t^2}{2} + \frac{t^5}{60} \right)$$

Ans

Page 8 216:—

$$\textcircled{1} y' + 5.2y = 10.4 \sin 2t ; y(0) = 0$$

$$\Rightarrow \mathcal{L}\{y' + 5.2y\} = \mathcal{L}\{10.4 \sin 2t\}$$

$$\Rightarrow \mathcal{L}\{y'\} + 5.2\mathcal{L}\{y\} = 10.4 \mathcal{L}\{\sin 2t\}$$

$$\Rightarrow [s\mathcal{L}\{y\} - y(0)] + 5.2\mathcal{L}\{y\} = 10.4 \times \frac{2}{s^2 + 2^2}$$

put the value $y(0) = 0 \Rightarrow$

$$\Rightarrow s\mathcal{L}\{y\} + 5.2\mathcal{L}\{y\} = \frac{38.8}{s^2 + 4}$$

$$\Rightarrow \mathcal{L}\{y\} (s + 5.2) = \frac{38.8}{s^2 + 4}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{38.8}{(s + 5.2)(s^2 + 4)} = \frac{A}{(s + 5.2)} + \frac{Bs + C}{(s^2 + 4)}$$

multiply $(s + 5.2)(s^2 + 4)$ on both sides:—

$$38.8 = A(s + 4) + (Bs + C)(s + 5.2) \quad \text{--- (1)}$$

$$\text{if } 38.8 = As^2 + 4A + Bs^2 + 5.2Bs + Cs + 5.2C \quad \text{--- (2)}$$

$$\Rightarrow s + 5.2 = 0$$

$$\therefore s = -5.2$$

put the value in eq (1) $\Rightarrow$

$$38.8 = A(4 - 5.2)^2 + 4A$$

$$\therefore A = 1.25$$

equating the co-efficients of $x^2, x^1, x^0 \Rightarrow$ from eq (2) —

$$38.8 = 4A + 5.2C$$

$$\therefore C = \frac{38.8 - 4 \times 1.25}{5.2}$$

$$= 6.5$$

Again:-

$$5.2B + c = 0$$

$$s(5.2B + c) = 0$$

$$\therefore B = \frac{-c}{5.2}$$

$$= \frac{-6.5}{5.2}$$

$$= -1.25$$

put the value of A, B, c in eq ①

$$L\{y\} = \frac{A}{s+5.2} + \frac{Bs+c}{s^2+4}$$

$$L\{y\} = \frac{1.5}{s+5.2} + \frac{-1.5s+6.5}{s^2+4}$$

$$L\{y\} = 1.5 \times \frac{1}{s+5.2} - \frac{1.5s}{s^2+4} + \frac{6.5}{s^2+4}$$

$$\therefore y = L^{-1}\left\{1.5 \times \frac{1}{s+5.2} - \frac{1.5s}{s^2+4} + \frac{6.5}{s^2+4}\right\}$$

$$y = 1.5e^{-5.2t} - 1.5 \cos 2t + 6.5 \times \frac{\sin 2t}{2}$$

$$y = 1.5e^{-5.2t} - 1.5 \cos 2t + 3.25 \sin 2t$$

Ans:-

$$\textcircled{2} \quad y' + 2y = 0 ; y(0) = 1.5$$

$$\Rightarrow \mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = 0$$

$$\Rightarrow [s\mathcal{L}\{y\} - y(0)] + 2\mathcal{L}\{y\} = 0$$

$$\Rightarrow s\mathcal{L}\{y\} - 1.5 + 2\mathcal{L}\{y\} = 0$$

$$\Rightarrow \mathcal{L}\{y\}(s+2) = 1.5$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{1.5}{s+2}$$

$$\Rightarrow y = \mathcal{L}^{-1}\left\{\frac{1.5}{s+2}\right\}$$

$$\Rightarrow y = 1.5 \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$\therefore y = 1.5 e^{-2t}$$

Ans:-

$$\textcircled{3} \quad y'' - y' - 6y = 0, y(0) = 11, y'(0) = 28$$

$$\Rightarrow \mathcal{L}\{y''\} - \mathcal{L}\{y'\} - 6\mathcal{L}\{y\} = 0$$

$$\Rightarrow [s^2\mathcal{L}\{y\} - sy(0) - y'(0)] - [s\mathcal{L}\{y\} - y(0)] - 6\mathcal{L}\{y\} = 0$$

$$\Rightarrow s^2\mathcal{L}\{y\} - 11s - 28 - s\mathcal{L}\{y\} + 11 - 6\mathcal{L}\{y\} = 0$$

$$\Rightarrow \mathcal{L}\{y\}(s^2 - s - 6) = 11s + 28 - 11$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{11s + 17}{s^2 - s - 6}$$

$$= \frac{11s + 17}{(s - \frac{1}{2})^2 - 6.25}$$

$$(s - \frac{1}{2})^2 - 6.25$$

P.T.O

$$\mathcal{L}\{y\} = \frac{-11(s-\frac{1}{2}) + 17 + \frac{11}{2}}{(s-\frac{1}{2})^2 - 6 \cdot 25}$$

$$y = 11 \mathcal{L}^{-1} \left\{ \frac{(s-\frac{1}{2})}{(s-\frac{1}{2})^2 - (2.5)^2} \right\} + 22.5 \mathcal{L}^{-1} \left\{ \frac{1}{(s-\frac{1}{2})^2 - (2.5)^2} \right\}$$

$$= 11 e^{\frac{1}{2}t} \cosh 2.5t + 22.5 \frac{e^{\frac{1}{2}t} \sinh 2.5t}{2.5}$$

$$\therefore y = 11 e^{\frac{1}{2}t} \cosh 2.5t + 9 e^{\frac{1}{2}t} \sinh 2.5t$$

Ans:-

④ $y'' + 0y = 10e^{-t}$, $y(0) = 0$, $y'(0) = 0$

$$\Rightarrow \mathcal{L}\{y''\} + 0\mathcal{L}\{y\} = 10 \mathcal{L}\{e^{-t}\}$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] + 0\mathcal{L}\{y\} = 10 \times \frac{1}{s+1}$$

$$\Rightarrow s^2 \mathcal{L}\{y\} + 0\mathcal{L}\{y\} = \frac{10}{s+1}$$

$$\Rightarrow \mathcal{L}\{y\}(s^2+0) = \frac{10}{s+1}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{10}{(s+1)(s^2+0)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+0} \quad \text{--- (1)}$$

Multiply $(s+1)(s^2+0)$ on both sides:-

$$10 = A(s^2+0) + (Bs+C)(s+1) \quad \text{--- (11)}$$

$$10 = As^2 + 0A + Bs^2 + Bs + Cs + C \quad \text{--- (12)}$$

If $s+1=0$

$\therefore s = -1$

Put the value of s in eq (11)

$A = 1$

from Equating the Co-efficients of $x^2, x^1, x^0 \Rightarrow$ in eq (11) $\Rightarrow$

$$0A + C = 10 \Rightarrow C = 1$$

$$Bs + Cs = 0$$

$$s(B+C) = 0$$

$$\therefore B = -C$$
$$= -1$$

put the value of A, B, C in eq (1) $\Rightarrow$

$$\Rightarrow \mathcal{L}^{-1}\{Y\} = \frac{1}{s+1} + \frac{-s+1}{s^2+9}$$

$$Y = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} + \mathcal{L}^{-1}\left\{\frac{-s+1}{s^2+9}\right\}$$
$$= e^{-t} - \cos 3t + \frac{\sin 3t}{3}$$

$$\therefore Y = e^{-t} - \cos 3t + \frac{1}{3} \sin 3t$$

Ans. -

$$\textcircled{5} \quad y'' - \frac{1}{4}y = 0, \quad y(0) = 12, \quad y'(0) = 0$$

$$\Rightarrow \mathcal{L}\{y''\} - \frac{1}{4}\mathcal{L}\{y\} = 0$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] - \frac{1}{4}\mathcal{L}\{y\} = 0$$

$$\Rightarrow s^2 \mathcal{L}\{y\} - 12 - \frac{1}{4}\mathcal{L}\{y\} = 0$$

$$\Rightarrow \mathcal{L}\{y\} \left(s^2 - \frac{1}{4} \right) = 12s$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{12s}{s^2 - \frac{1}{4}}$$

$$\Rightarrow y = \mathcal{L}^{-1} \left\{ \frac{12s}{s^2 - \left(\frac{1}{2}\right)^2} \right\}$$

$$= 12 \cosh \frac{1}{2}t$$

Ans:

$$\textcircled{6} \quad y'' - 6y' + 5y = 20 \cos 2t, \quad y(0) = 3.2, \quad y'(0) = 6.2$$

$$\Rightarrow \mathcal{L}\{y''\} - 6\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} = 20 \mathcal{L}\{\cos 2t\}$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] - 6[s \mathcal{L}\{y\} - y(0)] + 5\mathcal{L}\{y\} = 20 \times \frac{s}{s^2 + 4}$$

$$\Rightarrow s^2 \mathcal{L}\{y\} - 3.2s - 6.2 - 6s \mathcal{L}\{y\} + 10.2 + 5\mathcal{L}\{y\} = 20 \frac{s}{s^2 + 4}$$

$$\Rightarrow \mathcal{L}\{y\} (s^2 - 6s + 5) = \frac{20s}{s^2 + 4} + 3.2s + 6.2 - 10.2$$

$$\Rightarrow \text{L.H.S.} = \frac{20s}{(s^2+4)(s-1)(s-5)} + \frac{0.2s}{(s-1)(s-5)} - \frac{13}{(s-1)(s-5)}$$

$$= \frac{20s + 0.2s(s^2+4) - 13(s^2+4)}{(s-1)(s-5)(s^2+4)}$$

$$= \frac{20s + 0.2s^3 + 0.8s - 13s^2 - 52}{(s-1)(s-5)(s^2+4)}$$

$$= \frac{0.2s^3 + 41.8s - 13s^2 - 52}{(s-1)(s-5)(s^2+4)}$$

$$= \frac{0.2s^3 + 41.8s - 13s^2 - 52}{(s-1)(s-5)(s^2+4)}$$

$$\therefore \text{L.H.S.} = \frac{0.2s^3 + 41.8s - 13s^2 - 52}{(s-1)(s-5)(s^2+4)} = \frac{A}{(s-1)} + \frac{B}{(s-5)} + \frac{Cs+D}{s^2+4} \quad \text{--- (1)}$$

multiply (s-1)(s-5)(s^2+4) on both side, —

$$0.2s^3 + 41.8s - 13s^2 - 52 = A(s-5)(s^2+4) + B(s-1)(s^2+4) + (Cs+D)(s-1)(s-5) \quad \text{--- (1)}$$

Now: $s=5$

$$\therefore s=5$$

put the value in eq (1)

$$0.2(5)^3 + 41.8(5) - 13(5)^2 - 52 = B(5-1)(5^2+4)$$

$$\therefore B=2$$

Again: $s=1$

put the value in eq (1)

$$0.2 + 41.8 - 13 - 52 = A(1-5)(1+4)$$

$$\therefore A=2$$

Now:-

$$\Rightarrow 0.2s^3 - 13s^2 + 41.8s - 52 = A(s^2 + 4s - 5) + B(s^2 + 4s - 5) + (Cs + D)(s^2 - 6s + 5)$$

$$= As^3 + 4As^2 - 5As - 20A + Bs^3 + 4Bs^2 - 5B + Cs^3 - 6Cs^2 + 5Cs + DS^2 - 6DS + 5D$$

Equating like terms:-

$$As^3 + Bs^3 + Cs^3 = 0.2s^3$$

$$\Rightarrow s^3(A+B+C) = 0.2s^3$$

$$\Rightarrow C = 0.2 - (A+B)$$

$$= 0.2 - (1+2)$$

$$= 0.2$$

Again:-

$$-20A - 4B + 5D = -52$$

$$\therefore D = \frac{-52 + 20 \times 1 + 4 \times 2}{5}$$

$$= -4.8$$

Put the value of A, B, C, D in eq. (1)

$$d\{y\} = \frac{1}{s-1} + \frac{2}{s-5} + \frac{0.2s - 4.8}{s^2 + 2^2}$$

$$\therefore y = d^{-1}\left\{\frac{1}{s-1}\right\} + d^{-1}\left\{\frac{2}{s-5}\right\} + d^{-1}\left\{\frac{0.2s}{s^2+2^2}\right\} - d^{-1}\left\{\frac{4.8}{s^2+2^2}\right\}$$

$$= e^t + 2e^{5t} + 0.2 \cos 2t - 4.8 \times \frac{\sin 2t}{2}$$

$$= e^t + 2e^{5t} + 0.2 \cos 2t - 2.4 \sin 2t$$

Ans:-

$$(7) y'' + 7y' + 12y = 21e^{3x}, \quad y(0) = 3.5, \quad y'(0) = -10$$

$$\Rightarrow 21y'' + 7 \times 21y' + 12 \times 21y = 21 \times 21e^{3x}$$

$$\Rightarrow [21y' - 5y(0) - y'(0)] + 7[5 \times 21y' - y(0)] + 12 \times 21y = 21 \times \frac{1}{s-3}$$

$$\Rightarrow 21y' - 3.5s + 10 + 7 \times 5 \times 21y' - 24.5 + 12 \times 21y = \frac{21}{s-3}$$

$$\Rightarrow 21y'(s^2 + 7s + 12) = \frac{21}{s-3} + 3.5s + 14.5$$

$$\Rightarrow 21y' = \frac{21}{(s-3)(s+4)(s+3)} + \frac{3.5s}{(s+3)(s+4)} + \frac{14.5}{(s+3)(s+4)}$$

$$\Rightarrow 21y' = \frac{21 + 3.5s(s-3) + 14.5(s-3)}{(s-3)(s+4)(s+3)}$$

$$= \frac{21 + 3.5s^2 - 10.5s + 14.5s - 43.5}{(s-3)(s+4)(s+3)}$$

$$= \frac{21 + 3.5s^2 + 4s - 43.5}{(s-3)(s+4)(s+3)} = \frac{A}{s-3} + \frac{B}{s+4} + \frac{C}{s+3} \quad (1)$$

Now multiply (s-3)(s+4)(s+3) in both side!

$$21 + 3.5s^2 + 4s - 43.5 = A(s+4)(s+3) + B(s-3)(s+3) + C(s-3)(s+4) \quad (10)$$

Now! $s+3=0$
 $\therefore s = -3$

Put it in eq (10)

$$21 + 3.5(-3)^2 + 4(-3) - 43.5 = C(-3-3)(-3+4)$$

$$\therefore C = \frac{1}{2}$$

Again!

$$s+4=0$$

$$\therefore s = -4$$

$$21 + 3.5(-4)^2 + 4(-4) - 43.5 = B(-4-3)(-4+3)$$

$$\therefore B = \frac{5}{2}$$

Again - $s^2 - 4s + 4 = 0$
 $\therefore s = 2$

$$\therefore 21 + 9 \cdot 5(2)^2 + 4 \cdot 2 - 4 \cdot 9 \cdot 2 = A(2+4)(2+2)$$

$$\therefore A = \frac{1}{2}$$

put the value of A, B, C in eq (1)

$$\mathcal{L}\{y\} = \frac{1/2}{s-3} + \frac{5/2}{s+4} + \frac{1/2}{s+3}$$

$$\therefore y = \frac{1}{2} \mathcal{L}^{-1}\left(\frac{1}{s-3}\right) + \frac{5}{2} \mathcal{L}^{-1}\left(\frac{1}{s+4}\right) + \frac{1}{2} \mathcal{L}^{-1}\left(\frac{1}{s+3}\right)$$

$$= \frac{1}{2} e^{3t} + \frac{5}{2} e^{-4t} + \frac{1}{2} e^{-3t}$$

Ans.

⑧ $y'' - 4y' + 4y = 0$, $y(0) = 8.1$, $y'(0) = 3.2$

$$\Rightarrow \mathcal{L}\{y''\} - 4\mathcal{L}\{y'\} + 4\mathcal{L}\{y\} = 0$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - s y(0) - y'(0)] - 4[s \mathcal{L}\{y\} - y(0)] + 4\mathcal{L}\{y\} = 0$$

$$\Rightarrow s^2 \mathcal{L}\{y\} - 8.1s - 3.2 - 4s \mathcal{L}\{y\} + 8.1 + 4\mathcal{L}\{y\} = 0$$

$$\Rightarrow \mathcal{L}\{y\} (s^2 - 4s + 4) = 8.1s - 4.2$$

$$\therefore \mathcal{L}\{y\} = \frac{8.1s - 4.2}{(s-2)(s-2)}$$

$$= \frac{8.1s - 4.2}{(s-2)^2}$$

①

$$\therefore \mathcal{L}\{y\} = \frac{8.1(s-2) + 16.2}{(s-2)^2}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{8.1(s-2)}{(s-2)^2 + 0^2} + \frac{12}{(s-2)^2 + 0^2}$$

$$y = \mathcal{L}^{-1} \left\{ \frac{8.1(s-2)}{(s-2)^2 + 0^2} \right\} + \mathcal{L}^{-1} \left\{ \frac{12}{(s-2)^2} \right\}$$

$$= 8.1 e^{2t} \cos 0t + 12 e^{2t} t$$

$$= 8.1 e^{2t} + 12 e^{2t} t$$

⑨ $y'' - 4y' + 3y = 6t - 8$; $y(0) = 0$, $y'(0) = 0$

$$\Rightarrow \mathcal{L}\{y''\} - 4\mathcal{L}\{y'\} + 3\mathcal{L}\{y\} = \mathcal{L}\{6t - 8\}$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - s y(0) - y'(0)] - 4[s \mathcal{L}\{y\} - y(0)] + 3\mathcal{L}\{y\} = \frac{6}{s^2} - \frac{8}{s}$$

$$\Rightarrow s^2 \mathcal{L}\{y\} - 0 - 4s \mathcal{L}\{y\} + 3\mathcal{L}\{y\} = \frac{6}{s^2} - \frac{8}{s}$$

$$\Rightarrow \mathcal{L}\{y\} (s^2 - 4s + 3) = \frac{6 - 8s}{s^2}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{6 - 8s}{s^2(s-1)(s-3)} = \frac{A}{s-1} + \frac{B}{s-3} + \frac{Cs+D}{s^2} \quad \text{--- (1)}$$

Multiply $s^2(s-1)(s-3)$ on both side:—

$$6 - 8s = A(s^2+0)(s-3) + B(s-1)(s^2+0) + (Cs+D)(s-1)(s-3) \quad \text{--- (11)}$$

$$6 - 8s = As^2(s-3) + B(s^3-s^2) + (Cs+D)(s^2-4s+3)$$

$$= As^3 - 3As^2 + Bs^3 - Bs^2 + Cs^3 - 4Cs^2 + 3Cs + Ds^2 - 4Ds + 3D \quad \text{--- (12)}$$

Equating like terms:—

$$\therefore D = \frac{6}{3} = 2$$

Now, $s-3=0$
 $\therefore s=3$

Put the value in eq (iii)

$$6-8 \times 3 = B(3-1) \times 3^2$$

$$\therefore B = -1$$

Again, $s-1=0$

$$\therefore s=1$$

$$\therefore 6-8 \times 1 = A(1-3) \times 1^2$$

$$\therefore A = 1$$

from eq (iv)

$$3cs - 4ds = -8s$$

$$\therefore c = \frac{-8 + 4 \times 2}{3} = 0$$

Put the value of A, B, C, D in eq (i)

$$\mathcal{L}\{y\} = \frac{1}{s-1} + \frac{-1}{s-3} + \frac{2}{s^2}$$

$$\therefore \mathcal{L}^{-1}\left\{ \frac{1}{s-1} + \frac{-1}{s-3} + \frac{2}{s^2} \right\}$$

$$= e^t - e^{3t} + 2t$$

Ans

System

OF



System of ODES:-

Example:-

$$y_1' = 2y_1 - 3y_2 \dots \dots \dots \textcircled{i} \quad y_1(0) = 8$$

$$y_2' = y_2 - 2y_1 \dots \dots \dots \textcircled{ii} \quad y_2(0) = 3$$

⇒ From eq (i):-

$$L\{y_1'\} = L\{2y_1 - 3y_2\}$$

$$\Rightarrow sY_1 - y_1(0) = 2Y_1 - 3Y_2$$

$$\Rightarrow sY_1 - 2Y_1 + 3Y_2 = 8$$

$$\Rightarrow Y_1(s-2) + 3Y_2 = 8 \dots \dots \dots \textcircled{iii}$$

⇒ From eq (ii):-

$$L\{y_2'\} = L\{y_2 - 2y_1\}$$

$$\Rightarrow sY_2 - y_2(0) = Y_2 - 2Y_1$$

$$\Rightarrow sY_2 - Y_2 + 2Y_1 = 3$$

$$\Rightarrow (s-1)Y_2 + 2Y_1 = 3 \dots \dots \dots \textcircled{iv}$$

Now Solving two equation using cramer's rule:-

$$Y_1 = \frac{\begin{vmatrix} 8 & 3 \\ s-2 & 3 \end{vmatrix}}{\begin{vmatrix} s-2 & 3 \\ 2 & s-1 \end{vmatrix}} = \frac{8s-8-9}{s^2-3s+2-6} = \frac{8s-17}{s^2-3s-4} \dots \dots \dots \textcircled{v}$$

$$Y_2 = \frac{\begin{vmatrix} s-2 & 8 \\ 2 & 3 \end{vmatrix}}{\begin{vmatrix} s-2 & 3 \\ 2 & s-1 \end{vmatrix}} = \frac{3s-6-16}{s^2-3s-4} = \frac{3s-22}{s^2-3s-4} \dots \dots \dots \textcircled{vi}$$

from eq (V) =

$$Y_1 = \frac{8s-17}{s^2-3s-4} = \frac{A}{(s-4)} + \frac{B}{(s+1)} \quad \text{--- (VII)}$$

Using Coverup rule

$$s=4 ; A = \frac{8(4)-17}{4+1} = 3$$

$$s=-1 ; B = \frac{8(-1)-17}{-1-4} = 5$$

put the value in eq (VII) =

$$Y_1(s) = d^{-1} \left\{ \frac{3}{s-4} + d^{-1} \left\{ \frac{5}{s+1} \right\} \right\}$$

$$Y_1(t) = 3e^{4t} + 5e^{-t}$$

from eq (VI) =

$$Y_2 = \frac{3s-22}{s^2-3s-4} = \frac{A}{(s-4)} + \frac{B}{(s+1)} \quad \text{--- (VIII)}$$

Using Coverup rule

$$\therefore s=4 ; A = \frac{3(4)-22}{(4+1)} = -2$$

$$s=-1 ; B = \frac{3(-1)-22}{-1-4} = 5$$

put it in eq (VIII) =

$$Y_2(s) = d^{-1} \left\{ \frac{-2}{s-4} + \frac{5}{s+1} \right\}$$

$$\therefore Y_2 = 5e^{-t} - 2e^{4t}$$

Ans:-

Page 246:-

$$\textcircled{3} \quad Y_1' = -Y_1 + 4Y_2 \quad \dots \dots \textcircled{i} \quad Y_1(0) = 3$$

$$Y_2' = 3Y_1 - 2Y_2 \quad \dots \dots \textcircled{ii} \quad Y_2(0) = 4$$

$\Rightarrow$ From eq (i) $\Rightarrow$

$$s Y_1' = s(-Y_1 + 4Y_2)$$

$$\Rightarrow s Y_1 - Y_1(0) = -Y_1 + 4Y_2$$

$$\Rightarrow s Y_1 + Y_1 - 4Y_2 = 3$$

$$\Rightarrow Y_1(s+1) - 4Y_2 = 3 \quad \dots \dots \textcircled{iii}$$

from eq (ii) $\Rightarrow$

$$s Y_2' = s(3Y_1 - 2Y_2)$$

$$s Y_2 - Y_2(0) = 3Y_1 - 2Y_2$$

$$s Y_2 + 2Y_2 - 3Y_1 = 4$$

$$-3Y_1 + Y_2(s+2) = 4 \quad \dots \dots \textcircled{iv}$$

There is a problem with this math calculation

Just put S here and complete the calculation

Now Solving two equation using Cramer's rule:-

$$Y_1 = \frac{\begin{vmatrix} 3 & -4 \\ 4 & (s+2) \end{vmatrix}}{\begin{vmatrix} (s+1) & -4 \\ -3 & s+2 \end{vmatrix}} = \frac{3+6+16}{s^2+3s-12+2} = \frac{3+22}{s^2+3s-10} = \frac{25}{s^2+3s-10} \quad \dots \dots \textcircled{iii}$$

$$Y_2 = \frac{\begin{vmatrix} (s+1) & 3 \\ -3 & 4 \end{vmatrix}}{\begin{vmatrix} (s+1) & -4 \\ -3 & (s+2) \end{vmatrix}} = \frac{4s+4+9}{s^2+3s+2-12} = \frac{4s+13}{s^2+3s-10} \quad \dots \dots \textcircled{iv}$$

from eq (i) $\Rightarrow$

$$Y_1 = \frac{2s}{s^2 + 3s - 10} = \frac{A}{(s-2)} + \frac{B}{(s+5)} \quad \text{--- (v)}$$

Using Coverup rule:—

$$s=2; A = \frac{2s}{s+5} = \frac{2s}{7}$$

$$s=-5; B = \frac{2s}{s-2} = \frac{2s}{-7}$$

Put the value in eq (v) $\Rightarrow$

$$Y_1(s) = \frac{2s}{s^2 + 3s - 10} = \frac{2s/7}{s-2} + \frac{-2s/7}{s+5}$$

$$\therefore Y_1(s) = \frac{2s}{7} e^{2t} - \frac{2s}{7} e^{-5t}$$

from eq (ii) $\Rightarrow$

$$Y_2 = \frac{4s+13}{s^2+3s-10} = \frac{A}{(s-2)} + \frac{B}{(s+5)} \quad \text{--- (vi)}$$

Using Coverup rule:—

$$s=2; A = \frac{4s+13}{s+5} = 3$$

$$s=-5; B = \frac{4s+13}{s-2} = 1$$

Put in in eq (vi)

$$Y_2(s) = \frac{4s+13}{s^2+3s-10} = \frac{3}{s-2} + \frac{1}{s+5}$$

$$\therefore Y_2(s) = 3e^{2t} + e^{-5t}$$

Ans:—

$$⑥ \quad y_1' = 5y_1 + y_2 \quad \dots \dots \dots ① \quad y_1(0) = 1$$

$$y_2' = y_1 + 5y_2 \quad \dots \dots \dots ② \quad y_2(0) = -3$$

⇒ from eq ① ⇒

$$\mathcal{L}\{y_1'\} = \mathcal{L}\{5y_1 + y_2\}$$

$$\Rightarrow sY_1 - y_1(0) = 5Y_1 + Y_2$$

$$\Rightarrow sY_1 - 5Y_1 - Y_2 = +y_1(0)$$

$$\Rightarrow Y_1(s-5) - Y_2 = 1 \quad \dots \dots \dots ③$$

⇒ from eq ② ⇒

$$\mathcal{L}\{y_2'\} = \mathcal{L}\{y_1 + 5y_2\}$$

$$\therefore sY_2 - y_2(0) = Y_1 + 5Y_2$$

$$\Rightarrow sY_2 - 5Y_2 - Y_1 = y_2(0)$$

$$\Rightarrow -Y_1 + Y_2(s-5) = -3 \quad \dots \dots \dots ④$$

Now Solving the eq using Cramer's rule:-

$$Y_1 = \frac{\begin{vmatrix} 1 & -1 \\ -3 & (s-5) \end{vmatrix}}{\begin{vmatrix} (s-5) & -1 \\ -1 & (s-5) \end{vmatrix}} = \frac{s-5-3}{s^2-10s+25-1} = \frac{s-8}{s^2-10s+24} \quad \text{--- ⑤}$$

$$Y_2 = \frac{\begin{vmatrix} (s-5) & 1 \\ -1 & -3 \end{vmatrix}}{\begin{vmatrix} (s-5) & -1 \\ -1 & (s-5) \end{vmatrix}} = \frac{+1-3s+15}{s^2-10s+25-1} = \frac{-3s+16}{s^2-10s+24} \quad \text{--- ⑥}$$

from eq (V) $\Rightarrow$

$$Y_1 = \frac{s-8}{s^2-10s+24} = \frac{A}{(s-6)} + \frac{B}{(s-4)} \quad \text{--- (VII)}$$

using Coverup rule:—

$$s=6; \quad A = \frac{s-8}{s-4} = -1$$

$$s=4; \quad B = \frac{s-8}{s-6} = 2$$

put it in eq (VI) $\Rightarrow$

$$Y_1 = \frac{-1}{s-6} + \frac{2}{s-4}$$

$$\therefore \mathcal{G}_1(s) = \mathcal{L}^{-1} \left\{ \frac{2}{s-4} + \frac{-1}{s-6} \right\}$$

$$\boxed{y_1(t) = 2e^{4t} - e^{6t}}$$

from eq (VI) $\Rightarrow$

$$Y_2 = \frac{-3s+16}{s^2-10s+24} = \frac{A}{s-6} + \frac{B}{s-4} \quad \text{--- (VIII)}$$

using Coverup rule:—

$$s=6; \quad A = \frac{-3s+16}{s-4} = -1$$

$$s=4; \quad B = \frac{-3s+16}{s-6} = -2$$

put it in eq (VIII) $\Rightarrow$

$$Y_2 = \frac{-1}{s-6} + \frac{-2}{s-4}$$

$$\therefore \mathcal{G}_2(s) = \mathcal{L}^{-1} \left(\frac{-1}{s-6} + \frac{-2}{s-4} \right)$$

$$\boxed{y_2(t) = -e^{6t} - 2e^{4t}}$$

$$\textcircled{8} \quad y_1' = -2y_1 + 3y_2 \quad \dots \dots \textcircled{i} \quad y_1(0) = 4$$

$$y_2' = 4y_1 - y_2 \quad \dots \dots \textcircled{ii} \quad y_2(0) = 3$$

⇒ from eq. (i) ⇒

$$\Rightarrow d\{y_1'\} = d\{-2y_1 + 3y_2\}$$

$$\Rightarrow sY_1 - y_1(0) = -2Y_1 + 3Y_2$$

$$\Rightarrow sY_1 + 2Y_1 = y_1(0) + 3Y_2$$

$$\Rightarrow Y_1(s+2) - 3Y_2 = 4 \quad \dots \dots \textcircled{iii}$$

⇒ from eq. (ii) ⇒

$$d\{y_2'\} = d\{4y_1 - y_2\}$$

$$\Rightarrow sY_2 = 4Y_1 - Y_2 + y_2(0)$$

$$\Rightarrow -4Y_1 + sY_2 + Y_2 = y_2(0)$$

$$\Rightarrow -4Y_1 + Y_2(s+1) = 3 \quad \dots \dots \textcircled{iv}$$

Now Solving two eq using Cramers rule!

$$Y_1 = \frac{\begin{vmatrix} 4 & -3 \\ 3 & (s+1) \end{vmatrix}}{\begin{vmatrix} (s+2) & -3 \\ (s+1) & -4 \end{vmatrix}} = \frac{4s+4+9}{12-s^2-3s-2} = \frac{4s+13}{s^2+3s-10} \quad \textcircled{v}$$

$$Y_2 = \frac{\begin{vmatrix} (s+2) & 4 \\ -4 & 3 \end{vmatrix}}{\begin{vmatrix} (s+2) & -3 \\ -4 & (s+1) \end{vmatrix}} = \frac{+16+3s+6}{s^2+3s+2-12} = \frac{+3s+22}{s^2+3s-10} \quad \textcircled{vi}$$

from eq (V) $\Rightarrow$

$$Y_1 = \frac{4s+13}{s^2+9s-10} = \frac{A}{(s+10)} + \frac{B}{(s-2)} \quad \text{--- (VI)}$$

Using Cover-up rule, —

$$s = -10 ; A = \frac{4s+13}{s-2} = 1$$

$$s = 2 ; B = \frac{4s+13}{(s+10)} = 3$$

put it in eq (VI) $\Rightarrow$

$$Y_1 = \frac{1}{s+10} + \frac{3}{s-2}$$

$$H_1(s) = d^{-1} \left\{ \frac{1}{s+10} + \frac{3}{s-2} \right\}$$

$$Y_1(s) = e^{-10s} + 3e^{2s}$$

from eq (VI) $\Rightarrow$

$$Y_2 = \frac{+3s+22}{s^2+9s-10} = \frac{A}{s+10} + \frac{B}{s-2} \quad \text{--- (VII)}$$

Using Cover-up rule, —

$$s = -10 ; A = \frac{+3s+22}{s-2} = 1$$

$$s = 2 ; B = \frac{3s+22}{s+10} = 4$$

put it in eq (VII) $\Rightarrow$

$$Y_2 = \frac{1}{s+10} + \frac{4}{s-2}$$

$$\therefore H_2(s) = d^{-1} \left\{ \frac{1}{s+10} + \frac{4}{s-2} \right\}$$

$$Y_2(s) = -e^{-10s} + 4e^{2s}$$

$$\textcircled{i} \quad y_1' = 4y_1 + y_2 \quad \text{--- (i)} \quad y_1(0) = 3$$

$$\textcircled{ii} \quad y_2' = -y_1 + 2y_2 \quad \text{--- (ii)} \quad y_2(0) = 1$$

from eq (i) $\Rightarrow$

$$\mathcal{L}\{y_1'\} = \mathcal{L}\{4y_1 + y_2\}$$

$$\Rightarrow sY_1 - y_1(0) = 4Y_1 + Y_2$$

$$\Rightarrow sY_1 - 4Y_1 - Y_2 = y_1(0)$$

$$\Rightarrow Y_1(s-4) - Y_2 = 3 \quad \text{--- (iii)}$$

from eq (ii) $\Rightarrow$

$$\mathcal{L}\{y_2'\} = \mathcal{L}\{-y_1 + 2y_2\}$$

$$\Rightarrow sY_2 - y_2(0) = -Y_1 + 2Y_2$$

$$\Rightarrow Y_1 + sY_2 - 2Y_2 = y_2(0)$$

$$\Rightarrow Y_1 + Y_2(s-2) = 1 \quad \text{--- (iv)}$$

Solving two equation using Cramer's rule.

$$Y_1 = \frac{\begin{vmatrix} 3 & -1 \\ (s-4) & -1 \end{vmatrix}}{\begin{vmatrix} (s-4) & -1 \\ 1 & (s-2) \end{vmatrix}} = \frac{3s - 6 + 1}{s^2 - 6s + 8 + 1} = \frac{3s - 5}{s^2 - 6s + 9} \quad \text{--- (v)}$$

$$Y_2 = \frac{\begin{vmatrix} (s-4) & 3 \\ 1 & 1 \end{vmatrix}}{\begin{vmatrix} (s-4) & -1 \\ 1 & (s-2) \end{vmatrix}} = \frac{s - 4 - 3}{s^2 - 6s + 8 + 1} = \frac{s - 7}{s^2 - 6s + 9} \quad \text{--- (vi)}$$

from eq (V) =

$$Y_1 = \frac{3s-5}{s^2-6s+9} = \frac{A}{(s-3)} + \frac{B}{(s-3)} \quad (VI)$$

$$\Rightarrow Y_1 = \frac{3s-5}{(s-3)^2}$$

$$Y_1 = \frac{3(s-3)-5+9}{(s-3)^2+0^2}$$

$$\begin{aligned} Y_1(s) &= 3d^{-1} \left\{ \frac{s-3}{(s-3)^2+0^2} \right\} + 4d^{-1} \left\{ \frac{1}{(s-3)^2} \right\} \\ &= 3e^{3t} \cos 0t + 4te^{3t} \end{aligned}$$

$$y_1(t) = 3e^{3t} + 4te^{3t}$$

from eq (VI) =

$$Y_2 = \frac{s-7}{s^2-6s+9}$$

$$\Rightarrow Y_2 = \frac{(s-3)-7+3}{(s-3)^2}$$

$$\Rightarrow Y_2 = \frac{(s-3)-4}{(s-3)^2}$$

$$\begin{aligned} \therefore Y_2(s) &= d^{-1} \left\{ \frac{s-3}{(s-3)^2+0^2} \right\} - 4d^{-1} \left\{ \frac{1}{(s-3)^2} \right\} \\ &= e^{3t} \cos 0t - 4te^{3t} \end{aligned}$$

$$y_2(t) = e^{3t} - 4te^{3t}$$

Page: 252:-

$$\textcircled{35} \quad Y_1' = 2Y_1 - 4Y_2 \quad \text{--- (I)} \quad Y_1(0) = 3$$

$$Y_2' = Y_1 - 3Y_2 \quad \text{--- (II)} \quad Y_2(0) = 0$$

⇒ from eq (I) :-

$$\mathcal{L}\{Y_1'\} = \mathcal{L}\{2Y_1 - 4Y_2\}$$

$$\Rightarrow sY_1 - Y_1(0) = 2Y_1 - 4Y_2$$

$$\Rightarrow sY_1 - 2Y_1 + 4Y_2 = 3$$

$$\Rightarrow Y_1(s-2) + 4Y_2 = 3 \quad \text{--- (III)}$$

⇒ from eq (II) :-

$$\mathcal{L}\{Y_2'\} = \mathcal{L}\{Y_1 - 3Y_2\}$$

$$\Rightarrow sY_2 - Y_2(0) = Y_1 - 3Y_2$$

$$\Rightarrow -Y_1 + sY_2 + 3Y_2 = Y_2(0)$$

$$\Rightarrow -Y_1 + Y_2(s+3) = 0 \quad \text{--- (IV)}$$

Solve two equation using Cramer's rule:-

$$Y_1 = \frac{\begin{vmatrix} 3 & 4 \\ 0 & (s+3) \end{vmatrix}}{\begin{vmatrix} (s-2) & 4 \\ -1 & (s+3) \end{vmatrix}} = \frac{3(s+3) - 0}{s^2 + 3s - 2s - 6 + 4} = \frac{3(s+3)}{s^2 + s - 2} \quad \text{--- (V)}$$

$$Y_2 = \frac{\begin{vmatrix} (s-2) & 3 \\ -1 & 0 \end{vmatrix}}{\begin{vmatrix} (s-2) & 4 \\ -1 & (s+3) \end{vmatrix}} = \frac{0 - 3}{s^2 + 3s - 2s - 6 + 4} = \frac{-3}{s^2 + s - 2} \quad \text{--- (VI)}$$

from eq (v) =

$$Y_1 = \frac{3s+0}{s^2+s-2} = \frac{A}{(s-1)} + \frac{B}{(s+2)} \quad \text{vii}$$

using cover up rule:-

$$s=1; \quad A = \frac{3s+0}{s+2} = 4$$

$$s=-2; \quad B = \frac{3s+0}{s-1} = -1$$

Put the value in eq (vii) =

$$Y_1 = \frac{4}{s-1} + \frac{-1}{s+2}$$

$$\therefore \mathcal{L}^{-1}(Y_1) = \mathcal{L}^{-1}\left\{\frac{4}{s-1} + \frac{-1}{s+2}\right\}$$

$$y_1(t) = 4e^t - e^{-2t}$$

$$\begin{aligned} Y_2 - Y_1 &= (0)B - Y_2 \Leftrightarrow \\ Y_2 - Y_1 &= (0)B - Y_2 \Leftrightarrow \\ (0)B &= Y_2 + Y_2 + Y_1 - \Leftrightarrow \\ 0 &= (2+2)Y_2 + Y_1 - \Leftrightarrow \end{aligned}$$

from eq (vi) =

$$Y_2 = \frac{3}{s^2+s-2} = \frac{A}{(s-1)} + \frac{B}{(s+2)} \quad \text{viii}$$

using Cover up rule:-

$$s=1; \quad A = \frac{3}{s+2} = \frac{1}{(s+2)}$$

$$s=-2; \quad B = \frac{3}{s-1} = -1$$

put it in eq (viii) =

$$Y_2 = \frac{1}{s-1} + \frac{-1}{s+2}$$

$$y_2(t) = \mathcal{L}^{-1}\left\{\frac{1}{s-1} - \frac{1}{s+2}\right\}$$

$$\therefore y_2(t) = e^t - e^{-2t}$$

$$(36) \quad y_1' = 2y_1 + 4y_2 \quad \dots \dots \dots (i) \quad y_1(0) = -4$$

$$y_2' = y_1 + 2y_2 \quad \dots \dots \dots (ii) \quad y_2(0) = -4$$

from eq (i) =

$$d\{y_1\} = d\{2y_1 + 4y_2\}$$

$$\Rightarrow sy_1 - 2y_1 - 4y_2 = y_1(0)$$

$$\Rightarrow (s-2)y_1 - 4y_2 = -4 \quad \dots \dots \dots (iii)$$

from eq (ii) =

$$d\{y_2\} = d\{y_1 + 2y_2\}$$

$$\Rightarrow sy_2 - y_1 - 2y_2 = y_2(0)$$

$$\Rightarrow -y_1 + (s-2)y_2 = -4 \quad \dots \dots \dots (iv)$$

Solve two equation using crammers rule:-

$$y_1 = \frac{\begin{vmatrix} -4 & -4 \\ -4 & (s-2) \end{vmatrix}}{\begin{vmatrix} (s-2) & -4 \\ -1 & (s-2) \end{vmatrix}} = \frac{-4s+8-16}{s^2-2s-2s+4-4} = \frac{-4s-8}{s^2-4s} \quad \text{--- (v)}$$

$$y_2 = \frac{\begin{vmatrix} (s-2) & -4 \\ -1 & -4 \end{vmatrix}}{\begin{vmatrix} (s-2) & -4 \\ -1 & (s-2) \end{vmatrix}} = \frac{-4s+8-4}{s^2-2s-2s+4-4} = \frac{-4s+4}{s^2-4s} \quad \text{--- (vi)}$$

from eq (v) =

$$y_1 = \frac{-4s-8}{s^2-4s} = \frac{A}{s-4} + \frac{B}{s} \quad \text{--- (vii)}$$

Multiply $\{s(s-4)\}$ on both side in eq (vi) $\Rightarrow$

$$-4s - 8 = As + B(s-4)$$

$$-4s - 8 = As + Bs - 4B$$

equating like term:

$$-4B = -8$$

$$\therefore B = +2$$

$$As + Bs = -4s$$

$$A = -4 - B$$

$$= -4 - 2$$

$$= -2 - 4 = -6$$

Put the value in eq (vii) $\Rightarrow$

$$Y_1 = \frac{-6}{s-4} + \frac{+2}{s}$$

$$\therefore Y_1(s) = \mathcal{L}\left\{ \frac{-6}{s-4} + \frac{2}{s} \right\}$$

$$y_1(t) = -6e^{4t} + 2$$

from eq (vi) $\Rightarrow$

$$Y_2 = \frac{-4s+4}{s^2-4s} = \frac{A}{s-4} + \frac{B}{s} \quad \text{--- (viii)}$$

Multiply $\{s(s-4)\}$ on both side in eq (viii) $\Rightarrow$

$$-4s+4 = As + Bs - 4B$$

$\therefore$ equating like term:

$$-4B = 4$$

$$\therefore B = -1$$

$$As + Bs = -4s$$

$$\therefore A = -4 - B$$

$$= -4 + 1$$

$$= -3$$

Put the value in eq (viii) $\Rightarrow$

$$Y_2 = \frac{-3}{s-4} + \frac{-1}{s}$$

$$Y_2(s) = \mathcal{L}^{-1}\left\{ \frac{-3}{s-4} - \frac{1}{s} \right\}$$

$$\therefore y_2(t) = -3e^{4t} - 1$$

Dirac's Delta
Function

Dirac's delta (unit impulse) function: —

Formulas: —

$$\textcircled{1} \mathcal{L}\{\delta(t-a)\} = e^{-as}$$

$$\textcircled{2} \mathcal{L}\{\delta(t)\} = 1$$

$$\textcircled{3} \mathcal{L}\{f(t) \cdot \delta(t-a)\} = e^{-as} \cdot f(a)$$

Find the Laplace Transformation

Dirac's delta function: —

$$\textcircled{1} f(t) = \delta(t-2)$$

$$\mathcal{L}\{f(t)\} = e^{-2s}$$

$$\mathcal{L}\{\delta(t)\}$$

$$F(s) = e^{-2s} \cdot 1 \\ = e^{-2s}$$

$$\textcircled{2} f(t) = 2\delta(t-3)$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{2\delta(t-3)\}$$

$$F(s) = 2 \times \mathcal{L}\{\delta(t-3)\}$$

$$= 2 \times e^{-3s}$$

$$\mathcal{L}\{\delta(t)\}$$

$$= 2e^{-3s}$$

Ans: —

$$\textcircled{3} \quad f(t) = \sin t \cdot \delta(t-3)$$

$$\begin{aligned} \underline{\text{Now,}} \quad \mathcal{L}\{f(t)\} &= \mathcal{L}\{\sin t \cdot \delta(t-3)\} \\ &= e^{-3s} \cdot f(3) \\ &= e^{-3s} \sin 3 \end{aligned}$$

$$\therefore F(s) = e^{-3s} \cdot \sin 3$$

Ans:—

$$\textcircled{4} \quad f(t) = \cos \pi t \cdot \delta(t-3)$$

$$\begin{aligned} \underline{\text{Now,}} \quad \mathcal{L}\{f(t)\} &= \mathcal{L}\{\cos \pi t \cdot \delta(t-3)\} \\ &= e^{-3s} \cdot f(3) \\ &= e^{-3s} \cdot \cos 3\pi \\ &= e^{-3s} \cdot \cos 3\pi \end{aligned}$$

$$\therefore F(s) = e^{-3s} \cos 3\pi$$

Ans:—

Unit Step
Function.

#Unit Step function (Heaviside's function)

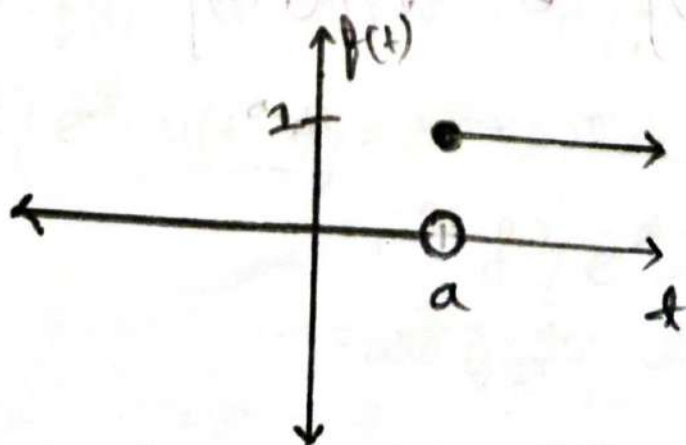
⇒ The unit step function is a typical "engineering function" made to measure for engineering application, which often involve functions (mechanical or electrical driving forces) that are either "off" or "on" multiplying functions with we can produce all sorts of effects.

So, The Unit Step or Heaviside's function is defined as follows:—

① $f(t) = u(t) = \begin{cases} 0; & t < 0 \\ 1; & t \geq 0 \end{cases}$



② $f(t) = u(t-a) = \begin{cases} 0; & t < a \\ 1; & t \geq a \end{cases}$

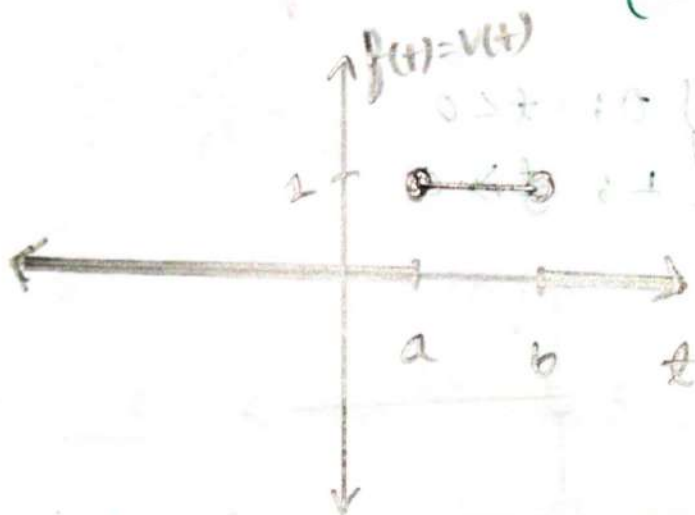


Rectangular Pulse:

⇒ A common situation in a circuit is for a voltage $v(t)$, to be applied at a particular time (let $t=a$) and removed later at (let $t=b$).

Then we write such a situation using unit step function as:—

$$\textcircled{3} \quad f(t) = v(t) = u(t-a) - u(t-b) = \begin{cases} 1 & ; a \leq t < b \\ 0 & ; \text{otherwise} \end{cases}$$



Formulae:

$$\textcircled{1} \quad \mathcal{L}\{u(t-a)\} = e^{-as}/s$$

$$\textcircled{2} \quad \mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \mathcal{L}\{f(t+a)\}$$

② $f(t) = t(0 < t < 2)$

$\Rightarrow f(t) = t[u(t-0) - u(t-2)]$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{t \cdot u(t-0)\} - \mathcal{L}\{t \cdot u(t-2)\}$

$= e^{-0s} \mathcal{L}\{t+0\} - e^{-2s} \mathcal{L}\{t+2\}$

$= \mathcal{L}\{t\} - e^{-2s} \mathcal{L}\{t+2\}$

$= \mathcal{L}\{t\} - e^{-2s} \mathcal{L}\{t\} - e^{-2s} \mathcal{L}\{2\}$

$= \frac{1}{s^2} - e^{-2s} \times \frac{1}{s^2} - e^{-2s} \frac{2}{s}$

Ans: -

③ $f(t) = t-2; (t > 2)$

$\Rightarrow f(t) = (t-2)[u(t-2)]$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{(t-2)u(t-2)\}$

$= e^{-2s} \mathcal{L}\{t+2\}$

$= e^{-2s} \mathcal{L}\{t+2-2\}$

$= e^{-2s} \mathcal{L}\{t\}$

$= e^{-2s} \times \frac{1}{s^2}$

④

$$f(t) = \cos 4t \quad (0 < t < \pi) \Rightarrow f(t) = (1) \cdot f(t) \quad (2)$$

$$\Rightarrow f(t) = \cos 4t [u(t-0) - u(t-\pi)] - (1) \cdot u(t-\pi) \quad (3)$$

$$\Rightarrow f(t) = \cos 4t u(t-0) - \cos 4t u(t-\pi) - (1) \cdot u(t-\pi)$$

$$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 4t u(t-0) - \cos 4t u(t-\pi) - (1) \cdot u(t-\pi)\}$$

$$= e^{-0s} \mathcal{L}\{f(t+0)\} - e^{-\pi s} \mathcal{L}\{f(t+\pi)\}$$

$$= \mathcal{L}\{\cos 4t\} - e^{-\pi s} \mathcal{L}\{\cos 4(t+\pi)\}$$

$$= \frac{s}{s^2 + 4^2} - e^{-\pi s} [\cos 4t \cdot \cos 4\pi - \sin 4t \cdot \sin 4\pi]$$

$$= \frac{s}{s^2 + 4^2} - e^{-\pi s} \cdot \cos 4\pi \times \frac{s}{s^2 + 4^2} + e^{-\pi s} \sin 4\pi \times \frac{4}{s^2 + 4^2}$$

Ans: $\frac{s}{s^2 + 4^2} - e^{-\pi s} \cdot \cos 4\pi \times \frac{s}{s^2 + 4^2} + e^{-\pi s} \sin 4\pi \times \frac{4}{s^2 + 4^2}$

⑤ $f(t) = e^t \quad (0 < t < \frac{\pi}{2})$

$$\therefore f(t) = e^t [u(t-0) - u(t-\frac{\pi}{2})]$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{e^t u(t-0) - e^t u(t-\frac{\pi}{2})\}$$

$$= e^{-0s} \mathcal{L}\{f(t+0)\} - e^{-\frac{\pi}{2}s} \mathcal{L}\{f(t+\frac{\pi}{2})\}$$

$$= \mathcal{L}\{e^t\} - e^{-\frac{\pi}{2}s} \mathcal{L}\{e^{t+\frac{\pi}{2}}\}$$

$$= \frac{1}{s-1} - e^{-\frac{\pi}{2}s} \cdot e^{\frac{\pi}{2}} \times \frac{1}{s-1}$$

$$\textcircled{7} \quad f(t) = e^{-\pi t} (2 < t < 4)$$

$$\Rightarrow f(t) = e^{-\pi t} [u(t-2) - u(t-4)]$$

$$= \mathcal{L}\{f(t)\} = \mathcal{L}\{e^{-\pi t} u(t-2) - e^{-\pi t} u(t-4)\}$$

$$= e^{-2s} \mathcal{L}\{f(t+2)\} - e^{-4s} \mathcal{L}\{f(t+4)\}$$

$$= e^{-2s} \mathcal{L}\{e^{-\pi(t+2)}\} - e^{-4s} \mathcal{L}\{e^{-\pi(t+4)}\}$$

$$= e^{-2s} e^{-2\pi} \mathcal{L}\{e^{-\pi t}\} - e^{-4s} e^{-4\pi} \mathcal{L}\{e^{-\pi t}\}$$

$$= e^{-2s} e^{-2\pi} \cdot \frac{1}{s+\pi} - e^{-4s} e^{-4\pi} \cdot \frac{1}{s+\pi}$$

Ans:-

$$\textcircled{10} \quad \sinh t \quad (0 < t < 2)$$

$$\therefore f(t) = \sinh t \quad (0 < t < 2)$$

$$f(t) = \sinh t [u(t-0) - u(t-2)]$$

$$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{\sinh t u(t-0) - \sinh t u(t-2)\}$$

$$= e^{-0s} \mathcal{L}\{\sinh t\} - e^{-2s} \mathcal{L}\{\sinh(t+2)\}$$

$$= \frac{1}{s^2-1} - e^{-2s} \mathcal{L}\{(\sin t \cdot \cos 2 + \cos t \cdot \sin 2)\}$$

$$= \frac{1}{s^2-1} - e^{-2s} \cdot \cos 2 \cdot \frac{1}{s^2+1} - e^{-2s} \sin 2 \cdot \frac{s}{s^2+1}$$

Ans:-

$$(11) f(t) = \sin t \left(-\frac{\pi}{2} < t < \pi \right)$$

$$\Rightarrow f(t) = \sin t \left\{ u\left(t + \frac{\pi}{2}\right) - u(t - \pi) \right\}$$

$$\begin{aligned} \therefore \mathcal{L}\{f(t)\} &= \mathcal{L}\left\{ \sin t \left[u\left(t + \frac{\pi}{2}\right) - u(t - \pi) \right] \right\} \\ &= e^{-\frac{\pi}{2}s} \mathcal{L}\{f(t + \frac{\pi}{2})\} - e^{-\pi s} \mathcal{L}\{f(t + \pi)\} \\ &= e^{-\frac{\pi}{2}s} \mathcal{L}\left\{ \sin\left(t + \frac{\pi}{2}\right) \right\} - e^{-\pi s} \mathcal{L}\left\{ \sin(t + \pi) \right\} \end{aligned}$$

$$\begin{aligned} &= e^{-\frac{\pi}{2}s} \mathcal{L}\left\{ \sin t \cdot \cos \frac{\pi}{2} + \cos t \cdot \sin \frac{\pi}{2} \right\} - e^{-\pi s} \mathcal{L}\left\{ \sin t \cdot \cos \pi + \cos t \cdot \sin \pi \right\} \\ &= e^{-\frac{\pi}{2}s} \cos \frac{\pi}{2} \mathcal{L}\left\{ \frac{1}{s^2+1} \right\} + e^{-\frac{\pi}{2}s} \sin \frac{\pi}{2} \mathcal{L}\left\{ \frac{s}{s^2+1} \right\} - e^{-\pi s} \cos \pi \mathcal{L}\left\{ \frac{1}{s^2+1} \right\} - e^{-\pi s} \sin \pi \mathcal{L}\left\{ \frac{s}{s^2+1} \right\} \\ &= 0 + e^{-\frac{\pi}{2}s} \cdot \frac{s}{s^2+1} + e^{-\frac{\pi}{2}s} \cdot \frac{1}{s^2+1} + 0 \end{aligned}$$

$$= e^{-\frac{\pi}{2}s} \cdot \frac{s}{s^2+1} + e^{-\pi s} \cdot \frac{1}{s^2+1}$$

Ans:-

Page: 251

(24) $f(t) = 16t^2 u(t - \frac{1}{4})$

$$\begin{aligned} \Rightarrow \mathcal{L}\{f(t)\} &= \mathcal{L}\{16t^2 u(t - \frac{1}{4})\} \\ &= e^{-\frac{1}{4}s} \cdot \mathcal{L}\{f(t + \frac{1}{4})\} \\ &= e^{-\frac{1}{4}s} \cdot \mathcal{L}\{16(t + \frac{1}{4})^2\} \\ &= 16e^{-\frac{1}{4}s} \mathcal{L}\{t^2 + 2t \cdot \frac{1}{4} + (\frac{1}{4})^2\} \\ &= 16e^{-\frac{1}{4}s} \left(\frac{2}{s^3} + \frac{1}{2} \cdot \frac{1}{s^2} + \frac{1}{16s} \right) \\ &= e^{-\frac{1}{4}s} \left(\frac{32}{s^3} + \frac{8}{s^2} + \frac{1}{s} \right) \end{aligned}$$

Ans: -

(25) $f(t) = e^{t/2} \cdot u(t-3)$

$$\begin{aligned} \Rightarrow \mathcal{L}\{f(t)\} &= \mathcal{L}\{e^{t/2} \cdot u(t-3)\} \\ &= e^{-\frac{3}{2}s} \cdot \mathcal{L}\{e^{\frac{t+9}{2}}\} \\ &= e^{-\frac{3}{2}s} \cdot \mathcal{L}\{e^{\frac{t}{2} + \frac{9}{2}}\} \\ &= e^{-\frac{3}{2}s} \cdot e^{\frac{9}{2}} \mathcal{L}\{e^{t/2}\} \\ &= e^{-\frac{3}{2}s + 9/2} \cdot \frac{1}{s - \frac{1}{2}} \end{aligned}$$

Ans: -

Q:- Find the Laplace Transformation of the Unit step function?

① $f(t) = (t+2)^2 u(t-1)$

We know:-

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \mathcal{L}\{f(t+a)\}$$

Now, $\Rightarrow \mathcal{L}\{(t+2)^2 \cdot u(t-1)\}$

$$= e^{-s} \mathcal{L}\{f(t+1)\}$$

$$\because f(t) = (t+2)^2$$

$$= e^{-s} \mathcal{L}\{(t+1+2)^2\}$$

$$= e^{-s} \mathcal{L}\{(t+3)^2\}$$

$$= e^{-s} \mathcal{L}\{t^2 + 6t + 9\}$$

$$= e^{-s} \left\{ \frac{2!}{s^3} + \frac{6}{s^2} + \frac{9}{s} \right\}$$

$$= e^{-s} \left(\frac{2}{s^3} + \frac{6}{s^2} + \frac{9}{s} \right)$$

Ans:-

② $f(t) = e^{-2t} \cdot u(t-3)$

We know:-

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \mathcal{L}\{f(t+a)\}$$

Now,

$$\Rightarrow \mathcal{L}\{e^{-2t} \cdot u(t-3)\} = e^{-3s} \mathcal{L}\{f(t+3)\}$$

$$= e^{-3s} \mathcal{L}\{e^{-2(t+3)}\}$$

$$= e^{-3s} \cdot \mathcal{L}\{e^{-2t} \cdot e^{-6}\}$$

$$= e^{-3s} \cdot e^{-6} \cdot \mathcal{L}\{e^{-2t}\}$$

$$= e^{-3s-6} \times \frac{1}{s+2}$$

$$= \frac{e^{-3s-6}}{s+2}$$

Ans:

Q3. $4 \cos t \cdot u(t-\pi) = f(t)$

We know:-
 $\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{f(t+a)\}$

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{f(t+a)\}$$

Now,

$$\mathcal{L}\{4 \cos t \cdot u(t-\pi)\} = e^{-\pi s} \cdot \mathcal{L}\{f(t+\pi)\}$$

$$= e^{-\pi s} \cdot \mathcal{L}\{4 \cos(t+\pi)\} \quad [\because \cos(a+b) = \cos a \cdot \cos b - \sin a \cdot \sin b]$$

$$= 4e^{-\pi s} \mathcal{L}\{\cos t \cdot \cos \pi - \sin t \cdot \sin \pi\}$$

$$= 4e^{-\pi s} \cos \pi \times \frac{s}{s^2+1} - 4e^{-\pi s} \sin \pi \times \frac{1}{s^2+1}$$

$$= \frac{4e^{-\pi s} \cos \pi s - 4e^{-\pi s} \sin \pi}{s^2+1}$$

$$= \frac{-4e^{-\pi s} s - 4e^{-\pi s} \times 0}{s^2+1}$$

$$= \frac{-4se^{-\pi s}}{s^2+1}$$

Ans:

④ $16t^2 \cdot u(t - \frac{1}{4}) = f(t)$

we know—
now

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{f(t+a)\}$$

Now

$$\Rightarrow \mathcal{L}\{16t^2 \cdot u(t - \frac{1}{4})\} \\ = e^{-\frac{1}{4}s} \mathcal{L}\{f(t + \frac{1}{4})\}$$

$$= e^{-\frac{1}{4}s} \mathcal{L}\{16(t + \frac{1}{4})^2\}$$

$$= e^{-\frac{1}{4}s} \mathcal{L}\{16t^2 + 8t + 1\}$$

$$= e^{-\frac{1}{4}s} \left(\frac{32}{s^3} + \frac{8}{s^2} + \frac{1}{s} \right)$$

Ans.—

⑤ $f(t) = e^{\frac{t}{2}} \cdot u(t-3)$

we know—
now

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{f(t+a)\}$$

Now

$$\Rightarrow \mathcal{L}\{e^{\frac{t}{2}} \cdot u(t-3)\} = e^{-3s} \cdot \mathcal{L}\{f(t+3)\}$$

$$= e^{-3s} \cdot \mathcal{L}\{e^{\frac{t+3}{2}}\}$$

$$= e^{-3s} \cdot \left(e^{\frac{3}{2}} \right) \mathcal{L}\{e^{\frac{t}{2}}\}$$

$$= e^{-3s + \frac{3}{2}} \cdot \frac{1}{s - \frac{1}{2}} = \frac{e^{-3s + \frac{3}{2}}}{s - \frac{1}{2}}$$

Ans.—

⑥ $f(t) = u(t - 2\pi) \sin t$

We know—
now

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{f(t+a)\}$$

Now,

$$\begin{aligned} &\Rightarrow \mathcal{L}\{u(t-2\pi) \sin t\} \\ &= e^{-2\pi s} \mathcal{L}\{f(t+2\pi)\} \\ &= e^{-2\pi s} \mathcal{L}\{\sin(t+2\pi)\} \quad [\because \sin(a+b) = \sin a \cos b + \cos a \sin b] \\ &= e^{-2\pi s} \mathcal{L}\{\sin t \cdot \cos 2\pi + \cos t \sin 2\pi\} \\ &= e^{-2\pi s} \cdot \cos 2\pi \times \frac{1}{s^2+1} + e^{-2\pi s} \cdot \sin 2\pi \times \frac{s}{s^2+1} \\ &= \frac{e^{-2\pi s}}{s^2+1} + 0 \\ &= \frac{e^{-2\pi s}}{s^2+1} \end{aligned}$$

Ans:-

⑦ $f(t) = t u(t-1)$

We know—
now

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{f(t+a)\}$$

$$\begin{aligned} \text{Now, } \mathcal{L}\{t u(t-1)\} &= e^{-s} \cdot \mathcal{L}\{f(t+1)\} \\ &= e^{-s} \mathcal{L}\{t+1\} \end{aligned}$$

$$\begin{aligned}
 &= e^{-s} \mathcal{L}\{t+1\} \\
 &= e^{-s} \left(\frac{1!}{s^2} + \frac{1}{s} \right) \\
 &= \frac{e^{-s}}{s} + \frac{e^{-s}}{s^2}
 \end{aligned}$$

Ans: →

⑧ $f(t) = (t-1)u(t-1)$

We know: —

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \mathcal{L}\{f(t+a)\}$$

Now,

$$\begin{aligned}
 \mathcal{L}\{(t-1) \cdot u(t-1)\} &= e^{-s} \mathcal{L}\{f(t+1)\} \\
 &= e^{-s} \mathcal{L}\{t+1-1\} \\
 &= e^{-s} \mathcal{L}\{t\} \\
 &= e^{-s} \frac{1!}{s^2} \\
 &= \frac{e^{-s}}{s^2}
 \end{aligned}$$

Ans: —

⑨ $f(t) = (t-1)^2 u(t-1)$

We know: —

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \mathcal{L}\{f(t+a)\}$$

Now,

$$\begin{aligned}
 \mathcal{L}\{(t-1)^2 \cdot u(t-1)\} &= e^{-s} \mathcal{L}\{f(t+1)\} \\
 &= e^{-s} \mathcal{L}\{(t-1+1)^2\}
 \end{aligned}$$

$$= e^{-s} \times \frac{2!}{s^3}$$

$$= \frac{2e^{-s}}{s^3}$$

Ans:-

(10.) $f(t) = e^{-3t} \cdot u(t-4)$

We know:-

$$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{f(t+a)\}$$

Now:- $\mathcal{L}\{e^{-3t} \cdot u(t-4)\} = e^{-4s} \mathcal{L}\{f(t+4)\}$

$$= e^{-4s} \mathcal{L}\{e^{-3(t+4)}\}$$

$$= e^{-4s} \mathcal{L}\{e^{-3t} \cdot e^{-12}\}$$

$$= e^{-4s} \cdot e^{-12} \mathcal{L}\{e^{-3t}\}$$

$$= e^{-4s} \cdot \frac{e^{-12} \cdot 1}{s+3}$$

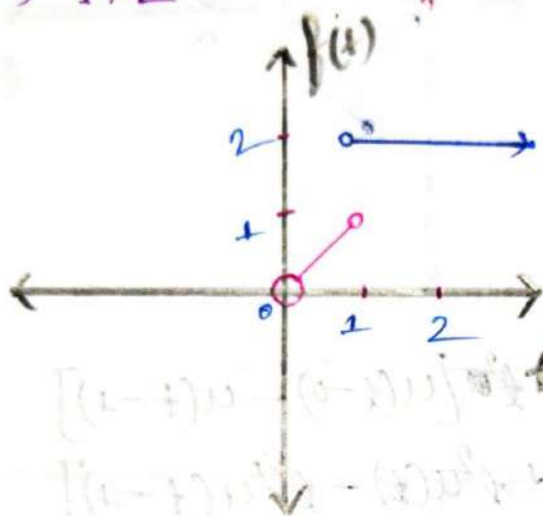
$$= \frac{e^{-4s} \cdot e^{-12}}{s+3}$$

$$= \frac{e^{-4s} \cdot e^{-12}}{s+3}$$

Ans:-

Express $f(t)$ in terms of unit
Step function and find its Laplace
Transformation! —————

① $f(t) = \begin{cases} t & ; 0 < t < 1 \\ 2 & ; t > 1 \end{cases}$



$$\Rightarrow f(t) = 2u(t-1) + t[u(t-0) - u(t-1)]$$

$$= 2u(t-1) + t^2u - u t(t-1)$$

$$f(t) = u(t-1)\{2-t\} + t \cdot u(t)$$

$$\therefore F(s) = \mathcal{L}\{(2-t)u(t-1)\} + \mathcal{L}\{t \cdot u(t)\}$$

$$= e^{-s} \mathcal{L}\{f(t+1)\} + e^0 \mathcal{L}\{f(t)\}$$

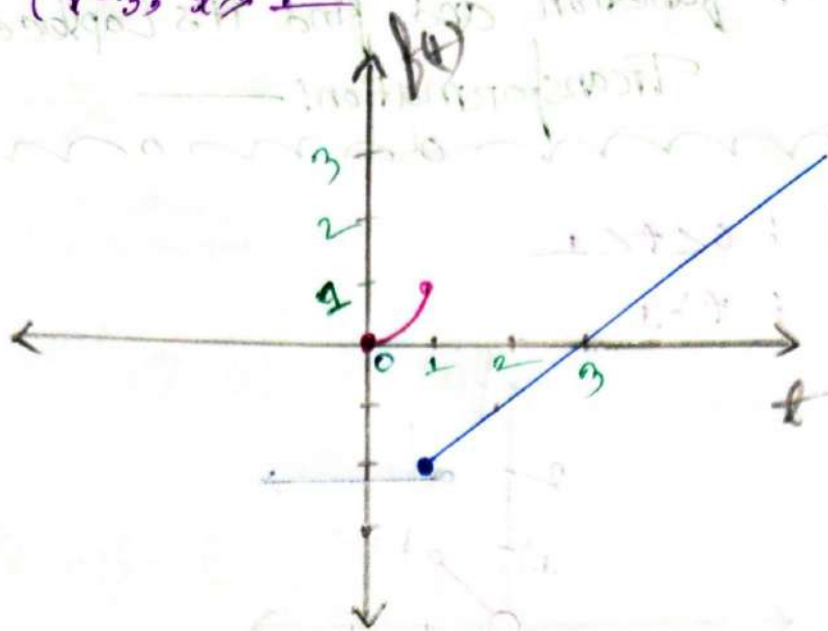
$$= e^{-s} \mathcal{L}\{(2-t+1)\} + \mathcal{L}\{t\}$$

$$= e^{-s} \times \left(\frac{-1}{s^2} + \frac{1}{s} \right) + \frac{1}{s^2}$$

$$= \frac{-e^{-s} + s e^{-s} + 1}{s^2}$$

Ans:

② $f(t) = \begin{cases} t^2; & 0 \leq t < 1 \\ t-3; & t \geq 1 \end{cases}$



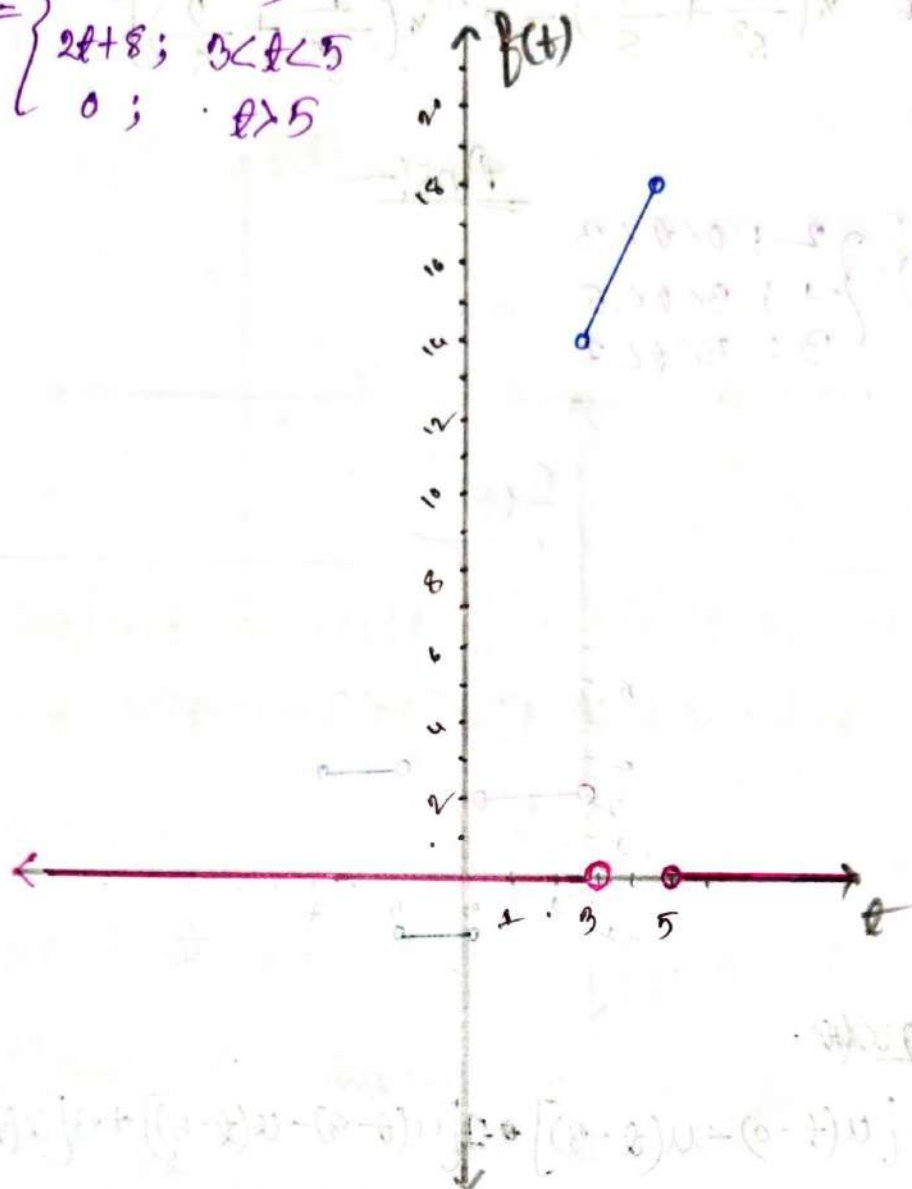
$$\begin{aligned} \therefore f(t) &= (t-3)[u(t-1)] + t^2 [u(t-0) - u(t-1)] \\ &= (t-3)[u(t-1)] + t^2 u(t) - t^2 u(t-1) \\ &= u(t-1)\{t-3\} - t^2 + t^2 u(t) \end{aligned}$$

Now,

$$\begin{aligned} F(s) &= \mathcal{L}\{(t-3-t^2)u(t-1)\} + \mathcal{L}\{t^2 u(t)\} \\ &= e^{-s} \mathcal{L}\{f(t+1)\} + e^0 \mathcal{L}\{f(t+0)\} \\ &= e^{-s} \mathcal{L}\{(t+1)-(t+1)^2-3\} + \mathcal{L}\{t^2\} \\ &= e^{-s} \mathcal{L}\{t+1-3-(t^2+2t+1)\} + \frac{2}{s^3} \\ &= e^{-s} \mathcal{L}\{t+1-3-t^2-2t-1\} + \frac{2}{s^3} \\ &= e^{-s} \mathcal{L}\{-t-3-t^2\} + \frac{2}{s^3} \\ &= \frac{2}{s^3} - \left(\frac{\bar{e}^s}{s^2} + \frac{3\bar{e}^s}{s} + \frac{2\bar{e}^s}{s^3} \right) \end{aligned}$$

(Answer)

Q. $f(t) = \begin{cases} 0; & t < 3 \\ 2t+8; & 3 < t < 5 \\ 0; & t > 5 \end{cases}$



$$\therefore f(t) = 0 \times u(t-5) + 0(t-3) + (2t+8)[u(t-3) - u(t-5)]$$

$$= (2t+8)[u(t-3)] - (2t+8)[u(t-5)]$$

Now,

$$F(s) = \int_0^\infty (2t+8) u(t-3) dt - \int_0^\infty (2t+8) u(t-5) dt$$

$$= e^{-3s} \int_0^\infty f(t+3) dt - e^{-5s} \int_0^\infty f(t+5) dt$$

$$= e^{-3s} \int_0^\infty 2(t+3)+8 dt - e^{-5s} \int_0^\infty 2(t+5)+8 dt$$

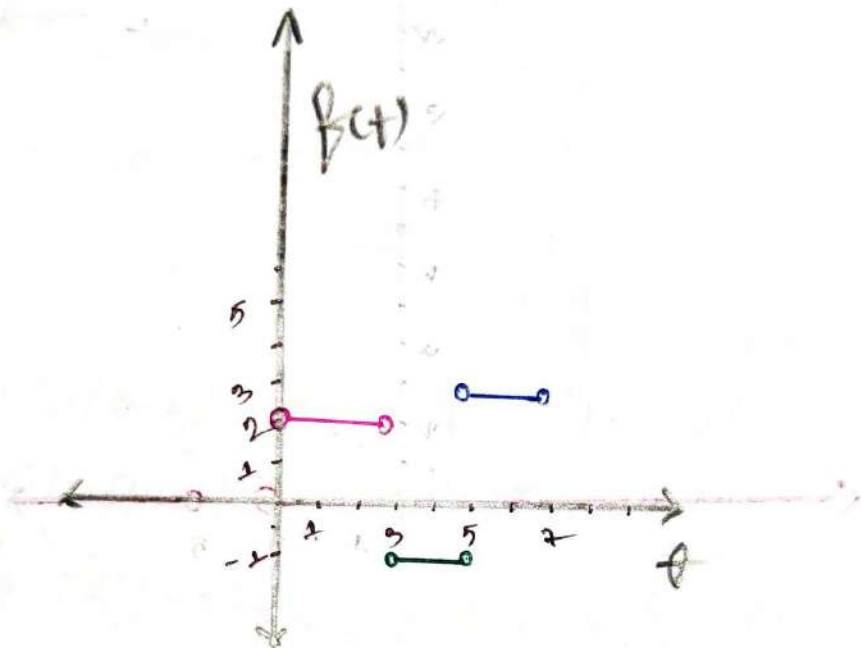
$$= e^{-3s} \int_0^\infty 2t+14 dt - e^{-5s} \int_0^\infty 2t+18 dt$$

$$= 2[e^{-3s} \int_0^\infty t+7 dt - e^{-5s} \int_0^\infty t+9 dt]$$

$$= 2 \left[e^{-3s} \times \left(\frac{1}{s^2} + \frac{7}{s} \right) - e^{-5s} \times \left(\frac{1}{s^2} + \frac{2}{s} \right) \right]$$

Ans:—

④ $f(t) = \begin{cases} 2; & 0 < t < 3 \\ -1; & 3 < t < 5 \\ 3; & 5 < t < 7 \end{cases}$



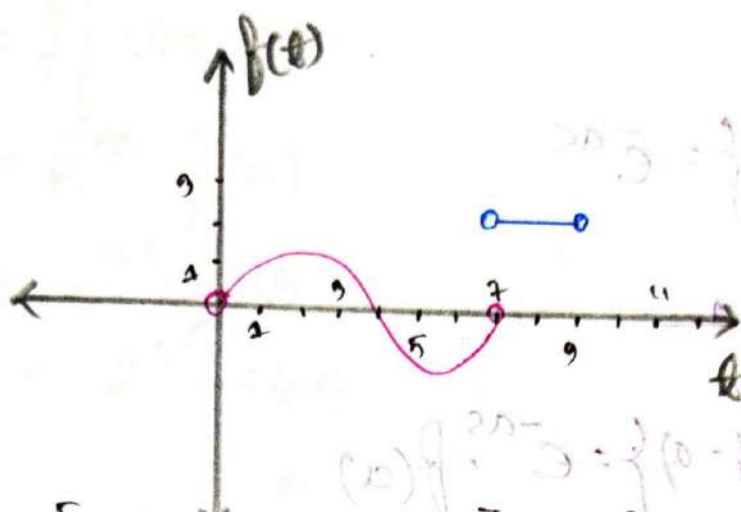
$$\begin{aligned} \therefore f(t) &= 2[u(t-0) - u(t-3)] - 1[u(t-3) - u(t-5)] + 3[u(t-5) - u(t-7)] \\ &= 2u(t) - 2u(t-3) - u(t-3) + u(t-5) + 3u(t-5) - 3u(t-7) \\ &= 2u(t) - 3u(t-3) + 4u(t-5) - 3u(t-7) \end{aligned}$$

Now,

$$\begin{aligned} F(s) &= 2 \mathcal{L}\{1 \cdot u(t)\} - 3 \mathcal{L}\{1 \cdot u(t-3)\} + 4 \mathcal{L}\{1 \cdot u(t-5)\} - 3 \mathcal{L}\{1 \cdot u(t-7)\} \\ &= 2 \times e^0 \mathcal{L}\{f(t)\} - 3 e^{-3s} \mathcal{L}\{f(t+3)\} + 4 e^{-5s} \mathcal{L}\{f(t+5)\} - 3 e^{-7s} \mathcal{L}\{f(t+7)\} \\ &= 2 \mathcal{L}\{1\} - 3 e^{-3s} \mathcal{L}\{1\} + 4 e^{-5s} \mathcal{L}\{1\} - 3 e^{-7s} \mathcal{L}\{1\} \\ &= \frac{2}{s} - \frac{3e^{-3s}}{s} + \frac{4e^{-5s}}{s} - \frac{3e^{-7s}}{s} \end{aligned}$$

Ans:—

$$\textcircled{5} f(t) = \begin{cases} \sin t & ; 0 < t < 7 \\ 3 & ; 7 < t < \infty \end{cases}$$



$$\therefore f(t) = \sin t [u(t-0) - u(t-7)] + 3[u(t-7) - u(t-\infty)]$$

$$= \sin t \cdot u(t-0) - \sin t \cdot u(t-7) + 3u(t-7) - 3u(t-\infty)$$

now

$$F(s) = \mathcal{L}\{\sin t \cdot u(t-0)\} - \mathcal{L}\{\sin t \cdot u(t-7)\} + 3\mathcal{L}\{u(t-7)\} - 3\mathcal{L}\{u(t-\infty)\}$$

$$= e^0 \mathcal{L}\{f(t+0)\} - e^{-7s} \mathcal{L}\{f(t+7)\} + 3e^{-7s} \mathcal{L}\{f(t+7)\} - 3e^{-\infty s} \mathcal{L}\{f(t+\infty)\}$$

$$= \mathcal{L}\{\sin t\} - e^{-7s} \mathcal{L}\{\sin(t+7)\} + 3e^{-7s} \mathcal{L}\{1\} - 3e^{-\infty s} \mathcal{L}\{1\}$$

$$= \frac{1}{s^2+1} - e^{-7s} \mathcal{L}\{\sin t \cdot \cos 7 + \cos t \cdot \sin 7\} + 3e^{-7s} \frac{1}{s} - 3e^{-\infty s} \frac{1}{s}$$

$$= \frac{1}{s^2+1} + \frac{3e^{-7s}}{s} - \frac{3e^{-\infty s}}{s} - e^{-7s} \cdot \cos 7 \mathcal{L}\{\sin t\} - e^{-7s} \cdot \sin 7 \mathcal{L}\{\cos t\}$$

$$= \frac{1}{s^2+1} + \frac{3e^{-7s}}{s} - \frac{3e^{-\infty s}}{s} - \frac{e^{-7s} \cdot \cos 7}{s^2+1} - \frac{e^{-7s} \cdot \sin 7}{s^2+1}$$

Ans. —

Inverse Unit
Step Function

Inverse Unit Step:-

Formula:-

$$\textcircled{1} \mathcal{L}^{-1}\{e^{-as} \cdot F(s)\} = f(t-a) \cdot u(t-a) \text{ where,}$$

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

Ex:-

$$F(s) = \frac{e^{-3s}}{(s-1)^3}$$

$$\Rightarrow \text{Here, } F_1(s) = \frac{1}{(s-1)^3}$$

$$\therefore \mathcal{L}^{-1}\{F_1(s)\} = e^t \cdot \frac{t^2}{2} = f_1(t)$$

$$\therefore \mathcal{L}^{-1}\{F(s)\} = f(t-3) \cdot u(t-3)$$

$$= e^{t-3} \cdot \frac{(t-3)^2}{2} \cdot u(t-3)$$

$$= \frac{e^{t-3}(t-3)^2}{2} \cdot u(t-3)$$

Ans:-

Page:- 2021-

$$\textcircled{22} f(s) = \frac{e^{-3s}}{(s-1)^2}$$

$$\therefore \text{Here, } F_1(s) = \frac{1}{(s-1)^2}$$

$$\mathcal{L}^{-1}\{F_1(s)\} = e^t \cdot \frac{t}{1!} = f_1(t)$$

$$\therefore \mathcal{L}^{-1}\{F(s)\} = f(t-3) \cdot u(t-3)$$

$$= e^{t-3} \cdot (t-3) \cdot u(t-3)$$

Ans.

$$(13) \quad F(s) = \frac{6(1 - e^{-\pi s})}{s^2 + 9}$$

Here, $F_1(s) = \frac{6}{s^2 + 9}$

$$= 6 \left(\frac{1}{s^2 + 3^2} \right)$$

$$\mathcal{L}^{-1}\{F_1(s)\} = 6 \times \mathcal{L}^{-1}\left(\frac{1}{s^2 + 3^2}\right)$$

$$= 6 \times \frac{1}{3} \sin 3t$$

$$= 2 \sin 3t = f(t)$$

$$\therefore \mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{6}{s^2 + 9}\right\} - \mathcal{L}^{-1}\left\{\frac{6e^{-\pi s}}{s^2 + 9}\right\}$$

$$= 2 \sin 3t - 2 \sin 3(t - \pi) u(t - \pi)$$

$$(14) \quad F(s) = \frac{4(e^{-2s} - 2e^{-5s})}{s}$$

$$\Rightarrow F(s) = \frac{4e^{-2s}}{s} - \frac{8e^{-5s}}{s}$$

$$\therefore \mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left(\frac{4e^{-2s}}{s} - \frac{8e^{-5s}}{s}\right)$$

$$f(t) = 4u(t - 2) - 8u(t - 5)$$

Ans -

$$(15) \quad F(s) = \frac{e^{-3s}}{s^4}$$

$$\text{Here } F_1(s) = \frac{1}{s^4}$$

$$\therefore \mathcal{L}^{-1}\{F_1(s)\} = \mathcal{L}^{-1}\left(\frac{1}{s^4}\right) = \frac{t^3}{3!} = \frac{1}{6} t^3 = f(t)$$

$$\begin{aligned} \text{Now, } \mathcal{L}^{-1}\{F(s)\} &= f(t-3) \cdot u(t-3) \\ &= \frac{1}{6} (t-3)^3 \cdot u(t-3) \\ &= \frac{1}{6} (t-3)^3 \cdot u(t-3) \end{aligned}$$

Ans:-

$$(16) \quad F(s) = 2(e^{-s} - 3e^{-3s})/s^2 - 4$$

$$\Rightarrow F(s) = \frac{2e^{-s} - 6e^{-3s}}{s^2 - 4}$$

$$\Rightarrow F(s) = \frac{2e^{-s}}{s^2 - 2^2} - \frac{6e^{-3s}}{s^2 - 4}$$

$$\text{here, } F_1(s) = \frac{2}{s^2 - 2^2}$$

$$\therefore \mathcal{L}^{-1}\{F_1(s)\} = \mathcal{L}^{-1}\left(\frac{2}{s^2 - 2^2}\right)$$

$$= \sinh 2t$$

$$= f_1(t)$$

$$\text{here, } F_2(s) = \frac{6}{s^2 - 2^2}$$

$$\mathcal{L}^{-1}\{F_2(s)\} = \mathcal{L}^{-1}\left(\frac{6}{s^2 - 2^2}\right)$$

$$= 6 \times \frac{1}{2} \sinh 2t$$

$$= 3 \sinh 2t$$

$$= f_2(t)$$

$$\therefore \mathcal{L}^{-1}\{F(s)\} = f(t-1) \cdot u(t-1) - f(t-3) \cdot u(t-3)$$

$$= \sinh 2(t-1) \cdot u(t-1) - 3 \sinh 2(t-3) \cdot u(t-3)$$

Ans:-

$$(13) \quad F(s) = \frac{(1 + e^{2\pi i(s+1)}) (s+1)}{((s+1)^2 + 1)}$$

$$\Rightarrow F(s) = \frac{(s+1)e^{2\pi i(s+1)}}{(s+1)^2 + 1} + \frac{s+1}{(s+1)^2 + 1}$$

Now, $F_1(s) = \frac{(s+1)e^{-2\pi i(s+1)}}{(s+1)^2 + 1}$

$$\therefore \mathcal{L}^{-1}\{F_1(s)\} = \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2 + 1}\right\} = e^{-t} \cdot \cos t$$

$$F_2(s) = \frac{s+1}{(s+1)^2 + 1}$$

$$\therefore \mathcal{L}^{-1}\{F_2(s)\} = \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2 + 1}\right\} = e^{-t} \cdot \cos t$$

So, $\mathcal{L}^{-1}\{F(s)\} = f(t-a) \cdot u(t-a)$

$$= f(t-2\pi) \cdot u(t-2\pi)$$

$$= e^{-2\pi} \cdot e^{-(t-2\pi)} \cdot \cos(t-2\pi) \cdot u(t-2\pi)$$

$$= e^{-2\pi} \cdot e^{-2\pi i} \cdot \cos(t-2\pi) \cdot u(t-2\pi)$$

$$\therefore \mathcal{L}^{-1}\{F(s)\} = e^{-2\pi} \cdot e^{-2\pi i} \cdot \cos(t-2\pi) \cdot u(t-2\pi) + e^{-t} \cdot \cos t$$

$$= e^{-t} \cdot \cos(t-2\pi) \cdot u(t-2\pi) + e^{-t} \cdot \cos t$$

Ans

Page 230

$$(3) \quad y'' + 4y = \delta(t - \pi) \quad ; \quad y(0) = 8, y'(0) = 0$$

$$\Rightarrow \mathcal{L}\{y'' + 4y\} = \mathcal{L}\{\delta(t - \pi)\}$$

$$\Rightarrow [\mathcal{L}\{y\} s^2 - sy(0) - y'(0)] + 4\mathcal{L}\{y\} = e^{-\pi s}$$

$$\Rightarrow [\mathcal{L}\{y\} (s^2 + 4)] = 8s + e^{-\pi s}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{8s + e^{-\pi s}}{s^2 + 4}$$

$$\therefore y(t) = \mathcal{L}^{-1} \left\{ \frac{8s}{s^2 + 4} + \frac{e^{-\pi s}}{s^2 + 4} \right\}$$

$$= 8 \cos 2t + \frac{1}{2} \sin 2(t - \pi) \cdot u(t - \pi)$$

Ans: -

$$(4) \quad y'' + 16y = 4\delta(t - 3\pi), \quad y(0) = 2, y'(0) = 0$$

$$\Rightarrow [\mathcal{L}\{y\} s^2 - sy(0) - y'(0)] + 16\mathcal{L}\{y\} = 4e^{-3\pi s}$$

$$\Rightarrow (s^2 + 16)\mathcal{L}\{y\} - 2s = 4e^{-3\pi s}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{2s + 4e^{-3\pi s}}{s^2 + 16}$$

$$\therefore \mathcal{L}^{-1}\{\mathcal{L}\{y\}\} = \mathcal{L}^{-1} \left\{ \frac{2s}{s^2 + 16} \right\} + \mathcal{L}^{-1} \left\{ \frac{4e^{-3\pi s}}{s^2 + 16} \right\}$$

$$= 2 \cos 4t + \sin 4(t - 3\pi) \cdot u(t - 3\pi)$$

Ans: -

$$\textcircled{5} \quad y'' + y = \delta(t - \pi) - \delta(t - 2\pi) : y(0) = 0, y'(0) = 1$$

$$\Rightarrow \mathcal{L}\{y'' + y\} = \mathcal{L}\{\delta(t - \pi) - \delta(t - 2\pi)\}$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] + \mathcal{L}\{y\} = e^{-\pi s} - e^{-2\pi s}$$

$$\Rightarrow (s^2 + 1)\mathcal{L}\{y\} = e^{-\pi s} - e^{-2\pi s} - 1$$

$$\therefore \mathcal{L}\{y\} = \frac{e^{-\pi s} - e^{-2\pi s} - 1}{s^2 + 1}$$

$$\therefore y(t) = \mathcal{L}^{-1}\left\{\frac{e^{-\pi s} - e^{-2\pi s} - 1}{s^2 + 1}\right\}$$

$$= \sin(t - \pi) \cdot u(t - \pi) - \sin(t - 2\pi) \cdot u(t - 2\pi) - \sin t$$

$$= \sin t \cdot u(t - \pi) - \sin t \cdot u(t - 2\pi) - \sin t$$

$$= \sin t \{u(t - \pi) - u(t - 2\pi) - 1\}$$

Ans:-

$$\textcircled{6} \quad y'' + 4y' + 5y = \delta(t - 1), y(0) = 0, y'(0) = 3$$

$$\Rightarrow \mathcal{L}\{y'' + 4y' + 5y\} = \mathcal{L}\{\delta(t - 1)\}$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] + 4[s \mathcal{L}\{y\} - y(0)] + 5\mathcal{L}\{y\} = e^{-s}$$

$$\Rightarrow s^2 \mathcal{L}\{y\} - 3 + 4s \mathcal{L}\{y\} + 5\mathcal{L}\{y\} = e^{-s}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{e^{-s} + 3}{s^2 + 4s + 5}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{e^{-s} + 3}{(s+2)^2 + 1}$$

$$\therefore y(t) = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{(s+2)^2 + 1} + \frac{3}{(s+2)^2 + 1}\right\}$$

$$= e^{-t} \cdot \sin(t-1) \cdot u(t-1) + 3e^{-2t} \cdot \sin t$$

Ans:-

$$(8) \quad y'' + 3y' + 2y = 10(\sin t + 8(t-1)), \quad y(0) = 1, \quad y'(0) = -1$$

$$\Rightarrow \mathcal{L}\{y'' + 3y' + 2y\} = \mathcal{L}\{10\sin t + 108(t-1)\}$$

$$\Rightarrow [s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] + 3[s \mathcal{L}\{y\} - y(0)] + 2\mathcal{L}\{y\} = \frac{10}{s^2+1} + 10e^{-s}$$

$$\Rightarrow s^2 \mathcal{L}\{y\} - s + 1 + 3s \mathcal{L}\{y\} - 3 + 2\mathcal{L}\{y\} = \frac{10}{s^2+1} + 10e^{-s}$$

$$\Rightarrow (s^2 + 3s + 2) \mathcal{L}\{y\} = s + 2 + \frac{10}{s^2+1} + 10e^{-s}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{s^3 + 2s^2 + s + 2 + 10 + (s^2+1)10e^{-s}}{(s^2+1)(s+2)(s+1)} = \frac{A}{(s+2)} + \frac{B}{(s+1)} + \frac{Cs+D}{(s^2+1)} \quad (1)$$

multiply $(s^2+1)(s+2)(s+1)$ both sides.

$$s^3 + 2s^2 + s + 12 + (s^2+1)10e^{-s} = A(s+1)(s^2+1) + B(s+2)(s^2+1) + (Cs+D)(s+2)(s+1) \quad (10)$$

if $s = -1$

$$\therefore (-1)^3 + 2(-1)^2 + (-1) + 12 + 10e^{-(-1)} = B(-1+2)(1+1)$$

$$\therefore B = 33.182$$

if $s = -2$

$$A = -75.80$$

now $s^3 + 2s^2 + s + 12 + (s^2+1)10e^{-s} = A(s^3 + s + s^2 + 1) + B(s^3 + s + 2s^2 + 2) + (Cs + D)(s^2 + 2s + 2)$

$$\Rightarrow s^3 + 2s^2 + s + 12 + 10e^{-s} = As^3 + As + As^2 + A + Bs^3 + Bs + 2Bs^2 + 2B + Cs^3 + Bs^2 + 2Cs + D + 2Ds + 2D$$

Equating like terms:

$$A + B + 2D = 12$$

$$2D = \frac{12 - A - B}{2} = \frac{12 + 75.80 - 33.182}{2} = 27.309$$

$$As^3 + Bs^3 + Cs^3 = s^3$$

$$0 = A + B + 2D = 27.309 - 33.182 = -42.708$$

put it in eq (2)

$$\mathcal{L}\{y\} = \frac{-75.80}{s+2} + \frac{33.182}{s+1} + \frac{42.708s + 27.754}{(s^2+1)}$$

$$y(t) = -75.80 e^{-2t} + 33.182 e^{-t} + 27.754 e^{\sin t} + 42.708 \cos t.$$

Ans: -

#Complex
Variables!

Complex Variables:-

$z = a + ib \rightarrow$ a complex number - c

where

$\text{Re}(z) \rightarrow$ real part of z is a

$\text{Im}(z) \rightarrow$ Imaginary part of z is b

for complex Numbers:-

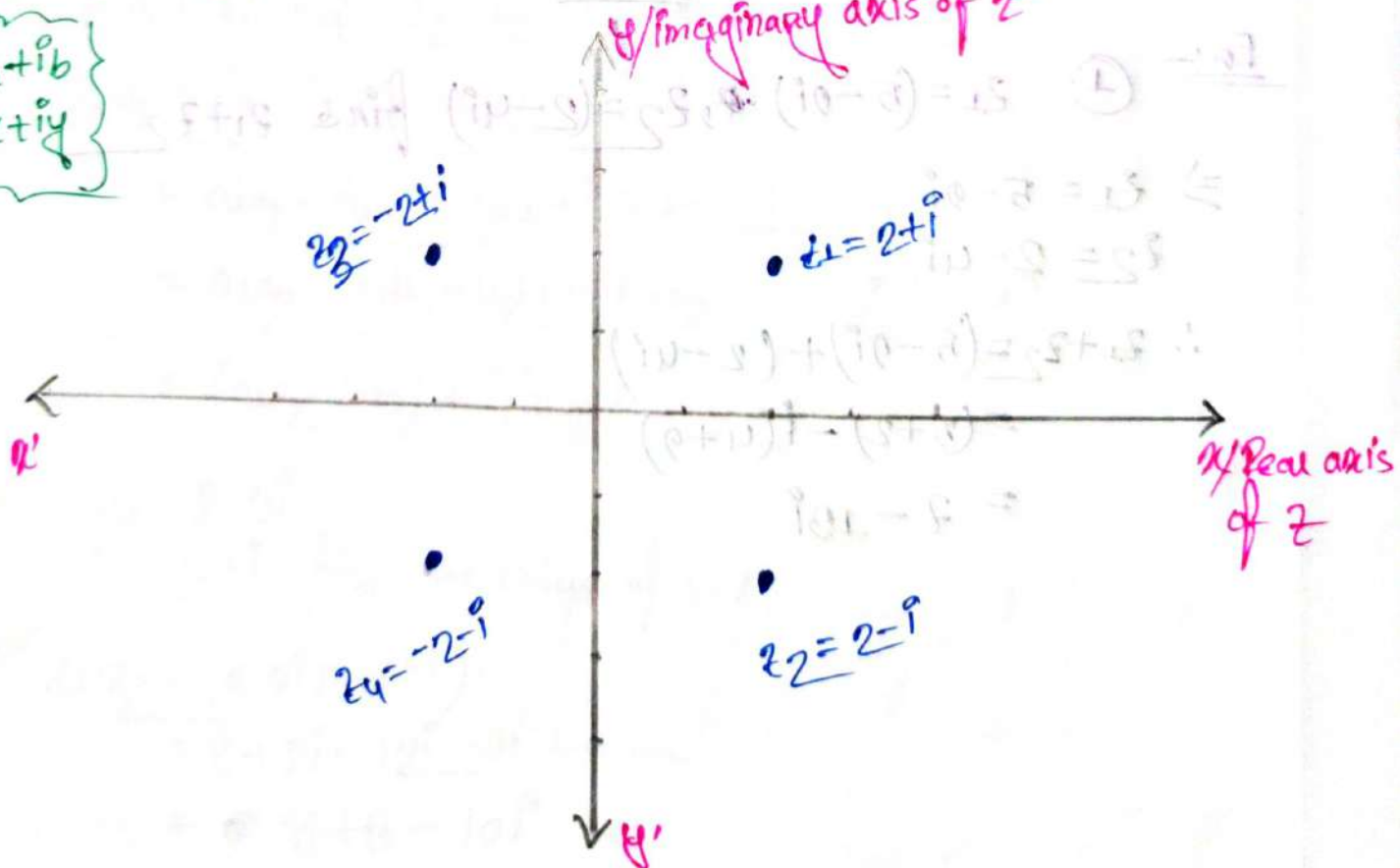
$\sqrt{-1} = i \rightarrow$ imaginary unit.

Graph of Complex Variables:-

$z_1 = 2 + i, z_2 = 2 - i,$

$z_3 = -2 + i, z_4 = -2 - i$

$$\begin{cases} z = a + ib \\ z = x + iy \end{cases}$$



properties of Complex numbers

① Addition.

② Subtraction

③ Multiplication

④ Division.

Addition:

* $z_1 = a_1 + ib_1$, $z_2 = a_2 + ib_2$ find $z_1 + z_2$

$$\Rightarrow z_1 + z_2 = (a_1 + ib_1) + (a_2 + ib_2) \\ = (a_1 + a_2) + i(b_1 + b_2)$$

Ans:-

Ex:- ① $z_1 = (5 - 0i)$, $z_2 = (2 - 4i)$ find $z_1 + z_2$

$$\Rightarrow z_1 = 5 - 0i$$

$$z_2 = 2 - 4i$$

$$\therefore z_1 + z_2 = (5 - 0i) + (2 - 4i) \\ = (5 + 2) - i(4 + 0) \\ = 7 - 4i$$

Subtraction:-

if, $z_1 = a_1 + ib_1$; $z_2 = a_2 + ib_2$ the $z_1 - z_2$ is

$$\Rightarrow z_1 - z_2 = (a_1 + ib_1) - (a_2 + ib_2) \\ = (a_1 - a_2) + i(b_1 - b_2)$$

Ex:- (1) $3(4-i) - 3(5+2i)$ find subtraction value.

$$\Rightarrow 12 - 3i - 15 - 6i$$

$$\Rightarrow -3 - 9i$$

Ans:-

Multiplication:-

if $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ then $z_1 \cdot z_2$ is

$$z_1 \cdot z_2 = (a_1 + ib_1)(a_2 + ib_2) \\ = a_1a_2 + ia_1b_2 + ia_2b_1 + i^2b_1b_2 \\ = a_1a_2 + ia_1b_2 + ia_2b_1 - b_1b_2 \quad [i^2 = -1] \\ = (a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)$$

Ex:- $z_1 = 2-3i$
 $z_2 = 4+i$ find the value of $z_1 \cdot z_2$

$$\Rightarrow z_1 \cdot z_2 = (2-3i)(4+i) \\ = 8 + 2i - 12i - 3i^2 \\ = 8 + 3 - 10i \\ = 11 - 10i$$

* $z_1 = i$.

$z_2 = (5+7i)$ find $z_1 \cdot z_2$

$\Rightarrow z_1 \cdot z_2 = i \cdot (5+7i)$

$= 5i + 7i^2$

$= -7 + 5i$

Ans:-

Division:-

if $z_1 = a_1 + ib_1$, $z_2 = a_2 + ib_2$ So, Find $\frac{z_1}{z_2}$

$\frac{z_1}{z_2} = \frac{a_1 + ib_1}{a_2 + ib_2}$

$= \frac{(a_1 + ib_1)(a_2 - ib_2)}{(a_2 + ib_2)(a_2 - ib_2)}$ [multiply $(a_2 - ib_2)$ on both side]

$= \frac{a_1 a_2 - i a_1 b_2 + i b_1 a_2 - i^2 b_1 b_2}{a_2^2 - i^2 b_2^2}$

$= \frac{(a_1 a_2 + b_1 b_2) + i(a_2 b_1 - a_1 b_2)}{a_2^2 + b_2^2}$

$= \frac{a_1 a_2 + b_1 b_2}{a_2^2 + b_2^2} + i \left(\frac{a_2 b_1 - a_1 b_2}{a_2^2 + b_2^2} \right)$

Ex-1 (1) $\frac{i}{1+i}$

$$= \frac{i(1-i)}{(1+i)(1-i)}$$

$$= \frac{i - i^2}{1 - i^2}$$

$$= \frac{i + 1}{1 + 1}$$

$$= \frac{1+i}{2}$$

Ans: 1-

(2)

$$\frac{1}{-i}$$

$$= \frac{1 \times i}{(-i) \times i}$$

$$= \frac{i}{-i^2}$$

$$= \frac{i}{1}$$

$$= i$$

Page: 07-

In problems write the given number in form $a+ib$.

(3) $(5-2i) \times (2-4i)$

$$\Rightarrow 10 - 20i - 18i + 36i^2$$

$$\Rightarrow 10 - 36 - i(20 + 18)$$

$$\Rightarrow -26 - 38i$$

$$\therefore -26 - 38i$$

Ans: -

(4) $3(4-i) - 3(5+2i)$

$$= 12 - 3i - 15 - 6i$$

$$= -3 - 9i$$

Ans: -

$$(5) i(5+7i)$$

$$\Rightarrow 5i + 7i^2$$

$$\Rightarrow 5i - 7$$

$$\Rightarrow -7 + 5i$$

Ans: -

$$(6) i(4-i) + 4i(1+2i)$$

$$\Rightarrow 4i - i^2 + 4i + 8i^2$$

$$\Rightarrow 8i + 7i^2$$

$$\Rightarrow -7 + 8i$$

Ans: -

$$(7) (2-3i)(4+i)$$

$$\Rightarrow 8 + 2i - 12i - 3i^2$$

$$\Rightarrow 8 + 3 - 10i$$

$$\Rightarrow 11 - 10i$$

Ans: -

$$(8) \left(\frac{1}{3} - \frac{4}{6}i\right)\left(\frac{2}{3} + \frac{5}{3}i\right)$$

$$\Rightarrow \frac{1}{3} + \frac{5}{6}i - \frac{4}{6}i - \frac{5}{12}i^2$$

$$\Rightarrow \frac{1}{3} + \frac{5}{12} + \left(\frac{5}{6} - \frac{4}{6}\right)i$$

$$\Rightarrow \frac{3}{4} + \frac{2}{3}i$$

Ans: -

$$(9) 3i + \frac{1}{2-i}$$

$$\Rightarrow \frac{3i(2-i) + 1}{2-i}$$

$$\Rightarrow \frac{6i - 3i^2 + 1}{2-i}$$

$$\Rightarrow \frac{6i + 3 + 1}{2-i}$$

$$\Rightarrow \frac{6i + 4}{2-i}$$

$$\Rightarrow \frac{(6i + 4)(2+i)}{(2-i)(2+i)}$$

$$\Rightarrow \frac{12i + 6i^2 + 8 + 4i}{2^2 - i^2}$$

$$\Rightarrow \frac{8 - 6 + 16i}{4 + 1}$$

$$\Rightarrow \frac{2}{5} + \frac{16i}{5}$$

Ans: -

$$(10) \frac{i}{1+i}$$

$$\Rightarrow \frac{i(1-i)}{(1+i)(1-i)}$$

$$\Rightarrow \frac{i - i^2}{1 - i^2}$$

$$\Rightarrow \frac{1 + i}{1 + 1}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{2}i$$

Ans: -

$$\textcircled{11} \quad \frac{2-4i}{3+5i}$$

$$\Rightarrow \frac{(2-4i)(3-5i)}{(3+5i)(3-5i)}$$

$$\Rightarrow \frac{6-10i-12i+20i^2}{3^2-5^2i^2}$$

$$\Rightarrow \frac{-14-22i}{34}$$

$$\Rightarrow -\frac{7}{17} - \frac{11}{17}i$$

Ans:-

$$\textcircled{12} \quad \frac{10-5i}{6+2i}$$

$$\Rightarrow \frac{(10-5i)(6-2i)}{(6+2i)(6-2i)}$$

$$\Rightarrow \frac{60-20i-30i+10i^2}{6^2-4i^2}$$

$$\Rightarrow \frac{50-50i}{36+4}$$

$$\Rightarrow \frac{5}{4} - \frac{5}{4}i$$

Ans:-

$$\textcircled{13} \quad \frac{(3-i)(2+3i)}{1+i}$$

$$\Rightarrow \frac{6+9i-2i-3i^2}{(1+i)(1-i)}$$

$$\Rightarrow \frac{(7+7i)(1-i)}{1-i+i-i^2}$$

$$\Rightarrow \frac{7-7i+7i-7i^2}{1+1}$$

$$\Rightarrow \frac{14-2i}{2}$$

$$\Rightarrow 7-i$$

Ans:-

$$\textcircled{14} \quad \frac{(1+i)(1-2i)}{(2+i)(4-3i)}$$

$$= \frac{1-2i+i-2i^2}{8-6i+4i-3i^2}$$

$$= \frac{3-i}{11-2i}$$

$$= \frac{(3-i)(11+2i)}{(11-2i)(11+2i)}$$

$$= \frac{33+6i-11i-2i^2}{(11)^2-4i^2}$$

$$= \frac{35i-5i}{125}$$

$$= \frac{7}{25} - \frac{1}{25}i \quad \text{Ans:-}$$

$$(15) \frac{(5-4i) - (3+2i)}{(4+2i) + (2-3i)}$$

$$\Rightarrow \frac{(2-11i)(6+i)}{(6-i)(6+i)}$$

$$\Rightarrow \frac{12+2i-66i-11i^2}{6^2-i^2}$$

$$\Rightarrow \frac{12+11-64i}{36+1}$$

$$\Rightarrow \frac{23}{37} - \frac{64}{37}i$$

Ans:-

$$(16) \frac{(4+5i) + 2i^3}{(2+i)^2}$$

$$\Rightarrow \frac{(4+5i+2i^3)}{4+4i+i^2}$$

$$\Rightarrow \frac{(4+5i-2i)(3-4i)}{(3+4i)(3-4i)}$$

$$\Rightarrow \frac{12-16i+9i+12i^2}{3^2-16}$$

$$\Rightarrow \frac{24-7i}{25}$$

$$\Rightarrow \frac{24}{25} - \frac{7}{25}i$$

Ans:-

Page: 8:-

$$(21) (2+3i)^2$$

$$= 4+12i+9i^2$$

$$= -5+12i$$

Find the Re(z) and Im(z)

$$(25) z = \left(\frac{i}{3-i} \right) \left(-\frac{1}{2+3i} \right)$$

$$= \frac{i}{6+0i-2i^2-3i^2}$$

$$= \frac{i(1-2i)}{(6+2i)(6-2i)}$$

$$= \frac{0i-2i^2}{36-40i^2}$$

$$= \frac{2+0i}{130}$$

$$= \frac{2}{130} + \frac{0}{130}i$$

Ans:-

$$\text{so, } \text{Re}(z) = \frac{2}{130}$$

$$\text{Im}(z) = \frac{0}{130}$$

Ans:-

Express the given quantity in terms of x and y .

(27) Let $z = x + iy$ find $\operatorname{Re}\left(\frac{1}{z}\right)$

$$\begin{aligned}\Rightarrow \frac{1}{z} &= \frac{1}{x+iy} \\&= \frac{(x-iy)}{(x+iy)(x-iy)} \\&= \frac{x-iy}{x^2 - i^2 y^2} \\&= \frac{x-iy}{x^2 + y^2} \\&= \frac{x}{x^2 + y^2} - i \left(\frac{y}{x^2 + y^2} \right)\end{aligned}$$

$$\therefore \operatorname{Re}\left\{\frac{1}{z}\right\} = \frac{x}{x^2 + y^2}$$

(28) Let $z = x + iy$ find $\operatorname{Re}(z^2)$

$$\begin{aligned}\Rightarrow z^2 &= (x+iy)^2 = x^2 + 2x(iy) + (iy)^2 \\&= x^2 - y^2 + 2ixy\end{aligned}$$

$$\therefore \operatorname{Re}(z^2) = x^2 - y^2$$

Ans.

29) Let $z = x + iy$ find $\text{Im}(2z + 4\bar{z} - 4i)$

$\Rightarrow$ is given:—
non

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$\begin{aligned}\therefore 2z + 4\bar{z} - 4i &= 2x + 2iy + 4x - 4iy - 4i \\ &= 6x - 2iy - 4i \\ &= 6x - i(2y + 4)\end{aligned}$$

$$\therefore \text{Im}(2z + 4\bar{z} - 4i) = 2y + 4$$

Ans:-

30) Let $z = x + iy$ find $\text{Im}(\bar{z}^2 + z^2)$

$$\therefore \bar{z} = x - iy$$

Sol

$$\begin{aligned}\bar{z}^2 &= x^2 - 2ixy + y^2i^2 \\ &= x^2 - y^2 - 2ixy\end{aligned}$$

$$\begin{aligned}z^2 &= x^2 + 2ixy + y^2i^2 \\ &= x^2 - y^2 + 2ixy\end{aligned}$$

$$\begin{aligned}\therefore \bar{z}^2 + z^2 &= x^2 - 2ixy - y^2 + x^2 - y^2 + 2ixy \\ &= 2(x^2 - y^2) \\ &= 2x^2 - 2y^2\end{aligned}$$

$$\therefore \text{Im}(\bar{z}^2 + z^2) = 0$$

Ans:-

(31) Let $z = x + iy$ Find $\text{Re}(iz)$

$$\Rightarrow iz = xi + i^2y$$
$$= -y + xi$$

$$\therefore \text{Re}(iz) = -y$$

Ans. -

(32) Let $z = x + iy$ Find $\text{Im}(iz)$

$$\Rightarrow iz = xi + i^2y$$

$$= xi - y$$

$$= -y + xi$$

So, $\text{Im}(iz) = x$

(33) Let $z = x + iy$ Find $\text{Im}\{(1+i)z\}$

$$\therefore (1+i)z = (1+i)(x+iy)$$

$$= x + iy + ix + i^2y^2$$

$$= x - y^2 + i(x+y)$$

So, $\text{Im}\{(1+i)z\} = x+y$

(34) Let $z = x + iy$ Find $\text{Re}(z^2)$

$$\Rightarrow z^2 = x^2 + 2ixy + y^2i^2$$

$$= x^2 - y^2 + 2ixy$$

$$\therefore \text{Re}\{z^2\} = x^2 - y^2$$

Ans. -

37) $2z = i(2+i)$

$$\Rightarrow z = \frac{1}{2}i(2+i)$$

$$= i + \frac{1}{2}i^2$$

$$= -\frac{1}{2} + i$$

38) $z - 2\bar{z} + 7 - 6i = 0$; $z = a+ib$

$$\Rightarrow (a+ib) - 2(a-ib) + 7 - 6i = 0$$

$$\Rightarrow a - 2a + 7 + ib + 2ib - 6i = 0$$

$\Rightarrow$ Now separate the real part;

$$-a - 2a + 7 = 0$$

$$\therefore a = 7$$

Now, Separate the Im(z);

$$ib - 2ib - 6i = 0$$

$$-ib - 6i = 0$$

$$i(b - 2) = 0$$

$$\text{So, } \left. \begin{array}{l} b - 2 = 0 \\ b = 2 \end{array} \right\} \begin{array}{l} \text{OR} \\ 3i \neq 0 \end{array}$$

$$\therefore z = 7 + 2i$$

Ans:-

39) $z^2 = i$; $z = a+ib$

$$\Rightarrow (a+ib)^2 = i$$

$$\Rightarrow a^2 + 2aib + i^2b^2 = i$$

$$\Rightarrow a^2 + 2aib - b^2 = i$$

$\Rightarrow$ Now Separate the real and Im part

$$\Rightarrow a^2 - b^2 = 0 \quad \text{--- (1)}$$

$$\Rightarrow 2aib = i \quad \text{--- (2)}$$

from eq (1)

$$a^2 - b^2 = 0$$

$$(a+b)(a-b) = 0 \quad \text{--- (3)}$$

$$\text{So, } \left. \begin{array}{l} a = b \\ a = -b \end{array} \right\}$$

$$\therefore a = \pm b$$

from eq (2)

$$2aib = i = (1i) \text{ on LHS}$$

$$2ab = 1$$

$$\therefore a = \frac{1}{2b}$$

$$(a+ib)(a+ib) = i \text{ on RHS}$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

$$(a+ib)(a+ib) = i$$

#Complex Plane!

Conjugate of Complex Number:-

$$* z = 3 + 2i \quad * \bar{z} = 3 + 2i = z$$

$$* \bar{z} = 3 - 2i$$

Ex:-
 $\therefore z \cdot \bar{z} = (3 + 2i)(3 - 2i)$

$$= 3^2 - 4i^2$$

$$= 9 + 4$$

$$= 13$$

Absolute Value/Modulus:-

If $z = x + iy$

Then, absolute value of z is $|z| = \sqrt{x^2 + y^2}$

Ex:- if $z = 1 + i$ then find the $|z + \bar{z}|$

$$\Rightarrow z = 1 + i$$

$$\bar{z} = 1 - i$$

$$\underline{\text{So,}} \quad z + \bar{z} = (1 + i) + 1 + (-i) \\ = 2$$

$$\underline{\text{Now,}} \quad |z + \bar{z}| = \sqrt{(2)^2 + (0)^2} \\ = \sqrt{4} \\ = 2$$

Ans:-

$$z + \bar{z} = 1 + i + 1 - i = 2$$

Page: 14:-

Find the modulus of the given Complex Numbers:

9. $(1-i)^2$

$$\Rightarrow 1 - 2i + i^2$$

$$\Rightarrow 1 - 2i - 1$$

$$\Rightarrow -2i$$

$$\text{So, } |-2i| = \sqrt{2^2} = 2$$

Ans:-

10. $i(2-i) - 4(1 + \frac{1}{4}i)$

$$\Rightarrow 2i - i^2 - 4 - i$$

$$\Rightarrow -4 + 1 + i$$

$$\Rightarrow -3 + i$$

$$\therefore |-3 + i| = \sqrt{3^2 + 1^2}$$

$$= \sqrt{10}$$

Ans:-

11. $\frac{2i}{3-4i}$

$$= \frac{2i(3+4i)}{(3-4i)(3+4i)}$$

$$= \frac{6i + 8i^2}{9 - 16i^2}$$

$$= \frac{-8 + 6i}{25}$$

$$= -\frac{8}{25} + \frac{6}{25}i$$

$$\therefore \left| -\frac{8}{25} + \frac{6}{25}i \right| = \sqrt{\left(\frac{-8}{25}\right)^2 + \left(\frac{6}{25}\right)^2} = \frac{2}{5}$$

12. $\frac{1-2i}{1+i} + \frac{2-i}{1-i}$

$$= \frac{(1-2i)(1-i) + (2-i)(1+i)}{(1+i)(1-i)}$$

$$= \frac{1-i-2i+2i^2 + 2+2i-i-i^2}{1-i^2}$$

$$= \frac{3-2i-2+1}{1+1}$$

$$= \frac{2}{2} - \frac{2}{2}i$$

$$= 1 - i$$

$$\therefore |1-i| = \sqrt{1^2 + 1^2}$$

$$= \sqrt{2}$$

Ans:-

Inequalities:

Formulae:

$$(1) |z_1 + z_2| \leq |z_1| + |z_2|$$

$$(3) |z_1 - z_2| \leq |z_1| + |z_2|$$

$$(2) |z_1 + z_2| \geq |z_1| - |z_2|$$

Ex8:

* if $|z| = 2$, then $|z + 6 + 8i| \leq 13$ prove it.

$\Rightarrow$ Is given:—

$$|z| = 2$$

$$\Rightarrow |z + 6 + 8i| \leq 13$$

$$\Rightarrow |z + 6 + 8i| \leq |z| + |6 + 8i| \leq 13$$

$$\Rightarrow 2 + |6 + 8i| \leq 13$$

$$\Rightarrow 2 + \sqrt{6^2 + 8^2} \leq 13$$

$$\Rightarrow 2 + 10 \leq 13$$

$$\Rightarrow 12 \leq 13$$

Proved

Example: 3

$\Rightarrow$ Find the Upper Bound for $\left(\frac{-1}{z^4 - 5z + 1}\right)$ if $|z| = 2$

$\Rightarrow$ Let,

$$\frac{-1}{|z^4 - 5z + 1|} \leq m \quad \text{--- (1)}$$

So,

$$|z^4 - 5z + 1| = |z^4 - (5z - 1)| \quad \text{--- (11)}$$

From eq (11) $\Rightarrow$

We know,

$$|z^4 - (5z - 1)| \geq ||z^4| - |5z - 1||$$

$$\text{or } |z^4 - (5z - 1)| \geq |12^4 - (5|z| - 1)|$$

$$= |2^4 - (5 \cdot 2 - 1)|$$

$$= 5$$

Now, from eq (1) $\Rightarrow$

$$\frac{-1}{|z^4 - 5z + 1|} \leq \frac{1}{5}$$

Ans:

Page: - 14

Establish the given inequality (1)

Q8 If $|z|=1$, then $1 \leq |z^2-3| \leq 4$

$\Rightarrow$ in given

$$|z|=1$$

$$\Rightarrow 1 \leq |z^2-3| \leq 4$$

$$\Rightarrow 1 \leq |z^2-3| \leq 4 \quad \text{--- (1)}$$

Now work left side on inequality.

$$1 \leq |z^2-3|$$

$$1 \leq |1-3|$$

$$1 \leq 2$$

Now, Right side on inequality.

$$|z^2-3| \leq 4$$

$$|1-3| \leq 4$$

$$|1-3| \leq 4$$

$$2 \leq 4$$

So, from eq (1) $\Rightarrow$

$$1 \leq |z^2-3| \leq 4$$

$$1 \leq 2 \leq 4$$

(Satisfied)

② Find an upper bound for the modulus of $3z^2 + 2z + 1$ if $|z| \leq 1$

$$\Rightarrow = |3z^2 + 2z + 1|$$

$$= |3z^2 + 2z + 2z - 2z + 1|$$

$$= |3z^2 + 4z + 1 - 2z|$$

$$= \left| \underbrace{(3z^2 + 1)}_{2z} - \underbrace{2z}_{2z} \right| \leq |(3z^2 + 1)(z + 1)| + |2z|$$

$$\leq |3z^2 + 1| |z + 1| + 2|z|$$

$$\leq (3|z| + 1)(|z| + 1) + 2|z|$$

$$\leq (3 + 1)(1 + 1) + 2 \times 1$$

$$\leq 4 + 2 + 2$$

$$\leq 10$$

$$\therefore |3z^2 + 2z + 1| \leq 10$$

Ans:-

Describe the graph and locus represented by each the following

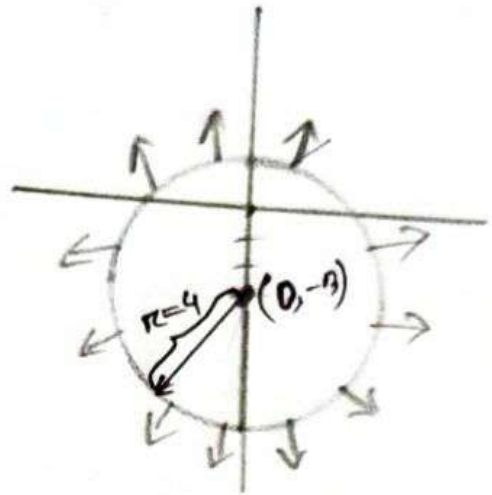
(a) $|z + 3i| > 4$,

$\Rightarrow |x + iy + 3i| > 4$

$\Rightarrow |x + i(y+3)| > 4$

$\Rightarrow [\sqrt{x^2 + (y+3)^2}] > 4$

$\Rightarrow x^2 + (y+3)^2 > 4^2$



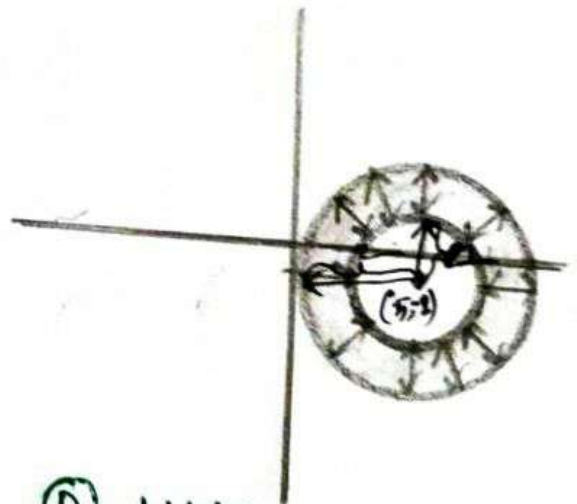
(b) $1 < |z - 5 + i| \leq 2$

$\Rightarrow 1 < |x + iy - 5 + i| \leq 2$

$\Rightarrow 1 < |(x-5) + i(y+1)| \leq 2$

$\Rightarrow 1 < \sqrt{(x-5)^2 + (y+1)^2} \leq 2$

$\Rightarrow 1 < (x-5)^2 + (y+1)^2 \leq 2^2$



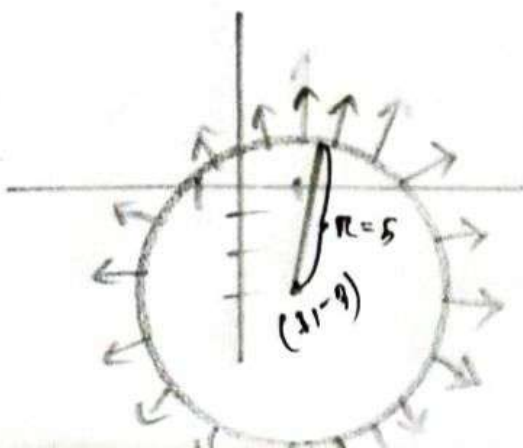
(c) $|z - 1 + 3i| > 5$,

$\Rightarrow |x + iy + 3i - 1| > 5$

$\Rightarrow |(x-1) + i(y+3)| > 5$

$\Rightarrow \sqrt{(x-1)^2 + (y+3)^2} > 5$

$\Rightarrow (x-1)^2 + (y+3)^2 > 5^2$

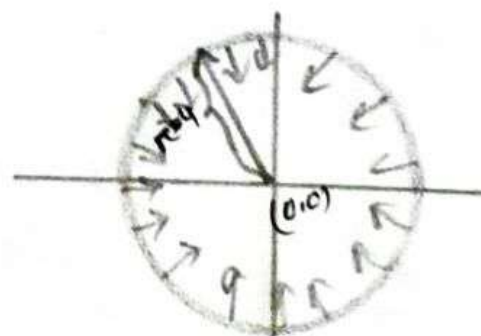


(d) $|z| \leq 4$

$\Rightarrow |x + iy| \leq 4$

$\Rightarrow \sqrt{x^2 + y^2} \leq 4$

$\Rightarrow x^2 + y^2 \leq 4^2$



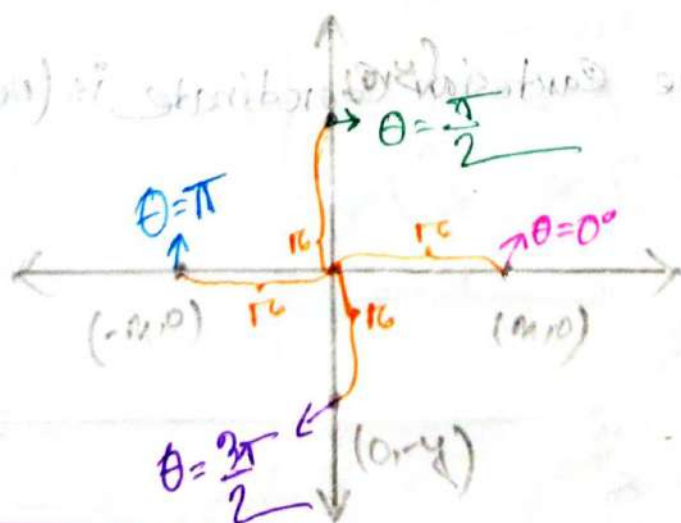
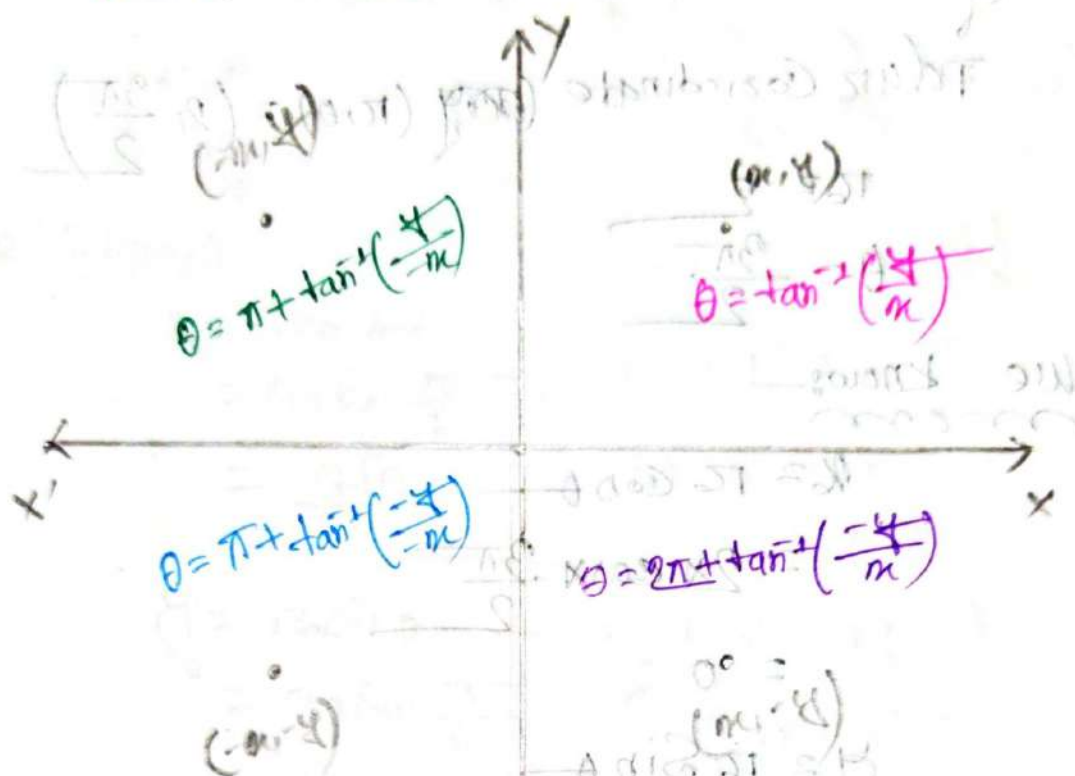
Find the modulus of the given complex number:

Problem

Polar Form of Complex Numbers

- ① $(1-i)^2$
 $= 1 - 2i + i^2 = 1 - 2i - 1 = -2i$
 $\therefore \sqrt{(-2)^2 + 0^2} = \sqrt{4} = 2$
- ② $(i+1)^2$
 $= i^2 + 2i + 1 = -1 + 2i + 1 = 2i$
 $\therefore \sqrt{0^2 + 2^2} = \sqrt{4} = 2$
- ③ $(1-i)(1+i)$
 $= 1 - i^2 = 1 - (-1) = 1 + 1 = 2$
 $\therefore \sqrt{2^2} = 2$
- ④ $(1-i)(1-i)$
 $= 1 - 2i + i^2 = 1 - 2i - 1 = -2i$
 $\therefore \sqrt{(-2)^2} = 2$
- ⑤ $(1-i)(1+i)$
 $= 1 - i^2 = 1 - (-1) = 1 + 1 = 2$
 $\therefore \sqrt{2^2} = 2$
- ⑥ $(1-i)(1-i)$
 $= 1 - 2i + i^2 = 1 - 2i - 1 = -2i$
 $\therefore \sqrt{(-2)^2} = 2$
- ⑦ $(1-i)(1+i)$
 $= 1 - i^2 = 1 - (-1) = 1 + 1 = 2$
 $\therefore \sqrt{2^2} = 2$
- ⑧ $(1-i)(1-i)$
 $= 1 - 2i + i^2 = 1 - 2i - 1 = -2i$
 $\therefore \sqrt{(-2)^2} = 2$
- ⑨ $(1-i)(1+i)$
 $= 1 - i^2 = 1 - (-1) = 1 + 1 = 2$
 $\therefore \sqrt{2^2} = 2$
- ⑩ $(1-i)(1-i)$
 $= 1 - 2i + i^2 = 1 - 2i - 1 = -2i$
 $\therefore \sqrt{(-2)^2} = 2$

Standard Position of Angles in the Coordinate plane:-



* Polar Coordinate (r, θ)

* Cartesian Coordinate (x, y)

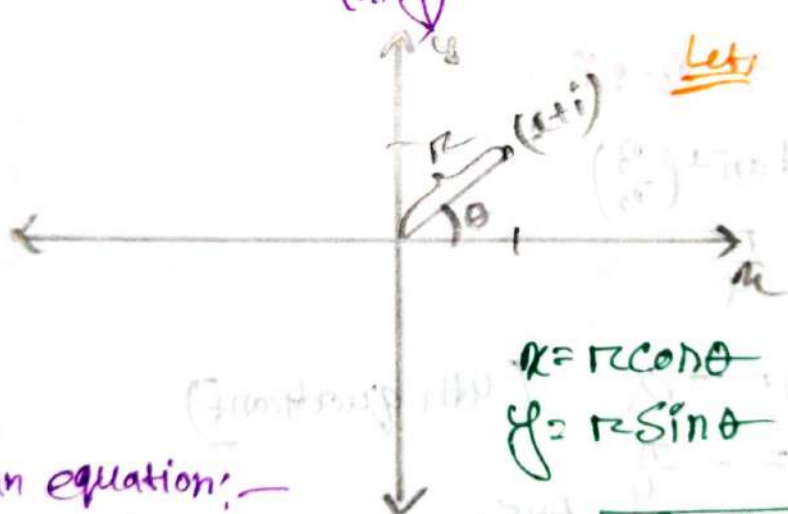
Principle Argument:-

- ① $0 < \theta \leq 360^\circ$ or
- ② $-\pi \leq \theta < \pi$ / $-\pi \leq \text{Arg}(z) \leq \pi$
- ③ $\arg(z) = \text{Arg}(z) + 2n\pi$; $n = 0, \pm 1, \pm 2, \pm 3, \dots$

Polar form & Cartesian form:-

* Polar form:-
(r, θ)

Cartesian form:-
(x, y)



Let $z = x + iy$
 $= 1 + i$

$\therefore$ Cartesian equation:-

$$z = x + iy$$

So, polar equation:-

$$z = r(\cos\theta + i\sin\theta)$$

$$x = r\cos\theta$$

$$y = r\sin\theta$$

$$\therefore r = \sqrt{x^2 + y^2}$$

$$\theta = \text{Arg}(z)$$

Argument:

$$z = x + iy$$

$$\therefore \alpha = \tan^{-1}\left(\frac{y}{x}\right)$$

Formula:

- (i) $\text{Arg}(z)$ for 1st = $\theta = \alpha$
- (ii) $\text{Arg}(z)$ for 2nd = $\theta = \pi - \alpha$
- (iii) $\text{Arg}(z)$ for 3rd = $\theta = -\pi + \alpha$
- (iv) $\text{Arg}(z)$ for 4th = $\theta = -\alpha$

Ex-1 If $z = 3 - 3i$

$$\therefore \alpha = \tan^{-1}\left(\frac{3}{3}\right)$$

$$= \frac{\pi}{4}$$

$$\therefore \text{Arg}(z) = -\alpha \quad [\text{4th quadrant}]$$

$$= -\frac{\pi}{4} \quad \underline{\text{Ans.}}$$

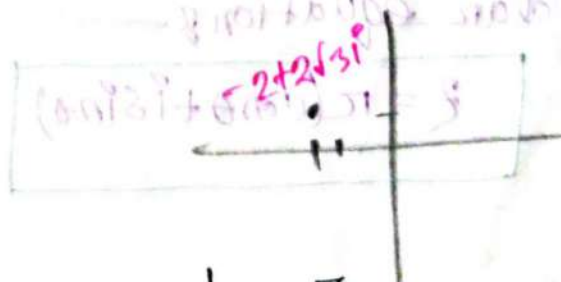
(ii) If $z = -2 + 2\sqrt{3}i$

$$\therefore \alpha = \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{3}$$

$$\therefore \text{Arg}(z) = \pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3} \quad \underline{\underline{\text{Ans.:-}}}$$



Ex:- Let $z = -2 + 2\sqrt{3}i$; Convert into polar form:—

$\Rightarrow$ is given:-

$$x = -2$$

$$y = 2\sqrt{3}$$

$$\therefore r = \sqrt{x^2 + y^2}$$

$$= \sqrt{(-2)^2 + (2\sqrt{3})^2}$$

$$= 4$$

$$\theta = \tan^{-1}\left(\frac{2\sqrt{3}}{-2}\right)$$

$$\alpha = \theta = \frac{\pi}{3}$$

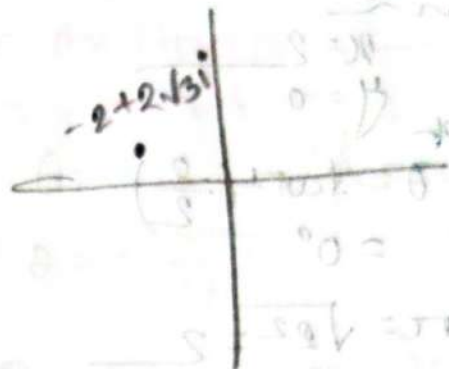
$$\begin{aligned}\text{So, Arg}(z) &= \pi - \alpha \\ &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3}\end{aligned}$$

$\therefore$ polar form:-

$$z = r(\cos\theta + i\sin\theta)$$

$$= 4\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

Ans:-



Page 21:-

① $z = 2$

we have:-

$$x = 2$$

$$y = 0$$

we know:-

$$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= 0^\circ$$

$$\therefore r = \sqrt{x^2 + y^2} = 2$$

for $\theta \neq \arg(z)$:-

$$z = 2(\cos 0 + i \sin 0)$$

for $\theta = \arg(z)$:-

$$z = 2(\cos 0 + i \sin 0)$$

② $z = -10$

we have:-

$$x = -10$$

$$y = 0$$

we know:-

$$\alpha = \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= 0$$

$$r = \sqrt{(-10)^2 + 0^2} = 10$$

for $\theta \neq \arg(z)$:-

$$z = 10(\cos 0 + i \sin 0)$$

for $\theta = \arg(z)$:-

$$z = 10(\cos 0 + i \sin 0)$$

③ $-3i$

we have:-

$$x = 0$$

$$y = -3$$

we know:-

$$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\alpha = \theta = \frac{\pi}{2}$$

$$r = \sqrt{(-3)^2 + 0^2} = 3$$

for $\theta \neq \arg(z)$:-

$$z = 3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$$

for $\theta = \arg(z)$

$$z = 3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$$

$$z = 3\left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2}\right)$$

④ $6i$

we have:-

$$x = 0$$

$$y = 6$$

we know:-

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}\left(\frac{6}{0}\right)$$

$$\alpha = \theta = \frac{\pi}{2}$$

$$r = 6$$

for $\theta \neq \arg(z)$:-

$$z = 6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$$

for $\theta = \arg(z)$:-

$$z = 6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$$

5) $z = 1 + i$

we have—

$x = 1$

$y = 1$

we know—

$r = \sqrt{1^2 + 1^2}$

$= \sqrt{2}$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$
 $= \frac{\pi}{4}$

For 1st condition $\theta \neq \text{Arg}(z)$:-

$z = r(\cos \theta + i \sin \theta)$
 $= \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$

For $\theta = \text{Arg}(z)$:-

$\theta = \alpha$ [for 1st quadrant]

$\therefore z = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$

Ans:-

6) $z = 5 - 5i$

we have—

$x = 1 + 4 = 5$

$y = -5$

we know—

$r = \sqrt{5^2 + 5^2}$

$= 5\sqrt{2}$

$\alpha = \theta = \tan^{-1}\left(-\frac{5}{5}\right)$
 $= \frac{\pi}{4}$

For $\theta \neq \text{Arg}(z)$:-

$\theta = 2\pi + \tan^{-1}\left(-\frac{5}{5}\right) = \frac{7\pi}{4}$

For $\theta \neq \text{Arg}(z)$:-

$z = 5\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$

For $\theta = \text{Arg}(z)$:-

$\theta = -\alpha$ [for 4th quadrant]

$\therefore \theta = -\frac{\pi}{4}$

$\therefore z = 5\sqrt{2}(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4})$

Ans:-

7) $-\sqrt{3} + i$

we have—

$x = \sqrt{3}$

$y = 1$

we know—

$r = \sqrt{(\sqrt{3})^2 + 1^2}$

$= 2$

$\alpha = \theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 $= \frac{\pi}{6}$

For $\theta \neq \text{Arg}(z)$

$\theta = \pi + \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 $= \frac{5\pi}{6}$

For $\theta = \text{Arg}(z)$:-

$z = 2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$

P.T.O

For $\theta = \text{Arg}(z)$:-

$$\theta = \pi - \frac{\pi}{6}$$

$$= \frac{5\pi}{6}$$

$$\therefore z = 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$

Ans:-

⑧ $-2 - 2\sqrt{3}i$

$\Rightarrow$ we have

$$x = -2$$

$$y = -2\sqrt{3}$$

we know

$$r = \sqrt{(-2)^2 + (-2\sqrt{3})^2}$$

$$= 4$$

$$\theta = \tan^{-1} \left(\frac{-2\sqrt{3}}{-2} \right)$$

$$= \frac{\pi}{3}$$

for $\theta \neq \text{Arg}(z)$:-

$$z = 4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

for $\theta = \text{Arg}(z)$:-

$$\theta = -\pi + \alpha$$

$$= -\pi + \frac{\pi}{3}$$

$$= \frac{-2\pi}{3}$$

$$\therefore z = 4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

Ans:-

when $\theta \neq \text{Arg}(z)$:-

$$x = -2$$

$$y = -2\sqrt{3}$$

$$\therefore \theta = \pi + \tan^{-1} \left(\frac{-2\sqrt{3}}{-2} \right) = \frac{4\pi}{3}$$

(Q3) $\frac{-i}{1+i}$ write the polar form in form of $a+ib$

$$\Rightarrow \frac{-i(1-i)}{(1+i)(1-i)}$$

$$\Rightarrow \frac{-i + i^2}{1^2 - (i)^2}$$

$$\Rightarrow \frac{-i - 1}{2}$$

$$\Rightarrow -\frac{1}{2} - \frac{1}{2}i$$

we have:

$$x = \frac{1}{2}$$

$$y = \frac{1}{2}$$

we know:

$$\begin{aligned} \alpha = \theta &= \tan^{-1}\left(\frac{y}{x}\right) \\ &= \tan^{-1}\left(\frac{1/2}{1/2}\right) \\ &= \frac{\pi}{4} \end{aligned}$$

and:

$$r = \sqrt{x^2 + y^2}$$

$$= \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$= \frac{1}{\sqrt{2}}$$

Note:

$$\text{Arg}(z) = \theta'$$

$$\theta' = -\pi + \alpha$$

$$\theta' = -\pi + \frac{\pi}{4}$$

$$= \frac{-4\pi + \pi}{4}$$

$$= -3\pi/4$$

$$\therefore z = r(\cos \theta' + i \sin \theta')$$

$$= \frac{1}{\sqrt{2}} \left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4} \right)$$

Ans:

Page 8.19

Multiplication & Division:—

Formulas:—

$$(i) \operatorname{Arg}(z_1 z_2) = \operatorname{Arg}(z_1) + \operatorname{Arg}(z_2)$$

$$(ii) \operatorname{Arg}(z_1/z_2) = \operatorname{Arg}(z_1) - \operatorname{Arg}(z_2)$$

Ex: 7

Q. if, $z_1 = i$, $z_2 = -\sqrt{3} - i$ and $\operatorname{Arg}(z_1) = \frac{\pi}{2}$, $\operatorname{Arg}(z_2) = -\frac{5\pi}{6}$
then find $\operatorname{Arg}(z_1 z_2)$ also $\operatorname{Arg}\left(\frac{z_1}{z_2}\right)$

$\Rightarrow$ It is given:—

$$\operatorname{Arg}(z_1) = \frac{\pi}{2}$$

$$\operatorname{Arg}(z_2) = -\frac{5\pi}{6}$$

We know:—

$$\begin{aligned} \operatorname{Arg}(z_1 z_2) &= \frac{\pi}{2} + \left(-\frac{5\pi}{6}\right) \\ &= \frac{3\pi - 5\pi}{6} \end{aligned}$$

$$= \frac{2\pi}{6} = \frac{\pi}{3}$$

Ans:—

$$\begin{aligned} \operatorname{Arg}\left(\frac{z_1}{z_2}\right) &= \frac{\pi}{2} - \left(-\frac{5\pi}{6}\right) \\ &= \frac{3\pi + 5\pi}{6} \end{aligned}$$

$$= \frac{4\pi}{3}$$

Ans:—

Integer power of $z(z^n)$

Formula:

$$(i) z^n = r^n (\cos n\theta + i \sin n\theta)$$

$$(ii) z^0 = 1$$

Ex: 8

Q:- if $z = -\sqrt{3} - i$ write the polar form of z^3 when $\theta \neq \text{Arg}(z)$ and find the value of z^3 .

⇒ In given:

$$z = -\sqrt{3} - i$$

$$\therefore x = -\sqrt{3}$$

$$y = -1$$

$$\therefore r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$$

when $\theta \neq \text{Arg}(z)$:

$$\theta = \pi + \tan^{-1}\left(\frac{-1}{-\sqrt{3}}\right)$$

$$= \frac{7\pi}{6}$$

sol $z = r^{\frac{1}{3}} (\cos 3\theta + i \sin 3\theta)$

$$z^3 = 2^3 \left(\cos 3 \times \frac{7\pi}{6} + i \sin 3 \times \frac{7\pi}{6} \right)$$

$$\therefore z^3 = -8i$$

Ans:-

De Moivre's Formula

Formula:

$$(1) z^n = (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$(2) z^n = r^n (\cos n\theta + i \sin n\theta) = r^n e^{in\theta} = (re^{i\theta})^n$$

Ex: 4:

Q:- $z = \cos \theta + i \sin \theta$ where $\theta = \frac{\pi}{6}$ find z^3 .

⇒ In given:

$$\theta = \frac{\pi}{6}$$

$$z = \cos \theta + i \sin \theta$$

$$z^3 = (\cos \theta + i \sin \theta)^3$$

We know:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$\therefore (\cos \theta + i \sin \theta)^3 = \cos 3 \times \frac{\pi}{6} + i \sin 3 \times \frac{\pi}{6}$$

$$= 0 + i \times 1$$

$$\therefore z^3 = (\cos \theta + i \sin \theta)^3 = i$$

Ans:-

#Power
and Roots

principal of nth Root:-

Formula:-

$$\textcircled{1} \omega_k = \sqrt[n]{r} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right]$$

Where,

$$k = 0, 1, 2, \dots, n-1$$

Examples

Q:- Find the three cube roots of $z = 1$

$\Rightarrow$ Is given:-

$$z = 1$$

$$\therefore \theta = 0$$

$$r = 1$$

$$\underline{\text{So,}} \quad r = \sqrt{1^2} = 1$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\arg(z) = \frac{\pi}{2}$$

Now,

$$\omega_k = \sqrt[3]{1} \left[\cos\left(\frac{\frac{\pi}{2} + 2k\pi}{3}\right) + i \sin\left(\frac{\frac{\pi}{2} + 2k\pi}{3}\right) \right]$$

$\underline{\text{So,}}$ The three roots are:-

$$k=0, \omega_0 = \sqrt[3]{1} \left[\cos\left(\frac{\pi/2 + 2 \times 0 \times \pi}{3}\right) + i \sin\left(\frac{\pi/2 + 2 \times 0 \times \pi}{3}\right) \right]$$
$$= \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

Ans:-

$$\begin{aligned}
 k=1; \quad w_1 &= \sqrt[3]{1} \left[\cos\left(\frac{\theta+2k\pi}{3}\right) + i \sin\left(\frac{\theta+2k\pi}{3}\right) \right] \\
 &= \left[\cos \frac{\frac{\pi}{2}+2\pi}{3} + i \sin \frac{\frac{\pi}{2}+2\pi}{3} \right] \\
 &= \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \\
 &= -\frac{\sqrt{3}}{2} + \frac{1}{2}i
 \end{aligned}$$

Ans:-

$$\begin{aligned}
 k=2; \quad w_2 &= \left[\cos \frac{\frac{\pi}{2}+4\pi}{3} + i \sin \frac{\frac{\pi}{2}+4\pi}{3} \right] \\
 &= \cos \frac{9\pi}{3} + i \sin \frac{9\pi}{3} \\
 &= -i
 \end{aligned}$$

Ans:-

Example: 2:-

Q:- Find the fourth roots of $z=1+i$

Is given:-

$$z = 1+i$$

$$\therefore x=1$$

$$y=1$$

So, $r = \sqrt{1^2+1^2} = \sqrt{2}$

$$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

We know,

$$w_k = \sqrt[4]{2} \left[\cos\left(\frac{\pi/4+2k\pi}{4}\right) + i \sin\left(\frac{\pi/4+2k\pi}{4}\right) \right]$$

So, the 4 roots are:

$$k=0, \omega_0 = \sqrt[4]{2} \left(\cos \frac{\pi/4}{4} + i \sin \frac{\pi/4}{4} \right)$$

$$\approx 1.1664 + 0.2320i$$

$$k=1, \omega_0 = \sqrt[4]{2} \left(\cos \frac{9\pi}{16} + i \sin \frac{9\pi}{16} \right)$$

$$\approx -0.2320 + 1.1664i$$

$$k=2, \omega_0 = \sqrt[4]{2} \left(\cos \frac{17\pi}{16} + i \sin \frac{17\pi}{16} \right)$$

$$\approx -1.1664 - 0.2320i$$

$$k=3, \omega_0 = \sqrt[4]{2} \left(\cos \frac{25\pi}{16} + i \sin \frac{25\pi}{16} \right)$$

$$\approx 0.2320 - 1.1664i$$

Page 26:-

Compute all roots, give the principle n th root in each case. Sketch the roots on an appropriate circle centered at the origin.

① $z = (8)^{\frac{1}{3}}$

$\Rightarrow$ Is given

$$\left. \begin{array}{l} x=8, y=0 \\ n=3 \\ \theta=0^\circ \\ \therefore k=0,1,2 \end{array} \right\} \begin{array}{l} r = \sqrt{x^2+y^2} \\ = \sqrt{8^2+0^2} \\ = 8 \end{array}$$

We know:

$$\omega_k = \sqrt[n]{r} \left[\cos \left(\frac{\theta + 2\pi k}{n} \right) + i \sin \left(\frac{\theta + 2\pi k}{n} \right) \right]$$

So, the 3 roots are:—

$$k=0, \omega_0 = \sqrt[3]{8} \left[\cos\left(\frac{0+0}{3}\right) + i \sin\left(\frac{0+0}{3}\right) \right]$$

$$= 2(1 + 0i)$$

$$= 2 + 0i$$

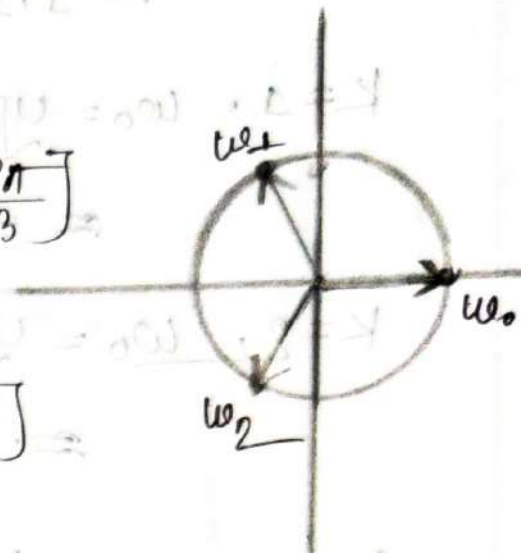
$$k=1, \omega_1 = 2 \left[\cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) \right]$$

$$= -1 + \sqrt{3}i$$

$$k=2, \omega_2 = 2 \left[\cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) \right]$$

$$= -1 - \sqrt{3}i$$

Ans:—



② $z = (-1)^{1/4}$

Is given:—

$$z = (-1)^{1/4}$$

$$\therefore n=4$$

$$k=0, 1, 2, 3$$

$$r = \sqrt{(-1)^2 + 0^2} = 1$$

$$\theta = \arg(z) = \pi$$

We know:—

$$\omega_k = \sqrt[n]{r} \left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i \sin\left(\frac{\theta + 2\pi k}{n}\right) \right]$$

$$\left[\frac{(-1)^{1/4}}{1} \right] \left[\cos\left(\frac{\pi + 2\pi k}{4}\right) + i \sin\left(\frac{\pi + 2\pi k}{4}\right) \right]$$

So, the 4 roots are

$$k=0; \omega_0 = \sqrt[4]{1} \left[\cos\left(\frac{\pi+0}{4}\right) + i \sin\left(\frac{\pi+0}{4}\right) \right]$$

$$= \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$$

$$= \frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$$

Ans.

$$k=1; \omega_1 = \sqrt[4]{1} \left[\cos\left(\frac{\pi+2\pi}{4}\right) + i \sin\left(\frac{\pi+2\pi}{4}\right) \right]$$

$$= -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$$

$$= -\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$$

Ans:-

$$k=2; \omega_2 = \sqrt[4]{1} \left[\cos\left(\frac{\pi+4\pi}{4}\right) + i \sin\left(\frac{\pi+4\pi}{4}\right) \right]$$

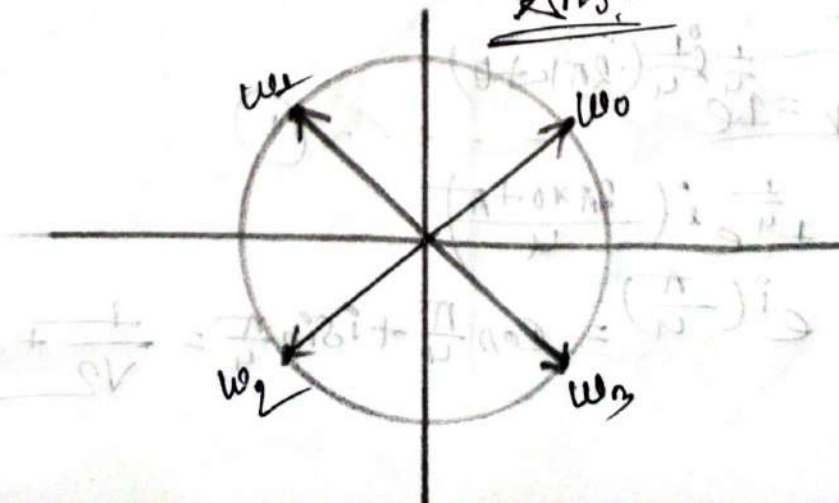
$$= -\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$$

Ans:-

$$k=3; \omega_3 = \sqrt[4]{1} \left[\cos\left(\frac{\pi+6\pi}{4}\right) + i \sin\left(\frac{\pi+6\pi}{4}\right) \right]$$

$$= \frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$$

Ans:-



2nd Approach for Finding a root

Formula:-

$$(i) z_k^n = r^n e^{in(\theta + 2\pi k)}$$

$$(ii) z_k^{\frac{1}{n}} = r^{\frac{1}{n}} e^{i\left(\frac{2\pi k + \theta}{n}\right)}$$

Page 86:-

$$(2) (-1)^{\frac{1}{4}}$$

$\Rightarrow$ In given:-

$$z = (-1)^{\frac{1}{4}}$$

$$\text{So, } \theta = -\pi, \quad y = 0$$

$$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right) = 0$$

$$\therefore \text{Arg}(z) = \pi - \theta$$

$$= \pi - 0$$

$$= \pi$$

Now,

$$r = \sqrt{(-1)^2} = 1$$

$$n = 4$$

$$k = 0, 1, 2, 3$$

We know:-

$$\therefore z_k = r^{\frac{1}{n}} e^{i\left(\frac{2\pi k + \theta}{n}\right)}$$

①

$$k=0, \quad z_0 = r^{\frac{1}{n}} e^{i\left(\frac{2\pi \times 0 + \pi}{4}\right)}$$

$$= e^{i\left(\frac{\pi}{4}\right)} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$$

$$k=1; z_1 = \frac{1}{4} e^{i(\frac{2\pi+\pi}{4})}$$

$$= e^{i(\frac{3\pi}{4})} = (\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}) = -\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$$

$$k=2; z_2 = \frac{1}{4} e^{i(\frac{4\pi+\pi}{4})}$$

$$= e^{i \frac{5\pi}{4}}$$

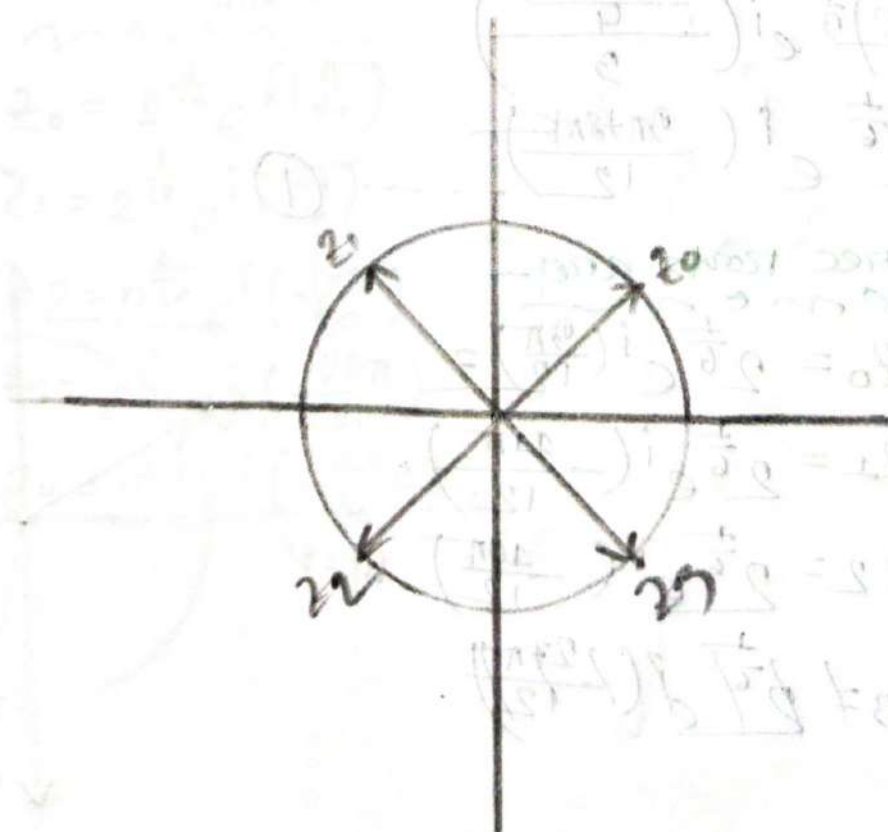
$$= \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$$

$$k=3; z_3 = \frac{1}{4} e^{i(\frac{6\pi+\pi}{4})}$$

$$= e^{i \frac{7\pi}{4}}$$

$$= \cos(\frac{7\pi}{4}) + i \sin \frac{7\pi}{4} = \frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$$

Ans:-



$$\textcircled{7} \quad z = (-1+i)^{\frac{1}{3}}$$

$\Rightarrow$ is given:-
non

$$y = 1$$

$$x = -1$$

$$\begin{aligned} \arg(z) = \theta &= \pi - \tan^{-1}\left(\frac{y}{x}\right) \\ &= \pi - \frac{\pi}{4} = \frac{3\pi}{4} \end{aligned}$$

$$\text{So, } r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$n = 3$$

$$k = 0, 1, 2.$$

we know:-
non

$$z^{\frac{1}{n}} = r^{\frac{1}{n}} e^{i\left(\frac{\theta + 2\pi k}{n}\right)}$$

$$\therefore = r^{\frac{1}{3}} e^{i\left(\frac{3\pi/4 + 2\pi k}{3}\right)}$$

$$= (\sqrt{2})^{\frac{1}{3}} e^{i\left(\frac{3\pi + 8\pi k}{12}\right)}$$

$$= 2^{\frac{1}{6}} e^{i\left(\frac{3\pi + 8\pi k}{12}\right)} \dots \dots \textcircled{1}$$

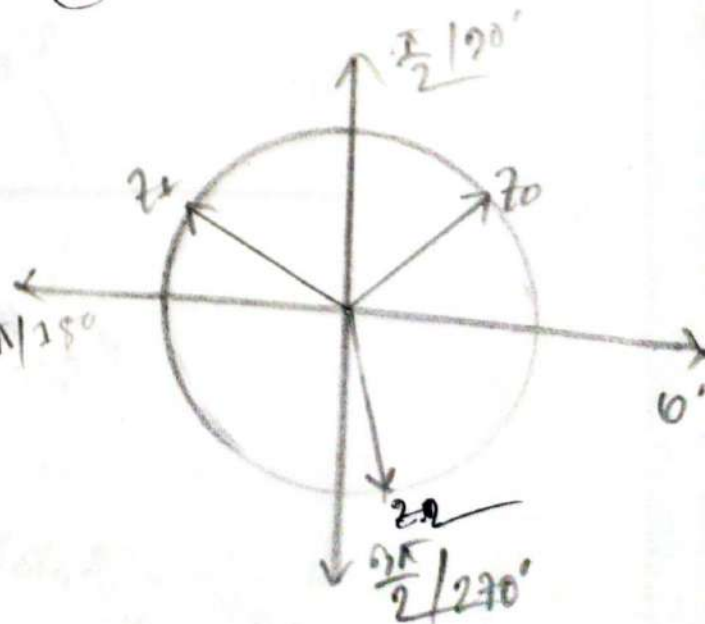
Now, the three roots are:-
non

$$k=0; \quad z_0 = 2^{\frac{1}{6}} e^{i\left(\frac{3\pi}{12}\right)} =$$

$$k=1; \quad z_1 = 2^{\frac{1}{6}} e^{i\left(\frac{11\pi}{12}\right)}$$

$$k=2; \quad z_2 = 2^{\frac{1}{6}} e^{i\left(\frac{19\pi}{12}\right)}$$

$$k=3; \quad z_3 = 2^{\frac{1}{6}} e^{i\left(\frac{27\pi}{12}\right)}$$



$$\textcircled{8} z = (1+i)^{\frac{1}{5}}$$

we have—

$$r = \sqrt{1+1} = \sqrt{2}$$

$$\text{Arg}(z) = \theta = \tan^{-1}\left(\frac{1}{1}\right) = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$n = 5$$

$$k = 0, 1, 2, 3, 4$$

we know—

$$z_k = r^{\frac{1}{n}} e^{i\left(\frac{\theta + 2k\pi}{n}\right)}$$

$$z_k = (\sqrt{2})^{\frac{1}{5}} e^{i\left(\frac{\theta + 2k\pi}{5}\right)}$$

$$= (\sqrt{2})^{\frac{1}{5}} e^{i\left\{\frac{(\pi/4 + 2k\pi)}{5}\right\}}$$

$$= 2^{\frac{1}{10}} e^{i\left(\frac{\pi + 8k\pi}{20}\right)} \dots \textcircled{1}$$

So, the five roots are:

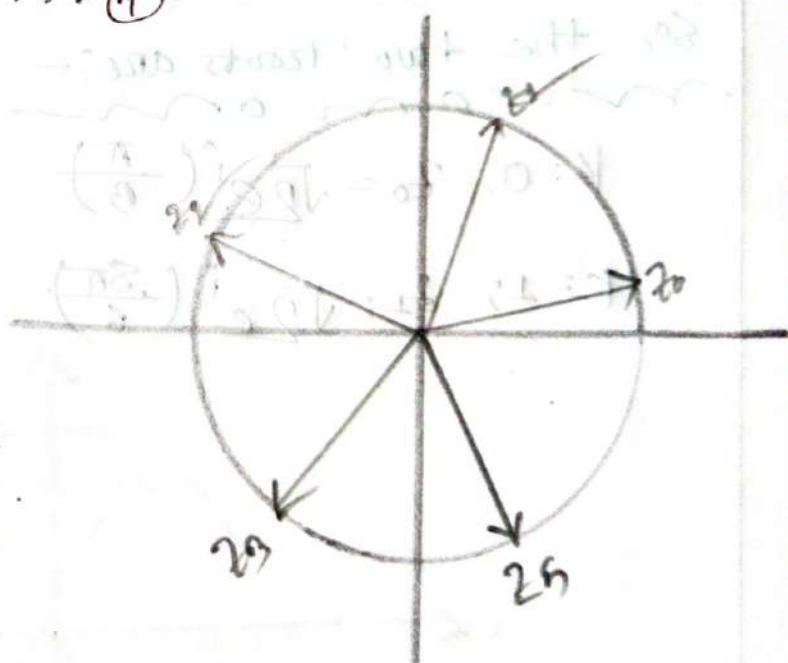
$$k=0; z_0 = 2^{\frac{1}{10}} e^{i\left(\frac{\pi}{20}\right)}$$

$$k=1; z_1 = 2^{\frac{1}{10}} e^{i\left(\frac{9\pi}{20}\right)}$$

$$k=2; z_2 = 2^{\frac{1}{10}} e^{i\left(\frac{17\pi}{20}\right)}$$

$$k=3; z_3 = 2^{\frac{1}{10}} e^{i\left(\frac{25\pi}{20}\right)}$$

$$k=4; z_4 = 2^{\frac{1}{10}} e^{i\left(\frac{33\pi}{20}\right)}$$



① $z = (-1 + \sqrt{3}i)^{\frac{1}{2}}$

⇒ we have:—

$$r = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2$$

$$\therefore \text{Arg}(z) = \theta = \pi - \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \pi - \frac{\pi}{3} = 2\pi/3$$

$$n = 2$$

$$k = 0, 1$$

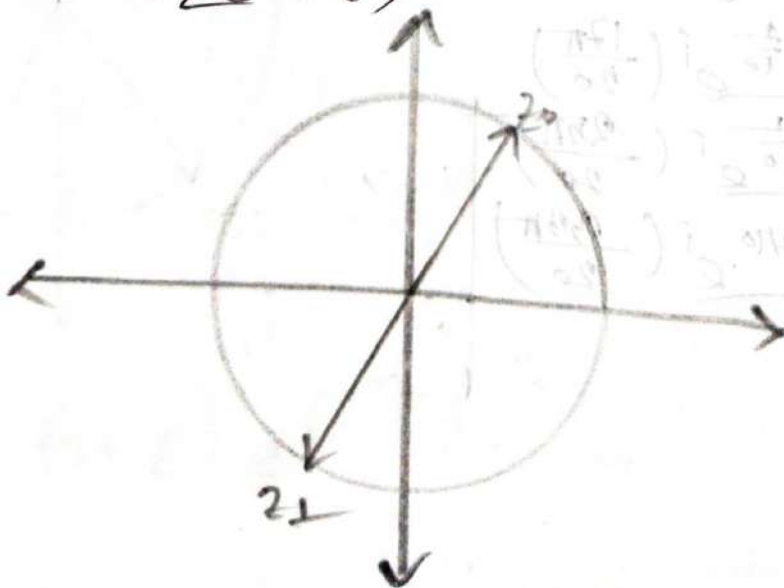
we know:—

$$\begin{aligned} z_k &= (-1 + \sqrt{3}i)^{\frac{1}{2}} \\ &= 2^{\frac{1}{2}} e^{i\left(\frac{\theta + 2\pi k}{2}\right)} \\ &= \sqrt{2} e^{i\left(\frac{2\pi + 6\pi k}{6}\right)} \end{aligned}$$

So, the two roots are:—

$$k=0, z_0 = \sqrt{2} e^{i\left(\frac{\pi}{3}\right)}$$

$$k=1, z_1 = \sqrt{2} e^{i\left(\frac{8\pi}{6}\right)}$$



10) $z = (-1 + \sqrt{3}i)^{\frac{1}{4}}$

we have:

$$r = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\text{Arg}(z) = \theta = -\pi + \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = -\pi + \frac{\pi}{3} = -\frac{2\pi}{3}$$

$$n = 4$$

$$k = 0, 1, 2, 3$$

we know:

$$z^{\frac{1}{n}} = r^{\frac{1}{n}} e^{i\left(\frac{\theta + 2\pi k}{n}\right)}$$

$$\begin{aligned} \therefore z_k &= 2^{\frac{1}{4}} e^{i\left(\frac{-2\pi/3 + 2\pi k}{4}\right)} \\ &= 2^{\frac{1}{4}} e^{i\left(\frac{-2\pi + 6\pi k}{12}\right)} \end{aligned}$$

①

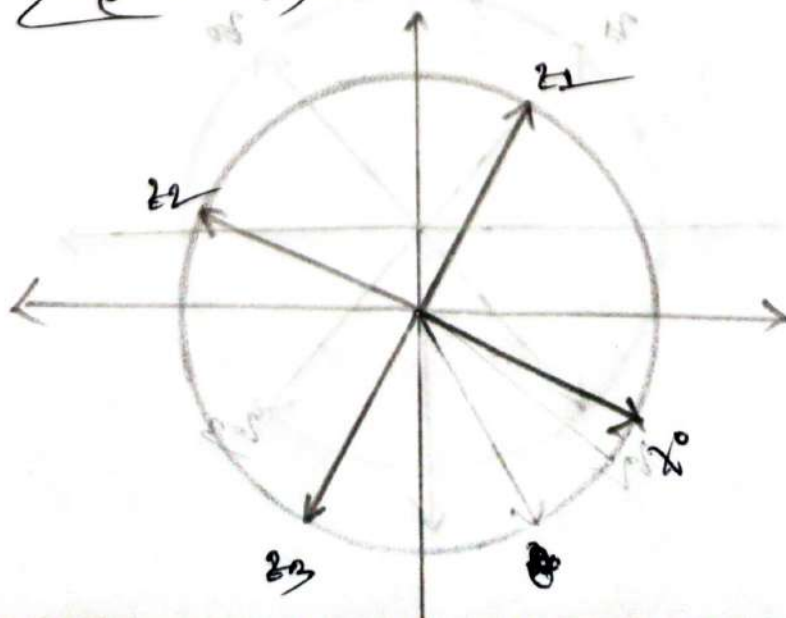
so the 4 roots are:

$$k=0, z_0 = 2^{\frac{1}{4}} e^{i\left(\frac{-\pi}{6}\right)}$$

$$k=1, z_1 = 2^{\frac{1}{4}} e^{i\left(\frac{4\pi}{12}\right)}$$

$$k=2, z_2 = 2^{\frac{1}{4}} e^{i\left(\frac{10\pi}{12}\right)}$$

$$k=3, z_3 = 2^{\frac{1}{4}} e^{i\left(\frac{16\pi}{12}\right)}$$



(17) find the all solutions of the equation $z^4 + 1 = 0$

Is given:

$$z^4 = -1$$

$$\text{So, } z = (-1)^{\frac{1}{4}}$$

$$r = \sqrt{(-1)^2} = 1$$

$$\text{Arg}(z) = \theta = \pi - \tan^{-1}\left(\frac{0}{-1}\right) = \pi - 0 = \pi$$

$$n = 4$$

$$k = 0, 1, 2, 3$$

we know:

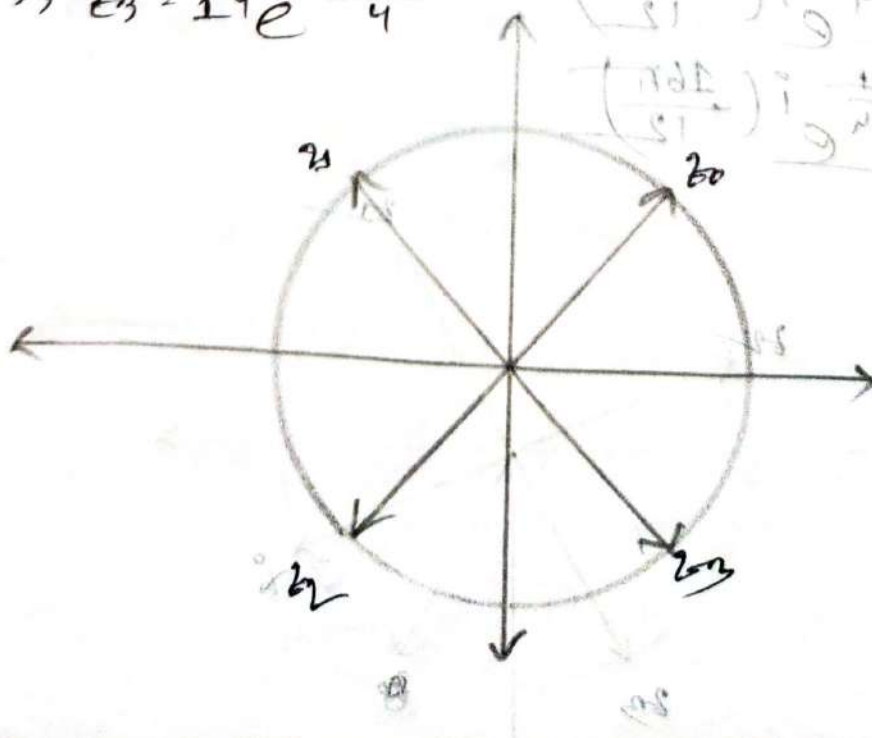
$$z_k = r^{\frac{1}{n}} e^{i\left(\frac{\theta + 2\pi k}{n}\right)}$$

$$\text{So, } k=0; z_0 = 1^{\frac{1}{4}} e^{i\left(-\frac{\pi}{4}\right)}$$

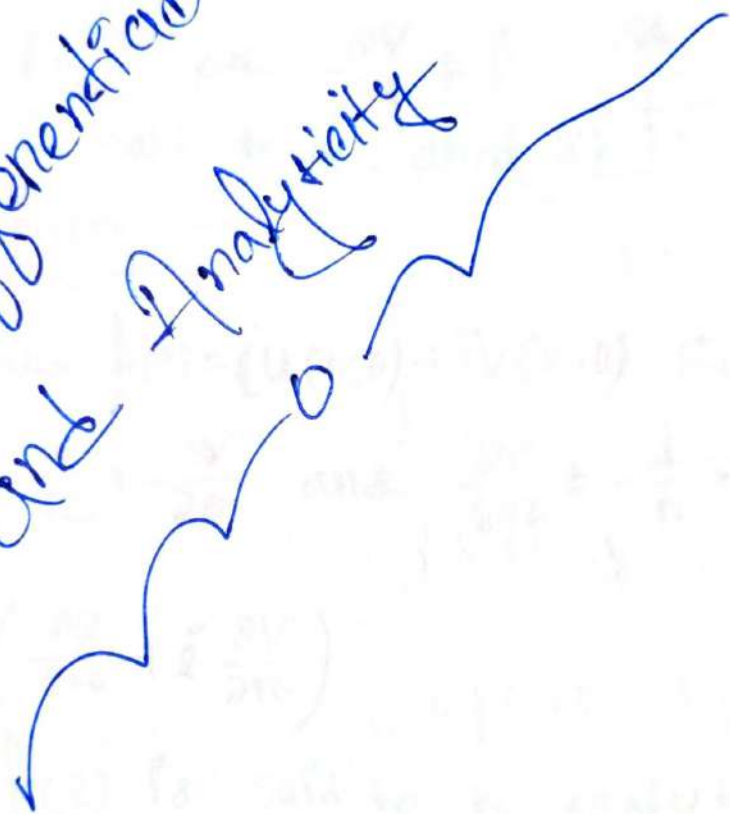
$$k=1; z_1 = 1^{\frac{1}{4}} e^{i\left(\frac{3\pi}{4}\right)}$$

$$k=2; z_2 = 1^{\frac{1}{4}} e^{i\left(\frac{5\pi}{4}\right)}$$

$$k=3; z_3 = 1^{\frac{1}{4}} e^{i\left(\frac{7\pi}{4}\right)}$$



Differentiability
and Analyticity



#Cauchy-Riemann Equations:—

For Rectangular form:—

⇒ If $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$ satisfies the (C-R)

that is $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \text{ or } \frac{\partial v}{\partial y} + i \frac{-\partial u}{\partial y}$$

Then $f(z)$ is said to be analytic

For polar form:—

⇒ If $z = re^{i\theta}$ and $f(z) = u(r, \theta) + iv(r, \theta)$ satisfies the (C-R)

that is $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$

$$f'(z) = e^{i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right)$$

Then $f(z)$ is said to be analytic

Ex—

Q. $z = x + iy$ show that $f(z) = z^2 + z$ function is analytic.
is given:—

$$z = x + iy$$

$$\therefore z^2 + z = (x + iy)^2 + x + iy$$

$$= x^2 + 2xiy + i^2 y^2 + x + iy$$

$$= \underbrace{x^2 - y^2 + x}_u + i \underbrace{(2xy + y)}_v$$

So, $u = x^2 + y^2$
 $v = 2xy$

Now

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

So, $\frac{\partial u}{\partial x} = 2x$; $\frac{\partial u}{\partial y} = 2y$

and $\frac{\partial v}{\partial x} = 2y$; $\frac{\partial v}{\partial y} = 2x$

So, $\frac{\partial u}{\partial x} = 2x$; $\frac{\partial v}{\partial y} = 2x$

$$\frac{\partial u}{\partial y} = 2y = -\frac{\partial v}{\partial x}$$

Then, $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$ or $\frac{\partial v}{\partial y} + i \frac{\partial u}{\partial y}$

$$\therefore f'(z) = 2x + i2y \text{ or } 2x + i2y$$

$$f'(z) = 2x + i2y \text{ or } 2x + i2y$$

Ans:-

So, the function is analytic

* $f(z) = \frac{\cos \theta}{r} - i \sin \theta / r$ is analytic or not show that.

$\Rightarrow$ here, $u = \cos \theta / r$

$v = -\sin \theta / r$

we know

$$\frac{\partial u}{\partial r} = \frac{1}{r^2} \frac{\partial v}{\partial \theta} \text{ and } \frac{\partial v}{\partial r} = -\frac{1}{r^2} \frac{\partial u}{\partial \theta}$$

so $\frac{\partial u}{\partial r} = \frac{-1}{r^2} \cos \theta ; \frac{\partial v}{\partial r} = +\frac{1}{r^2} \sin \theta$

and $\frac{\partial v}{\partial \theta} = \frac{-1}{r} \sin \theta ; \frac{\partial u}{\partial \theta} = -\frac{1}{r} \sin \theta$

$$\left(\begin{aligned} \frac{d}{d\theta} \sin \theta &= \cos \theta \\ \frac{d}{d\theta} \cos \theta &= -\sin \theta \end{aligned} \right)$$

so $\frac{\partial u}{\partial r} = \frac{1}{r^2} \frac{\partial v}{\partial \theta} = \frac{1}{r^2} \cos \theta = \frac{1}{r^2} \cos \theta$

$$\Rightarrow \frac{1}{r^2} \cos \theta = \frac{1}{r^2} \cos \theta$$

Again:- $\frac{\partial v}{\partial r} = \frac{1}{r^2} \frac{\partial u}{\partial \theta} \Rightarrow +\frac{1}{r^2} \sin \theta = -\frac{1}{r^2} \sin \theta$

$$\Rightarrow \frac{1}{r^2} \sin \theta = \frac{1}{r^2} \sin \theta$$

so, the function is analytic.

$$\begin{aligned} f'(z) &= e^{-i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) \\ &= e^{-i\theta} \left(-\frac{1}{r^2} \cos \theta + i \frac{1}{r^2} \sin \theta \right) \end{aligned}$$

Agg.

$$\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} = \frac{1}{r^2} \cos \theta + \frac{1}{r^2} \sin \theta = \frac{1}{r^2} (\cos \theta + i \sin \theta)$$

Page 167

† Show that the function $z = x + iy$ are analytic at any point.

① $f(z) = z^3$

⇒ So, $f(x + iy) = (x + iy)^3$

$$= x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3$$

$$= x^3 + 3x^2iy - 3xy^2 - y^3i$$

$$= (x^3 - 3xy^2) + i(3x^2y - y^3)$$

now

$$u = x^3 - 3xy^2$$

$$v = 3x^2y - y^3$$

So

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\text{and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial u}{\partial x} = 3x^2 - 3y^2 ; \quad \frac{\partial v}{\partial y} = 3x^2 - y^3$$

$$\frac{\partial v}{\partial x} = 6xy$$

$$; \quad \frac{\partial u}{\partial y} = -6xy$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 3x^2 - y^3$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 6xy$$

∴ $f(z)$ satisfies the Cauchy-R equation at every point

$$\therefore f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 3x^2 - y^3 + i6xy$$

$$\textcircled{2} f(z) = 3z^2 + 5z - 6i$$

$$\therefore f(u+iv) = 3(u+iv)^2 + 5(u+iv) - 6i$$

$$= 3(u^2 + i2uv + v^2i^2) + 5u + 5iv - 6i$$

$$= 3u^2 - 3v^2 + i6uv + 5u + 5iv - 6i$$

$$\Rightarrow (3u^2 - 3v^2 + 5u) + i(6uv + 5v - 6)$$

$$\therefore u = 3u^2 - 3v^2 + 5u$$

$$v = 6uv + 5v - 6$$

we know

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial x} \quad \frac{\partial v}{\partial x} = - \frac{\partial v}{\partial y}$$

$$\therefore \frac{\partial u}{\partial x} = 6u + 5 \quad ; \quad \frac{\partial v}{\partial y} = 6u + 5$$

$$\frac{\partial v}{\partial x} = 6y + 0 \quad ; \quad \frac{\partial u}{\partial y} = -6y$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 6u + 5$$

$$\frac{\partial v}{\partial x} = - \frac{\partial u}{\partial y} = -6y$$

The equation Satisfy @-R equation.

$$\text{now, } f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= \frac{6u+5}{1} + i6y$$

$$= 6u+5+i6y$$

Ans:

Show that the function $z = x + iy$
are not analytic at any point

③ $f(z) = \operatorname{Re}(z)$

$\therefore f(x+iy) = \operatorname{Re}(z) = x$

$u = x$
 $v = 0$

we know — $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$\therefore \frac{\partial u}{\partial x} = 1, \frac{\partial v}{\partial y} = 0$

$\frac{\partial v}{\partial x} = 0, \frac{\partial u}{\partial y} = 0$

$\therefore$ so, $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$

$\therefore$ The C-R equation are not satisfied.

So, the function is not analytic at any point

④ $f(z) = y + ix$

$\therefore u = y$
 $v = x$

we know — $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$\therefore \frac{\partial u}{\partial x} = 0, \frac{\partial v}{\partial y} = 0; \frac{\partial u}{\partial y} = 1, \frac{\partial v}{\partial x} = 1$

Since $\frac{\partial v}{\partial x} \neq \frac{\partial u}{\partial y}$

So, The function $e-z$ are not satisfied.

The function is not analytic at any point.

(5) $f(z) = 4z - 6\bar{z} + 3$

$$\begin{aligned} \therefore f(x+iy) &= 4(x+iy) - 6(x-iy) + 3 \\ &= 4x + 4iy - 6x + 6iy + 3 \\ &= -2x + 10iy + 3 \\ &= -2x + 3 + 10iy \end{aligned}$$

So, $u = -2x + 3$

$v = 10y$

We know

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial u}{\partial x} = -2, \quad \frac{\partial v}{\partial y} = 10, \quad \frac{\partial v}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = 0$$

Since $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$

The function is not analytic at any point.

(6) $f(z) = \bar{z}^2$

$$\begin{aligned} \Rightarrow f(x+iy) &= (x-iy)^2 = x^2 - 2xyi + i^2y^2 \\ &= x^2 - y^2 - i2xy \end{aligned}$$

$\therefore u = x^2 - y^2$

$v = -2xy$

We know-

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = -2x, \quad \frac{\partial v}{\partial x} = -2y; \quad \frac{\partial u}{\partial y} = +2y$$

$$\text{Since } \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

So The function is not analytic at any point.

⑦ $f(z) = x^2 + y^2$

$$\therefore u = x^2 + y^2$$

$$v = 0$$

We know-

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\therefore 2x = \frac{\partial u}{\partial x}, \quad \frac{\partial v}{\partial y} = 0, \quad \frac{\partial v}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = 2y$$

$$\text{Since } \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

So The function is not analytic at any point.

⑧ $f(z) = \frac{x}{x^2 + y^2} + i \frac{y}{x^2 + y^2}$

$$\therefore u = \frac{x}{x^2 + y^2}$$

$$v = \frac{y}{x^2 + y^2}$$

We know—

$$\frac{\partial u}{\partial m} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial m} = -\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial u}{\partial m} = \frac{\partial \left(\frac{m}{m^2 + y^2} \right)}{\partial m} =$$

$$\frac{\partial v}{\partial y} = \frac{\partial \left(\frac{y}{m^2 + y^2} \right)}{\partial y} =$$

$$\frac{\partial v}{\partial m} = \frac{\partial \left(\frac{y}{m^2 + y^2} \right)}{\partial m} =$$

$$\frac{\partial u}{\partial y} = \frac{\partial \left(\frac{m}{m^2 + y^2} \right)}{\partial y} =$$

Since $\frac{\partial u}{\partial y} \neq \frac{\partial v}{\partial x}$

So, the function is not Analytic at any point

☑ Show that the given function is analytic or not:—

① $f(z) = e^{-x} \cos y - i e^{-x} \sin y$

$\therefore u = e^{-x} \cos y$

$v = -e^{-x} \sin y$

We know—

$$\frac{\partial u}{\partial m} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial m} = -\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial u}{\partial m} = -e^{-x} \cos y; \quad \frac{\partial v}{\partial y} = -e^{-x} \cos y$$

$$\frac{\partial v}{\partial m} = e^{-x} \sin y; \quad \frac{\partial u}{\partial y} = -e^{-x} \sin y$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = -e^{-x} \cos y$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = +e^{-x} \sin y$$

so, $f(z)$ is analytic at any point

$$\therefore f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ = -e^{-x} \cos y + i e^{-x} \sin y$$

$$(10) f(z) = x + \sinh x \cdot \cosh y + i(y + \cosh x \cdot \sinh y)$$

$$\therefore u = x + \sinh x \cdot \cosh y$$

$$v = y + \cosh x \cdot \sinh y$$

We know,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (x + \sinh x \cdot \cosh y) = 1 + \cosh x \cdot \cosh y$$

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x} (y + \cosh x \cdot \sinh y) = \sinh y$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} (y + \cosh x \cdot \sinh y) = 1 + \cosh x \cdot \cosh y$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x + \sinh x \cdot \cosh y) = \sinh x \cdot \sinh y$$

Now $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 1 + \cos x \cdot \cosh y$

$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -\sin x \cdot \sinh y$

So, $f(z)$ is analytic at any point

$\therefore f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$
 $= 1 + \cos x \cdot \cosh y - i \sin x \cdot \sinh y$

(11) $f(z) = e^{x^2-y^2} \cdot \cos 2xy + i e^{x^2-y^2} \sin 2xy$

$\therefore u = e^{x^2-y^2} \cdot \cos 2xy$

$v = e^{x^2-y^2} \cdot \sin 2xy$

We know,

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} e^{x^2-y^2} \cdot \cos 2xy = 2x e^{x^2-y^2} \cdot \cos 2xy - e^{x^2-y^2} \cdot \sin 2xy \cdot 2y$
 $= 2e^{x^2-y^2} (x \cos 2xy - y \sin 2xy)$

$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x} e^{x^2-y^2} \cdot \sin 2xy = 2x e^{x^2-y^2} \cdot \sin 2xy + e^{x^2-y^2} \cdot \cos 2xy \cdot 2y$
 $= 2e^{x^2-y^2} (x \sin 2xy + y \cos 2xy)$

$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} e^{x^2-y^2} \cdot \sin 2xy = -2y e^{x^2-y^2} \cdot \sin 2xy + e^{x^2-y^2} \cdot \cos 2xy \cdot 2x$
 $= 2e^{x^2-y^2} (-y \sin 2xy + x \cos 2xy)$

$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} e^{x^2-y^2} \cdot \cos 2xy = -2y e^{x^2-y^2} \cdot \cos 2xy - e^{x^2-y^2} \cdot \sin 2xy \cdot 2x$
 $= -2e^{x^2-y^2} (y \cos 2xy + x \sin 2xy)$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

so $f(z)$ is analytic at any point.

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 2e^{x^2-y^2}(-y \sin 2xy + x \cos 2xy) + i 2e^{x^2-y^2}(x \sin 2xy + y \cos 2xy)$$

Ans:-

$$(10) f(z) = 4x^2 + 5x - 4y^2 + 9 + i(8xy + 5y - 1)$$

$$u = 4x^2 + 5x - 4y^2 + 9$$

$$v = 8xy + 5y - 1$$

We know

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\therefore \frac{\partial u}{\partial x} = 8x + 5, \quad \frac{\partial v}{\partial y} = 8x + 5$$

$$\frac{\partial v}{\partial x} = 8y; \quad \frac{\partial u}{\partial y} = -8y$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 8x + 5$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 8y$$

$\therefore f(z)$ is analytic at any point

$$\text{so, } f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= 8x + 5 + i 8y$$

$$(16) f(z) = 5r \cos \theta + r^4 \cos 4\theta + i(5r \sin \theta + r^4 \sin 4\theta) \quad (1)$$

$$\Rightarrow u = 5r \cos \theta + r^4 \cos 4\theta \quad (2)$$

$$v = 5r \sin \theta + r^4 \sin 4\theta \quad (3)$$

we know:-

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

$$\therefore \frac{\partial u}{\partial r} = 5 \cos \theta + 4r^3 \cos 4\theta$$

$$\frac{\partial v}{\partial \theta} = 5r \cos \theta + 4r^4 \cos 4\theta$$

$$\frac{\partial v}{\partial r} = 5 \sin \theta + 4r^3 \sin 4\theta$$

$$\frac{\partial u}{\partial \theta} = -5r \sin \theta + r^4 (-\sin 4\theta \times 4)$$

$$\therefore \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} = 5 \cos \theta + 4r^3 \cos 4\theta$$

$$\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta} = 5 \sin \theta + 4r^3 \sin 4\theta$$

$\therefore f(z)$ is analytic at any point.

$$\therefore z' = e^{i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial \theta} \right)$$

$$= e^{i\theta} (5 \cos \theta + 4r^3 \cos 4\theta + i(5r \sin \theta + 4r^4 \sin 4\theta))$$

Ans:-

(15/05/19)

25) $f(z) = e^z = e^x \cos y + i e^x \sin y$

(a) Show that $f(z) = e^z$ is an entire function.

(b) Show that $f'(z) = f(z)$

Is given:-

$$f(z) = e^z = e^x \cos y + i e^x \sin y$$

$$u = e^x \cos y$$

$$v = e^x \sin y$$

we know:-

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\frac{\partial u}{\partial x} = e^x \cos y, \quad \frac{\partial v}{\partial x} = e^x \sin y$$

$$\frac{\partial v}{\partial y} = e^x \cos y, \quad \frac{\partial u}{\partial y} = -e^x \sin y$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = e^x \cos y$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = e^x \sin y$$

(b)

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= e^x \cos y + i e^x \sin y$$

$$= f(z)$$

(Proved)



The End